



SWINBURNE UNIVERSITY OF TECHNOLOGY

COS20044 - Introduction to Quantum Computation and Applications of Quantum Technologies

Quantum Labs 1 - 6 Portfolio

Computational Laboratory Artefacts: Weeks 1–6

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Student Name	Karamjot Singh
Student ID	104809887
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Week 1 Artefact - Moore's Law

Historical context, numerical investigation, physical interpretation, and the scaling path from genAI to AGI.

Historical context

Digital computing milestones

1. ENIAC (1945)

ENIAC was the first large-scale electronic general-purpose computer. It used around 17,500 vacuum tubes, weighed about 27 tonnes, took up roughly 170 m² of floor space, and drew about 150 kW of power. It could do around 5,000 additions per second.

The speed is not really the point. What mattered is that it proved a machine could replace rooms full of people doing calculations by hand. Before ENIAC, the word "computer" meant a person with a desk calculator. ENIAC was built to calculate artillery firing tables, work that took a human days per table.

Its big weakness was programming. Changing what ENIAC did meant physically rewiring plugboards, which took days. The idea that fixed this came a few years later: store the program in the same memory as the data. That stored-program design (EDVAC, and the Manchester Baby in 1948) is arguably a bigger conceptual leap than ENIAC itself, and every computer since has used it.

2. IBM System/360 (1964)

The System/360 mattered for a commercial reason rather than a physical one. It separated the architecture from the implementation. IBM sold a family of machines at different prices, and a program written for the cheapest one would run on the most expensive one without changes.

This sounds obvious now. At the time it was not. Before the System/360, buying a faster computer meant rewriting all your software, so upgrading was painful and expensive. IBM reportedly spent around \$5 billion on the project, which in today's money is more than the Manhattan Project cost.

What I take from this is that Moore's law was never purely a physics story. Doubling the transistors every two years only makes business sense if someone will buy the extra transistors. Software compatibility is what created that demand. If every new chip had broken all existing software, the industry would not have kept investing.

3. Intel 4004 (1971)

The 4004 had 2,300 transistors on a 10 μm pMOS process, ran at 740 kHz, and was 4-bit. Its raw performance was roughly comparable to ENIAC, but it fitted on a chip of about 12 mm².

I use 1971 as the starting point for my dataset in Section 2 because the 4004 was the first commercial single-chip CPU. From this point on, the whole processor shrinks with each lithography improvement, so "transistors per chip" becomes a number that actually means something and can be tracked over time.

4. Where are we today (2024–2025)

Leading-edge chips now have around 2×10^{11} transistors. NVIDIA's Blackwell B200 is about 208 billion transistors across two joined dies, and Apple's M1/M2 Ultra is 114–134 billion. Manufacturing is at TSMC's 3 nm class node.

One thing I did notice: the "3 nm" name is marketing, not a measurement. Nothing on the chip is 3 nm across. The actual gate length is closer to 16 nm and the metal pitch is around 21 nm. Node names stopped matching physical dimensions years ago.

I think that naming drift is itself evidence. When the number on the label stops describing anything real, it usually means the thing it used to describe has stopped happening. Classic geometric shrinking ended some time ago, and modern progress comes from architecture (FinFET, gate-all-around nanosheets), packaging (chiplets, 3D stacking), and specialised chips rather than from making everything smaller.

Impossible, then unremarkable

What strikes me most is how fast "impossible" turns into "invisible":

➤ Numerical weather prediction

- In 1922 Lewis Fry Richardson estimated it would take 64,000 human computers working non-stop just to keep up with the weather as it happened. Today a phone shows a ten-day forecast that beats a 1980s three-day forecast for accuracy, and nobody finds it interesting.

➤ Personal photography without scarcity

- Film cost money per shot, so every photo was a decision. That constraint vanished so completely that the problem flipped. It went from "I can't afford another photo" to something like "I can't find the photo among thousands."

➤ Real-time video communication

- Demonstrated at the 1964 World's Fair as an expensive curiosity, commercially dead for forty years, then became free and in 2020 became how a lot of the world worked.
- **Turn-by-turn navigation**
 - Needs GPS *and* enough cheap local computing power to search a national road network in real time. GPS is fundamentally an atomic clock technology.
- **Machine translation and speech recognition**
 - Both were the classic "always ten years away" AI problems for about fifty years, then became usable within roughly a decade of GPU deep learning becoming affordable.

I picked the things that worked. People were equally confident about fusion power, household robots, and self-driving cars being a year away since about 2015. Those did not arrive on an exponential schedule, because they are not limited by transistor count alone. Faster chips are necessary for a lot of things but not sufficient for everything.

Numerical investigation

Exponential growth on linear and logarithmic scales

I looked at exponential growth two ways: with the provided widget, and with my own plots.

➤ The widget

The widget shows a square grid where the number of coloured cells doubles every τ seconds. The grid has a fixed number of cells, so you can watch the doubling run out of space.



Figure 1: Exponential growth widget, small grid part-way through. At this stage the growth looks gentle and nothing suggests the grid is about to fill

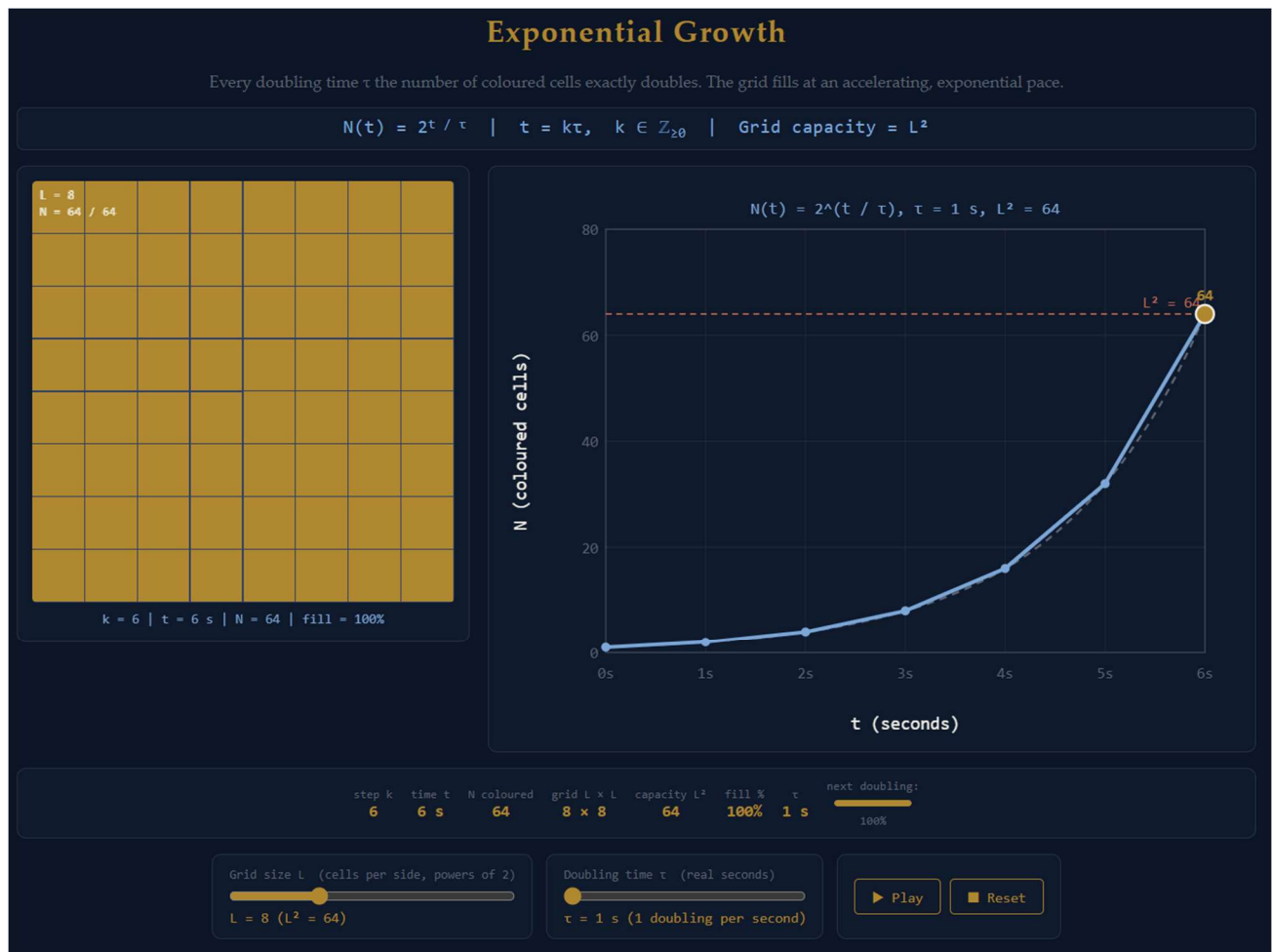


Figure 2: The same 64-cell grid, full after exactly 6 doublings



Figure 3: The largest grid: over a million cells, full after exactly 20 doublings

The rule the widget shows is that a grid of $L = 2^n$ cells per side has $L^2 = 2^{2n}$ cells and fills in $2n$ doublings. So:

Grid	Cells	Doublings to fill
8 x 8	64	6
32 x 32	1,024	10
1024 x 1024	1,048,576	20

This is the result I found most useful in this week. Going from 64 cells to over a million cells is 16,000 times more room, and it only buys 14 extra doublings. Making the resource massively bigger barely delays the problem, because the resource grows in a straight line while the demand grows exponentially.

I also ran the same grid ($L = 32$) at $\tau = 1 \text{ s}$ and $\tau = 5 \text{ s}$. It still took 10 doublings both times, just 10 seconds versus 50 seconds. Changing the doubling time changes the deadline but not the shape of the problem.

My own plots

I plotted $N(t) = 2^{(t/2)}$ on linear and log axes.

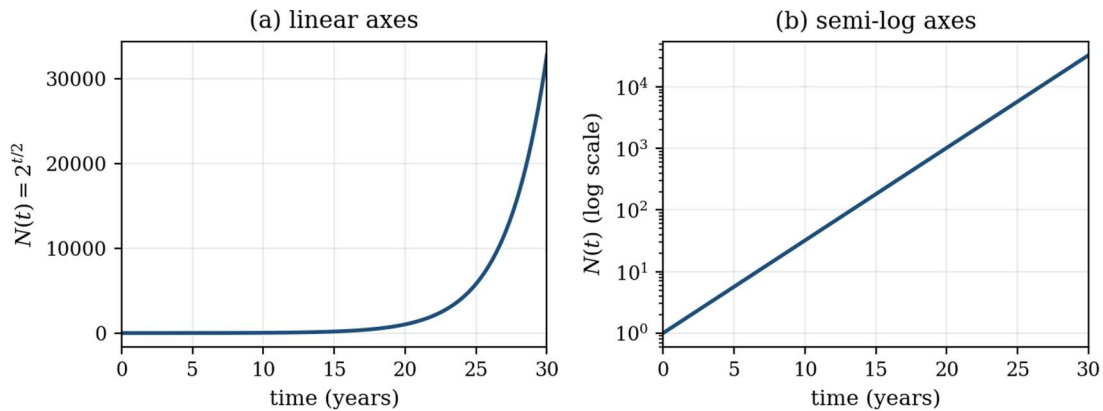


Figure 4: The same exponential $N(t)=2^{(t/2)}$ plotted on (a) linear and (b) semi-logarithmic axes. On linear axes the first 20 years appear flat and the eye cannot extract a growth rate; on log axes the gradient $d(\log_2 N)/dt=1/2$ directly gives the doubling time of 2 years

A straight line on a log plot is a claim you can test: it says $\log_2 N$ is linear in time, so any curve in the data means the growth law is changing. Every plot after this one uses log axes for that reason.

Fitting Moore's law to real data

I built a dataset of 23 flagship processors from the Intel 4004 (1971) to the NVIDIA B200 (2024) and fitted

$$\log_2 N(t) = \frac{t - t_0}{T_2} + \log_2 N_0, \quad \text{or} \quad N(t) = N_0 \cdot 2^{(t-t_0)/T_2}$$

using least squares on $\log_2 N$ with $t_0 = 1971$.

Fitting in log space matters. If I had fitted in linear space, the last three points would completely dominate the fit, because they are eight orders of magnitude bigger than the first one. The early data would effectively be ignored.

My results:

Range	Doubling time T_2	R^2
1971–2024 (all)	2.03 years	0.9935
1971–2002	2.18 years	0.9891
2003–2024	2.11 years	0.9588

The fitted law is

$$N(t) = 2.05 \times 10^3 \cdot 2^{(t-1971)/2.03}$$

Over 53 years that is a factor of 9.0×10^7 , or 26.4 doublings.

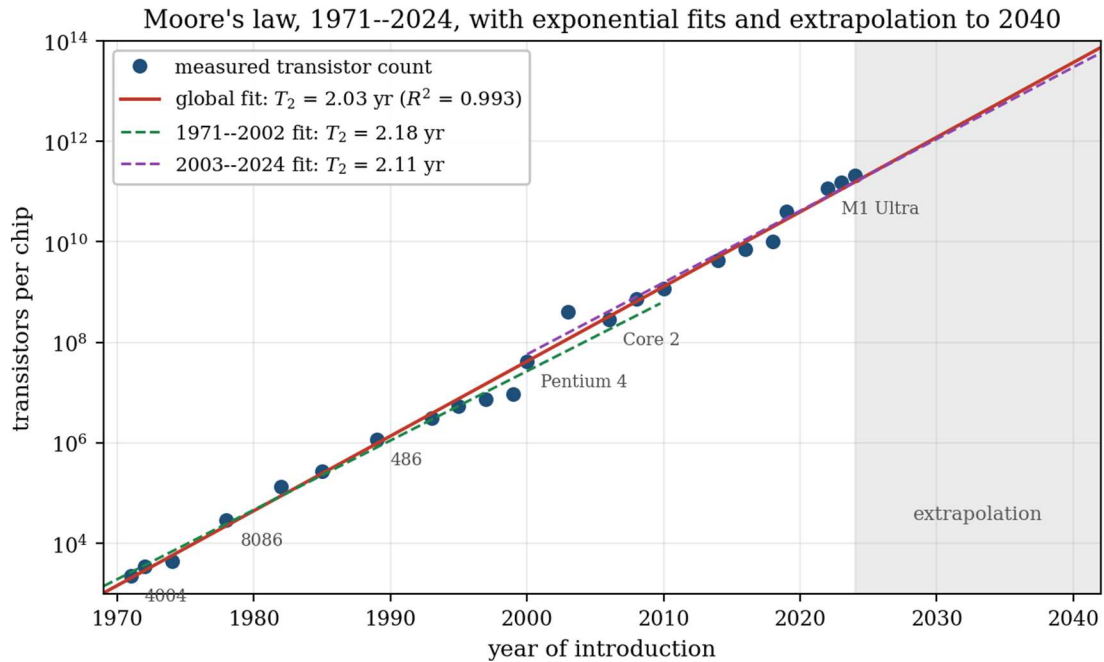


Figure 5: Transistor count of flagship microprocessors, 1971–2024, with least-squares exponential fits. The global fit gives a doubling time of 2.03 years and explains 99.35% of the variance in $\log_2 N$. The shaded region is the extrapolation examined.

What the residuals reveal

An R^2 of 0.9935 is easy to over-read, so I plotted the residuals. This is the most interesting result in the section.

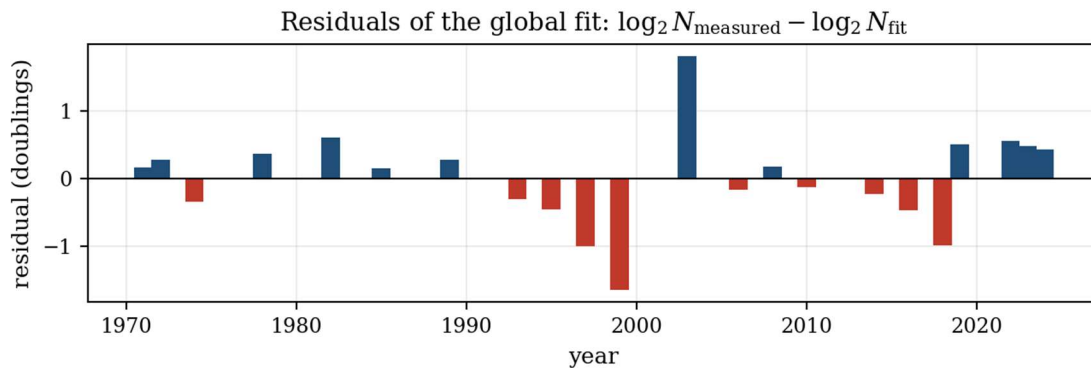


Figure 6: Residuals of the global fit, in doublings. The pattern is structured rather than random. The largest single residual is 1.81 doublings (a factor of 3.5) for the Itanium 2.

Three observations came out of this.

First, the residuals are not random noise. There are runs of several points in a row on the same side of the line. That breaks the independence assumption behind ordinary least squares, so the R^2 overstates how good the fit really is as a predictor.

Second, the biggest outlier is the Itanium 2 in 2003, which sits 1.81 doublings above the line. That is not a manufacturing breakthrough. Most of that chip is cache SRAM, which packs far more densely than logic because its layout is simple and repetitive. This shows a real weakness in the metric: "transistors per chip" quietly mixes logic and memory, and those two things scale differently. I would not have noticed this without plotting the residuals.

Third, and this is the part I think matters most: a good fit does not make Moore's law a law of nature. It was an industry roadmap that companies deliberately aimed at. The ITRS and later IRDS roadmaps set the two-year cadence as a target and companies invested to hit it. So the straight line is partly self-fulfilling, which is exactly why it could stop when the economics changed rather than when the physics ran out.

Issue encountered – What I got wrong at first

My first attempt just fitted the whole range and reported $R^2 = 0.99$ as if that settled it. Splitting the range showed the doubling times before and after 2003 are 2.18 and 2.11 years, which is basically the same, but R^2 drops from 0.989 to 0.959. The *rate* barely changed. The *scatter* grew. That is the real signal, and I would have missed it completely by quoting one number. I think the growing scatter reflects the industry splitting apart after 2005, when GPUs, phone chips and server CPUs stopped being comparable products and "the flagship chip" stopped being a well-defined thing.

What if Moore's law continues to 2040?

Extrapolating my fit to 2040 gives:

$$N(2040) = 2.05 \times 10^3 \cdot 2^{(2040 - 1970)/2.03} = 3.70 \times 10^{13} \text{ transistors.}$$

To test whether that is physically possible, I compared it to the number of atoms available. Silicon has a lattice constant of 0.543 nm, so one atomic layer holds $1/(0.543 \text{ nm})^2 = 3.4 \times 10^{18}$ atoms per m^2 . On a generous 800 mm^2 die, that is 2.71×10^{15} surface atoms.

Area per transistor, 2024 (208 billion on 800 mm^2)	$3.85 \times 10^{-15} \text{ m}^2$, a 62 nm square
Area per transistor, 2040 (extrapolated)	$2.16 \times 10^{-17} \text{ m}^2$, a 4.65 nm square
Surface silicon atoms per transistor in 2040	73
Year the fit reaches one atom per transistor	2053

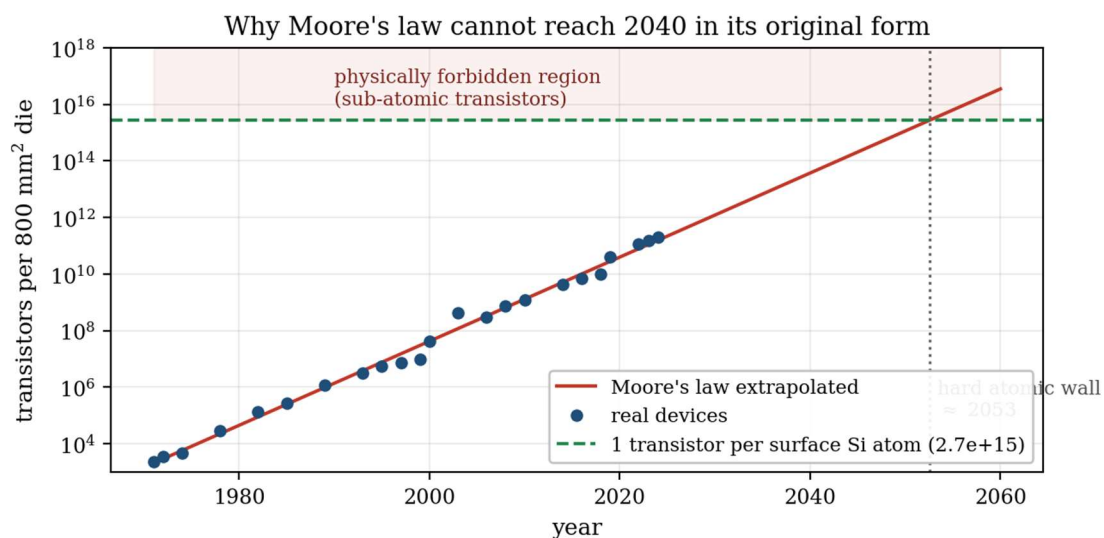


Figure 7: Extrapolated Moore's law against the atomic budget of an 800 mm^2 die. The line reaches one transistor per surface silicon atom in 2053, but devices become unworkable well before that.

I wish to be careful here, because the simple version of this argument is wrong and I made it first. The 2040 number is not literally sub-atomic. It allows about 73 surface atoms per device, roughly a 9×9 square of atoms.

But a working transistor needs a source, a drain, a channel long enough to actually switch off, a gate stack, and isolation from its neighbours. At 9 atoms across, none of those regions can be more than two or three atoms wide. The channel could not hold off leakage. Dopant fluctuation would make every device different, because at that volume a transistor contains zero or one dopant atoms, so the threshold voltage becomes a random number. The gate oxide would be about two atoms thick.

A counter-argument I had to take seriously

My area argument assumes a flat, two-dimensional chip, and that assumption is already out of date. 3D NAND stacks over 200 layers. Chiplet packaging (AMD's EPYC, Intel's Foveros) already puts several dies in one package. CFET designs stack nMOS on top of pMOS. If the metric is transistors *per package* and stacking continues, the 2D atomic wall is not the binding limit. Heat is, because stacked layers have no path to a heatsink.

So, my honest conclusion is not "Moore's law dies at the atom." It is that Moore's law *as Moore described it* - shrinking a flat circuit at constant cost per transistor - ended around the mid-2010s. What continues is a more expensive, messier growth in device count that uses different tricks.

Two further reasons the 2040 extrapolation is very unlikely:

➤ Rock's law (Moore's second law)

- The capital cost of a leading-edge fab has itself grown exponentially, roughly doubling every four years: a 1970s fab cost ~\$4M; a modern 3 nm fab costs \$20B or more. Rock's law and Moore's law are on a collision course - at some point the number of companies that can afford the next node falls below one. It has already fallen to three (TSMC, Samsung, Intel), from more than twenty in 2001.

➤ Cost per transistor stopped falling

- The historically essential half of Moore's original claim was cost. Below roughly the 28 nm node, multi-patterning and EUV made cost per transistor flat or rising for many designs. An exponential in *count* that is no longer an exponential in *value* does not sustain the investment that drove it.

Gordon Moore's banana flies

If one fly in 1965 doubles weekly with unlimited food, then after t years the population is $2^{t \cdot 365.2425/7}$. From 1965 to 2026 is $61 \times 365.2425/7 = 3182.8$ weeks, giving

$$N_{\text{flies}} = 2^{3183} = 10^{958.1}$$

Taking about 1 mg for a fruit fly:

$$M = N_{\text{flies}} \times 10^{-6} \text{ kg} = \mathbf{10^{952} \text{ kg.}}$$

For comparison, the Earth is 5.97×10^{24} kg and the observable Universe roughly 1.5×10^{53} kg. The fly biomass therefore exceeds the mass of the observable Universe by a factor of 10^{899} .

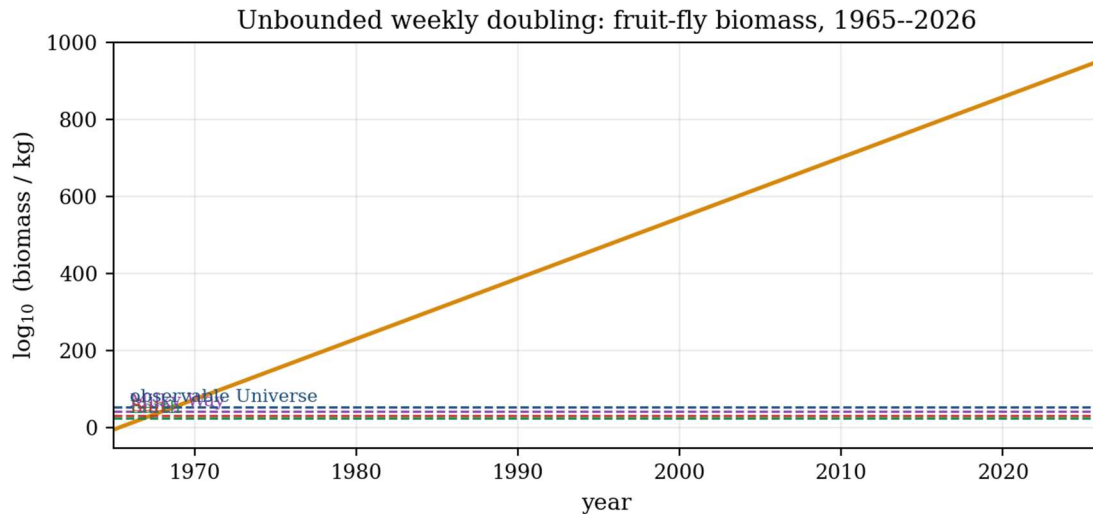


Figure 8: Fruit-fly biomass under unbounded weekly doubling from a single fly in 1965. The vertical axis is $\log_{10}$ of the mass in kg, so the straight line is on axes that are already logarithmic.

Physical interpretation: why it worked, and what stopped it

Dennard scaling: why it worked so long

Robert Dennard's 1974 scaling rules are the engine underneath Moore's law. If every linear dimension and the supply voltage are divided by a factor $\kappa > 1$, and doping is scaled up by κ , then for a MOSFET:

$$\begin{aligned} \text{gate capacitance } C &= \frac{\epsilon_{\text{ox}} WL}{t_{\text{ox}}} \rightarrow C/\kappa, \\ \text{delay } \tau &\sim \frac{CV_{dd}}{I_{\text{on}}} \rightarrow \tau/\kappa \Rightarrow f \rightarrow \kappa f, \\ \text{dynamic power } P &= \alpha CV_{dd}^2 f \rightarrow \frac{1}{\kappa} \cdot \frac{1}{\kappa^2} \cdot \kappa P = P/\kappa^2, \\ \text{power density } P/A &\rightarrow \frac{P/\kappa^2}{A/\kappa^2} = \text{constant}. \end{aligned}$$

This is the remarkable result: every generation gave κ^2 times more transistors, κ times faster, *for the same power per unit area*. Performance was genuinely free.

I think this is the key thing to understand about the whole period. Dennard scaling, not Moore's observation, is why the exponential lasted thirty years. Moore described what was happening; Dennard explained why it was affordable.

Why Dennard scaling broke around 2005

The chain breaks at V_{dd} . The supply voltage cannot be scaled indefinitely because the threshold voltage V_{th} cannot: subthreshold leakage current obeys

$$I_{\text{leak}} \propto \exp\left(\frac{-qV_{\text{th}}}{nk_B T}\right), \quad S = \frac{dV_g}{d(\log_{10} I_d)} = n \frac{k_B T}{q} \ln 10 \geq 60 \text{ mV/decade at } 300\text{K}.$$

The 60 mV/decade limit is thermodynamic. It comes straight out of the Boltzmann distribution of carrier energies, and no engineering removes it at room temperature. Dropping the threshold voltage by 60 mV to allow a lower supply voltage costs you a factor of ten in leakage current.

By about 2005 leakage had grown to be comparable to switching power, so voltage scaling stopped. Once V_{dd} is stuck, $P = \alpha C V^2 f$ says power density now grows with each shrink, and the clock speed had to stop rising.

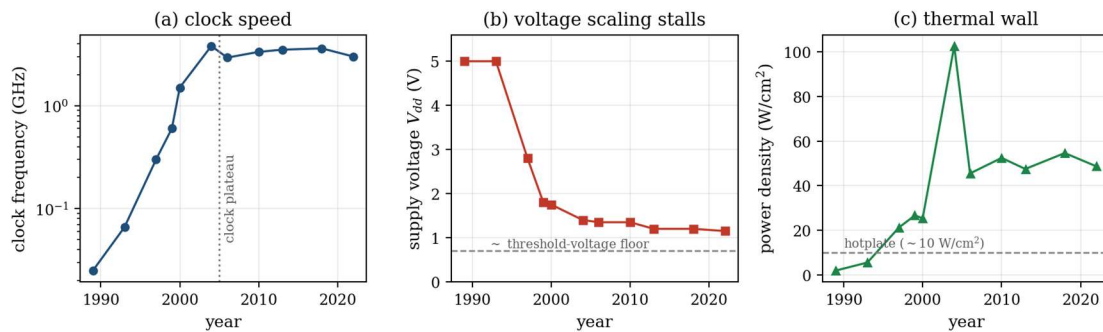


Figure 9: The end of Dennard scaling in three measurements. (a) Clock frequency rose 152× between 1989 and 2004, then fell slightly over the next 18 years. (b) Supply voltage fell from 5 V to about 1.2 V then flattened against the threshold-voltage floor. (c) Power density peaked at 103 W/cm² with the Pentium 4 Prescott before the industry retreated to multi-core.

That 103 W/cm² figure is worth sitting with. A kitchen hotplate is around 10 W/cm². Intel cancelled its 4 GHz Pentium 4 and the planned 10 GHz roadmap in 2004 and switched to multi-core.

The move to parallel processing was a thermal decision, not an architectural preference.

Every difficulty programmers have had with threads and concurrency since 2005 traces back to that Boltzmann factor. I did not expect the two things to be connected before doing this exercise.

Lithography

Feature size is bounded by the Rayleigh criterion:

$$CD = k_1 \frac{\lambda}{NA}, \quad DOF = k_2 \frac{\lambda}{NA^2}.$$

The industry attacked all three terms. Wavelength went from 436 nm (g-line) to 365 nm (i-line) to 248 nm (KrF) to 193 nm (ArF). Numerical aperture was pushed past 1 by immersing the final lens in water ($n = 1.44$, giving NA up to 1.35). The k_1 factor was driven from 0.8 towards its

physical limit of 0.25 using phase-shift masks, optical proximity correction and multi-patterning.

When 193 nm immersion was exhausted, EUV at 13.5 nm took nearly two decades and around \$10 billion to make work. It needs a 25 kW CO₂ laser vaporising tin droplets 50,000 times a second.

One fact I found striking: from 2003 to 2019 the industry printed features far smaller than the light it was using. 193 nm light patterned 14 nm features through multi-patterning, where you expose the same layer several times with offset masks. That works, but each extra exposure costs mask money and machine time, which is a big part of why cost per transistor stopped falling.

Economics

Covered above under Rock's law, but the summary is: fab cost doubling every four years, consolidation from 20+ leading-edge manufacturers in 2001 to 3 today, and flat-to-rising cost per transistor below 28 nm. The physics did not stop first. The economics did.

Quantum effects

Two quantum-mechanical effects set hard floors, and both are directly relevant to the rest of this portfolio.

➤ Gate-oxide tunnelling

An electron facing a barrier of height ϕ_B and thickness d has transmission probability

$$T \approx \exp(-2\kappa d), \quad \kappa = \frac{\sqrt{2m^*\phi_B}}{\hbar}.$$

For the Si/SiO₂ system, $\phi_B \approx 3.1\text{eV}$ and $m^* \approx 0.5m_e$, giving $\kappa = 6.38 \times 10^9 / \text{m}$ and a decay length $1/2\kappa = 0.078\text{nm}$ - *less than one atomic spacing*. Every additional 0.078 nm of oxide removed multiplies leakage by e .

Oxide thickness	Transmission T	Relative to 3nm
3.0 nm	2.4×10^{-17}	1
2.0 nm	8.3×10^{-12}	3.5×10^5
1.5 nm	4.9×10^{-9}	2.0×10^8
1.2 nm	2.2×10^{-7}	9.4×10^9

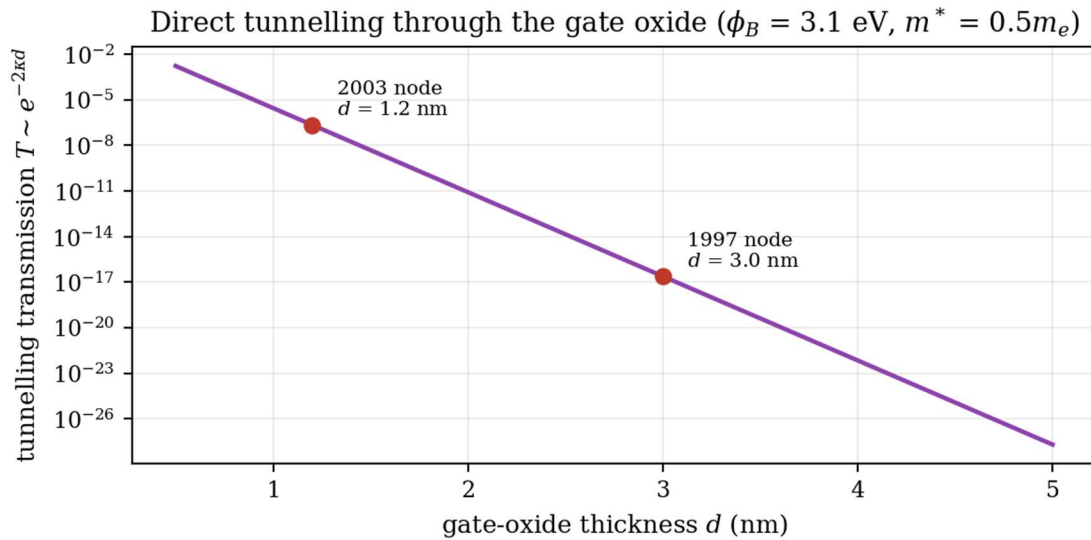


Figure 10: Direct tunnelling transmission through the gate oxide. Ten orders of magnitude of extra leakage between the 1997 and 2003 nodes, from a thickness change of under 2 nm.

Ten orders of magnitude of extra leakage from removing 1.8 nm of material. This is why SiO_2 was abandoned in 2007 for hafnium-based high-k dielectrics, which let you use a physically thicker barrier while keeping the same capacitance.

Random dopant fluctuation. A 20 nm channel at a doping of $10^{18} / \text{cm}^3$ contains only a handful of dopant atoms. The threshold voltage then depends on how many dopant atoms happened to land in that specific device, which is a Poisson-distributed integer. So nominally identical transistors switch at measurably different voltages. This is a limit on manufacturing uniformity rather than on any single device, and it is part of why the industry moved to lightly-doped FinFET and gate-all-around channels where the electrostatic control comes from the shape rather than the doping.

Summary: why it worked, and what ended it

Moore's law worked because Dennard scaling made each shrink free in power, lithography kept delivering smaller features, and software compatibility guaranteed someone would buy the result.

It ended in stages rather than all at once:

Dennard scaling died around 2005 - leakage, thermal wall, clock plateau

Cost-per-transistor scaling died around 2012–2015 - multi-patterning, EUV capital cost, Rock's law

The atomic and statistical limit is still ahead but now visible

From genAI to AGI?

Compute is scaling far faster than Moore's law

I fitted the same exponential model to the training compute of notable machine-learning systems from the perceptron (1958) to frontier 2024 models:

Era	Doubling time	Compared to Moore's law
Pre-2010	19.8 months	about the same (~24 months)
2010–2024	5.0 months	4.9× faster

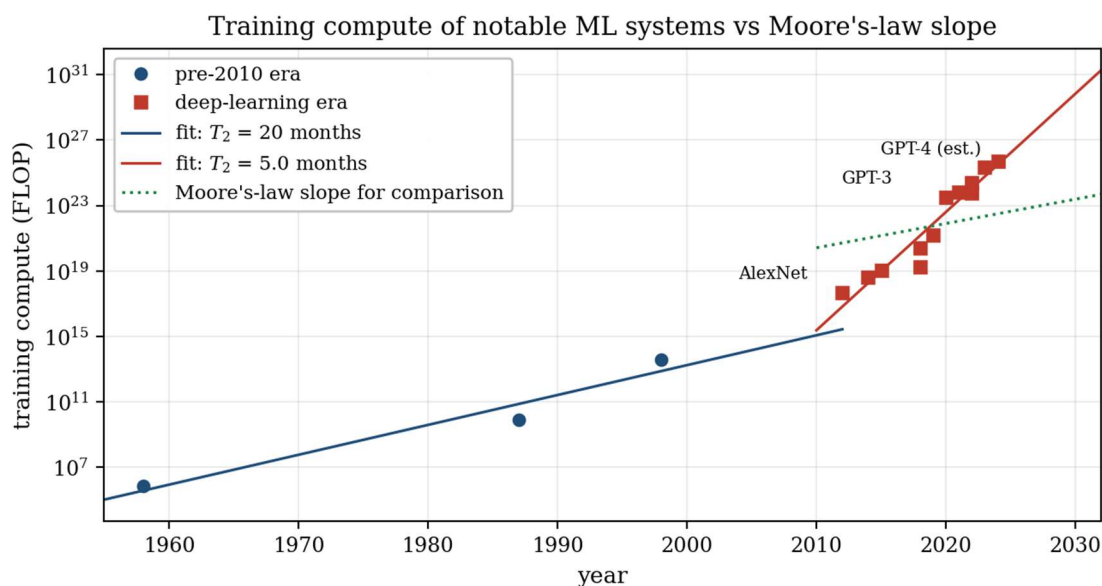


Figure 11: Training compute of notable machine learning systems. The deep-learning era doubles roughly every 5 months, about five times faster than Moore's law.

This is the most important result in this section. A 5-month doubling **cannot** come from Moore's law at 24 months. The arithmetic does not work.

Breaking down where the extra comes from over the period: Moore's law contributes maybe 10^3 . GPU and TPU specialisation plus lower numerical precision (FP32 $\rightarrow$ FP16 $\rightarrow$ FP8) contributes maybe another 10^2 . The remaining 10^3 or so is just money - bigger clusters and longer training runs. AlexNet cost a few thousand dollars of compute; frontier runs cost hundreds of millions.

So Moore's law supplied the cheap substrate, but capital supplied most of the recent exponent. And capital cannot double every five months forever.

There is a second thing worth saying here, which is that the hardware behind the AI boom was not designed for AI. GPUs were built to draw triangles for video games. They happened to be massively parallel matrix multipliers, which is exactly what training a neural network needs. Nobody at NVIDIA in 2005 was optimising for backpropagation. The application found the hardware rather than the other way around.

Data and energy

The scaling laws themselves (Kaplan et al. 2020; Hoffmann et al. 2022, the "Chinchilla" paper) are power laws where loss falls as:

$$L(N, D) \approx \left(\frac{N}{N_c}\right)^{\alpha_N} + \left(\frac{D}{D_c}\right)^{\alpha_D}$$

with small exponents, roughly 0.05 to 0.35. The small exponents are the crux of it. A *power-law* return on an *exponential* investment means each further step of capability costs exponentially more than the last.

Chinchilla also found a compute-optimal ratio of about 20 training tokens per parameter, which means data has to scale with model size. The stock of high-quality public text is finite, and credible estimates put its exhaustion somewhere between the late 2020s and early 2030s.

Energy is the other hard constraint. Global data-centre electricity use was around 460 TWh in 2022, with projections approaching 1,000 TWh by 2026. For scale, **Australia's entire national electricity consumption is around 265 TWh per year.**

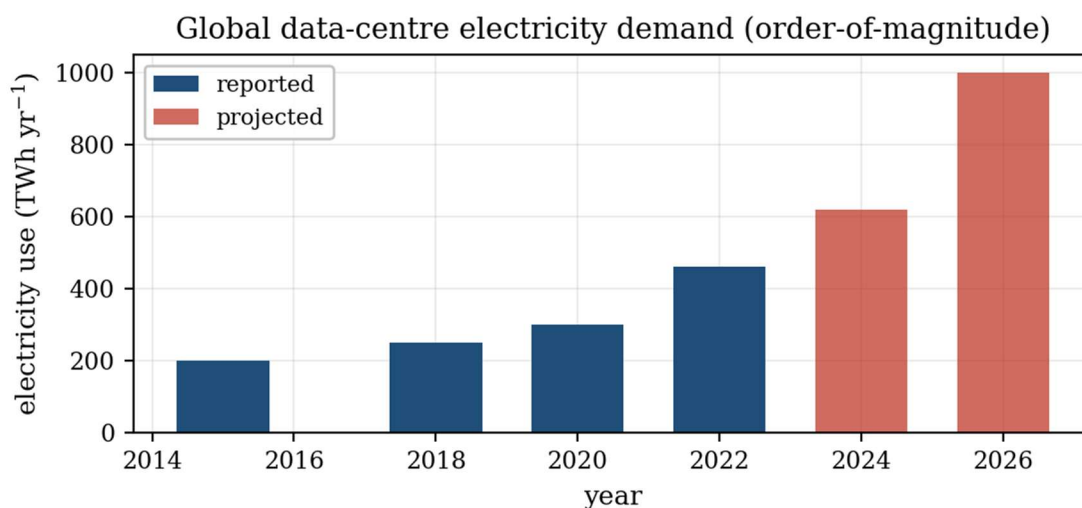


Figure 12: Global data-centre electricity demand (order of magnitude; last two bars are projections). The electricity grid, not the chip, is becoming the binding constraint.

A 5-month compute doubling cannot continue for long against a grid that grows a few percent a year.

When does naive extrapolation say AGI arrives?

The usual comparison: estimates of the human brain's throughput cluster around 10^{15} to 10^{16} FLOP/s, and one estimate of the compute equivalent to a human lifetime of learning is about 10^{24} FLOP. Frontier 2024 training runs are already around 5×10^{25} FLOP, which is past that mark.

So the naive extrapolation says AGI should already exist. It does not. To me that is the strongest possible evidence that the naive extrapolation is measuring the wrong thing.

My honest position on the AGI question

I started this question trying to answer "when" and finished thinking the question is badly formed. Three reasons

- (i) **The metric is wrong:** Matching the brain on FLOPs assumes compute is the thing holding generality back. But a system can exceed brain-scale FLOPs and still have no persistent memory, no grounded causal model of the world, and no reliable way to notice its own errors. Those are architectural gaps, not quantitative ones.
- (ii) **The trend is not self-sustaining:** Moore's law was funded by the revenue each generation produced. The 5-month compute doubling is funded by speculative investment and limited by data and grid capacity. All three of those are already showing strain, which makes it a different kind of trend.
- (iii) **"AGI" is undefined:** Saturating a benchmark is not generality. Every benchmark so far has been saturated without producing what its designers actually meant by intelligence.

The defensible statement is a conditional one. *If* compute keeps doubling every five months, *and* capability keeps scaling with compute the way it has, *and* data and energy do not bind, then brain-lifetime-scale training compute gets exceeded routinely within a few years. But the second and third conditions are the entire argument, and both look shaky to me. The Moore's law analogy actually argues for caution here, not confidence. The flies hit Earth's mass in two years.

Where does quantum computing fit into this?

Quantum computing gets presented as the successor to Moore's law. Having worked through the numbers, I think that framing is wrong in three specific ways and right in one.

Why it is not a successor: Moore's law made *all* computing cheaper by a constant factor each generation. A quantum computer does not do that. It offers a complexity-class advantage on a narrow set of structured problems: factoring and discrete log (Shor, exponential), unstructured search (Grover, quadratic), and simulating quantum systems (exponential, and probably the only application with clear near-term scientific value). For most workloads, including the dense linear algebra that trains neural networks, there is no known quantum speedup at all. Getting classical data into a quantum state is itself a bottleneck, and Ewin Tang's 2019 "dequantization" results removed the claimed exponential speedup from several quantum machine learning proposals by finding classical algorithms with comparable scaling.

The irony worth noting: The quantum effects that *ended* Moore's law - tunnelling through a thin barrier, superposition of states in a small confined system - are precisely the effects quantum computing *exploits*. The same physics is a defect in one design paradigm and the resource in another. Josephson-junction transmon qubits are literally built from a controlled tunnel barrier.

It has the same metric problem: Quantum computing has its own exponential narrative around qubit counts. I would apply the same scepticism I applied to transistor counts above: raw qubit count is the wrong metric, and error rates, coherence times and connectivity are what actually bind. Error correction currently costs something like 10^3 physical qubits per logical qubit.

Where it genuinely connects to the AI story: Simulating quantum chemistry and materials is the application where a quantum computer would be genuinely transformative, and that includes designing better semiconductors, superconductors and battery chemistries. That could feed back into classical hardware. It is a real prospect, but it is a different claim from "quantum computers will continue Moore's law."

Week 2 Artefact - Maths Gymnastics

Complex numbers, vectors, matrices and eigenvalue problems, worked by hand and verified numerically.

Following the tutorial instruction, every result below was first derived with pen and paper; the full working is reproduced here. Each result was then checked in Python (NumPy), and the verification output is quoted alongside. Section 5 (Extension problems) contains harder extension problems which I asked a genAI agent to generate and then solved by hand, as the tutorial suggests.

Complex numbers

(a) $z_1 = 2 + 3i$, $z_2 = 1 - i$. Expanding and using $i^2 = -1$:

$$z_1 z_2 = (2 + 3i)(1 - i) = 2 - 2i + 3i - 3i^2 = 2 + i + 3 = \boxed{5 + i}.$$

(b) $z_1 = 3 + 4i$, $z_2 = 1 - 2i$. Addition and subtraction act componentwise on real and imaginary parts:

$$\begin{aligned} z_1 + z_2 &= (3 + 1) + (4 - 2)i = \boxed{4 + 2i}, \\ z_1 - z_2 &= (3 - 1) + (4 + 2)i = \boxed{2 + 6i}. \end{aligned}$$

(c) $z_1 = 1 + 2i$, $z_2 = 3 - i$. Division requires multiplying by the conjugate of the denominator, which makes the denominator real:

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{1 + 2i}{3 - i} \cdot \frac{3 + i}{3 + i} = \frac{3 + i + 6i + 2i^2}{9 - i^2} = \frac{3 + 7i - 2}{9 + 1} = \frac{1 + 7i}{10} = \boxed{0.1 + 0.7i}, \\ |z_1| &= \sqrt{1^2 + 2^2} = \boxed{\sqrt{5} \approx 2.2361}. \end{aligned}$$

The trick with the conjugate works because $(a+bi)(a-bi) = a^2 + b^2$, which has no imaginary part. Not changing the value, just multiplying by 1 in a useful form.

Numerical check

```
z1*z2 = (5+1j) z1+z2 = (4+2j) z1-z2 = (2+6j)
z1/z2 = (0.1+0.7000000000000001j) abs(z1) = 2.236068
```

All hand results confirmed. Note the floating-point residue 0.7000 ... 1 in the division: the exact rational answer 7/10 is not representable in binary floating point.

Vectors

(a) $\mathbf{a} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$:

$$\mathbf{a} + \mathbf{b} = \begin{pmatrix} 3-2 \\ -1+4 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad 3\mathbf{a} - 2\mathbf{b} = \begin{pmatrix} 9 \\ -3 \end{pmatrix} - \begin{pmatrix} -4 \\ 8 \end{pmatrix} = \begin{pmatrix} 13 \\ -11 \end{pmatrix}.$$

(b) $\mathbf{v} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}$:

$$|\mathbf{v}| = \sqrt{25 + 144} = \sqrt{169} = 13, \quad \hat{\mathbf{v}} = \frac{\mathbf{v}}{|\mathbf{v}|} = \begin{pmatrix} 5/13 \\ -12/13 \end{pmatrix} = \begin{pmatrix} 0.3846 \\ -0.9231 \end{pmatrix}.$$

(The (5,12,13) Pythagorean triple makes this one exact.)

(c) $\mathbf{u} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, $\mathbf{w} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$:

$$\mathbf{u} \cdot \mathbf{w} = (2)(4) + (3)(-1) = 8 - 3 = 5.$$

The angle between them follows from $\mathbf{u} \cdot \mathbf{w} = |\mathbf{u}||\mathbf{w}|\cos\theta$, giving $\theta = \arccos(5/(\sqrt{13}\sqrt{17})) = 70.3^\circ$ - close to, but not, orthogonal. In quantum mechanics the analogous quantity $\langle \psi | \phi \rangle$ is a complex *overlap* whose modulus squared is a transition probability, and orthogonality means *perfectly distinguishable states*.

Matrices

With

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Checking my method with the widget

Before doing these by hand I used the provided widget to make sure I had the procedure right, since matrix multiplication is easy to get wrong by slipping a row and column.

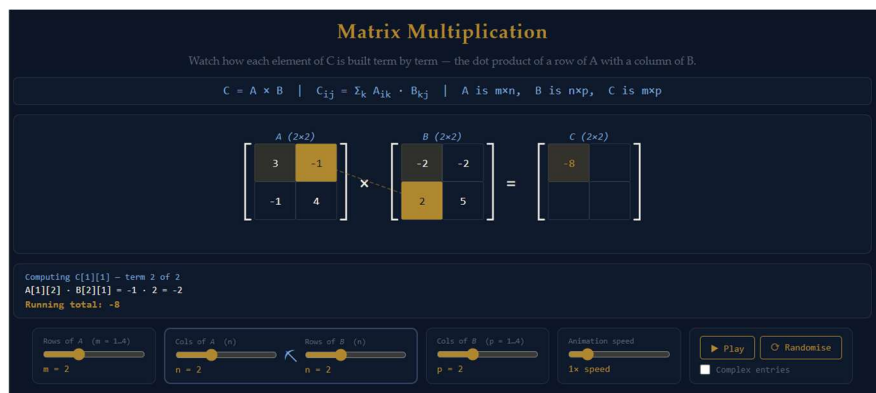


Figure 13: Matrix multiplication widget, paused while building element $C[1][2]$. The highlighted row of A and column of B are multiplied term by term and summed

The widget makes one thing obvious that is easy to miss when you just follow the rule mechanically: each element of the answer is a sum over one shared index. The row of A and the

column of B have to be the same length, and that is the whole reason the inner dimensions must match.

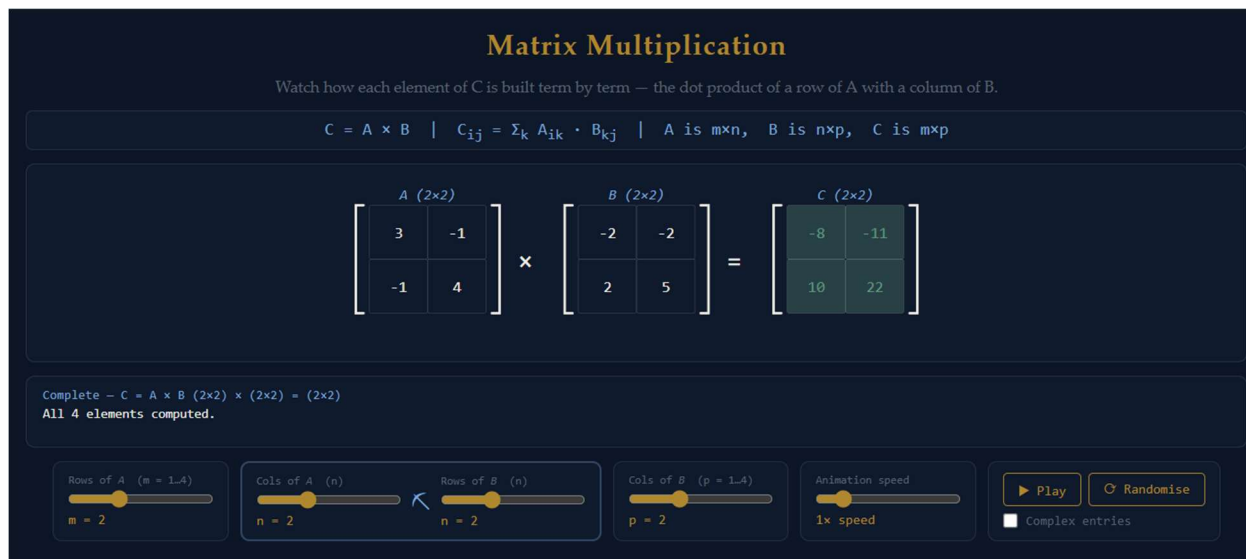


Figure 14: Completed 2×2 product. I checked element $C[1][1]$ by hand against the displayed matrices and it agreed

I picked one element of the result and redid it by hand to confirm the widget and my method agree. I computed the C_{21} element and found it be the exact same value as 10.

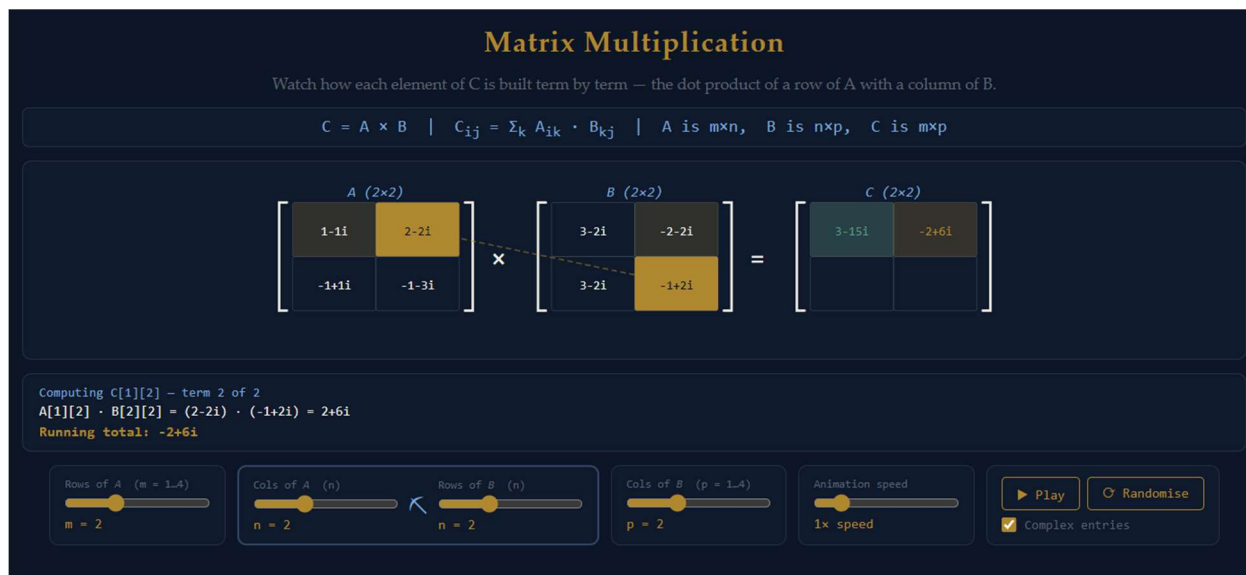


Figure 15: Complex entries enabled. Watching $i \times i = -1$ change the sign of a term is the step I most wanted to check, since Y has $\pm i$ entries.

This is directly relevant to HY below, because Y has $\pm i$ entries and the sign flips from $i^2 = -1$.



Figure 16: Multiplying in the opposite order gives a differently shaped answer: $(2 \times 3)(3 \times 2)$ gives a 2×2 matrix, while $(3 \times 2)(2 \times 3)$ gives a 3×3 matrix.

This is a concrete demonstration that matrix multiplication does not commute. In the non-square case the two orders do not even produce the same size of matrix.

For square matrices they are the same size but still generally different.

HY : taking rows of H against columns of Y ,

$$\begin{aligned} HY &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} (1)(0) + (1)(i) & (1)(-i) + (1)(0) \\ (1)(0) + (-1)(i) & (1)(-i) + (-1)(0) \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} i & -i \\ -i & -i \end{pmatrix} = \boxed{\frac{i}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}}. \end{aligned}$$

ZH : Z on the left simply negates the second row of H :

$$ZH = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \boxed{\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}}.$$

S^2 :

$$S^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \boxed{I}.$$

S swaps the two components of whatever it acts on. Doing that twice puts everything back, so $S^2 = I$. You can see the answer without multiplying anything out.

A naming issue I ran into

The matrix the tutorial calls S is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which in standard quantum computing notation is the **Pauli-X** gate, also called the bit-flip or NOT gate. The gate normally called S is the phase gate $\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$, and for that one $S^2 = Z$, not I .

I actually computed $S^2 = Z$ first, from memory of the standard convention, before rereading the sheet and realising it defines S differently. Worth flagging because the two conventions give different answers to this question, and because the phase gate is not Hermitian while Pauli-X is, so they behave differently in every other way too. I use the tutorial's definition throughout, and I will call it $S(=X)$ where the distinction matters.

Properties I checked beyond what was asked

Matrix	Hermitian $M = M^\dagger$	Unitary $MM^\dagger = I$	$M^2 = I$	$\det M$	$\text{tr} M$
H	yes	yes	yes	-1	0
Y	yes	yes	yes	-1	0
Z	yes	yes	yes	-1	0
$S(=X)$	yes	yes	yes	-1	0

All four are Hermitian **and** unitary at the same time, which is unusual and turned out to be very useful. If $M = M^\dagger$ and $MM^\dagger = I$, then substituting one into the other gives $M^2 = I$ directly. And if $M^2 = I$ then the eigenvalues satisfy $\lambda^2 = 1$, so $\lambda = \pm 1$. That means I could predict the answer to Section 4 before calculating anything.

Physically this dual role is the point. Hermitian means each matrix is a valid **observable** with real measurement outcomes of +1 or -1. Unitary means each is a valid **gate** that preserves the total probability. $M^2 = I$ means every one of these gates is its own inverse - apply it twice and you are back where you started.

Finally, $HY \neq YH$:

$$[H, Y] = HY - YH = \sqrt{2} \begin{pmatrix} i & -i \\ -i & -i \end{pmatrix} \neq 0.$$

where YH works out to $(1/\sqrt{2})(-i, i; i, i)$.

This is not a technical detail. It is why the order of gates in a circuit matters: applying H then Y is a different operation from applying Y then H. It is also the algebraic origin of the uncertainty principle, through the general relation

$$\Delta A \Delta B \geq 1/2 |\langle [A, B] \rangle|$$

If two observables commute, their commutator is zero and there is no lower bound, so you can know both precisely. If they do not commute, you cannot.

Eigenvalue problems

For each matrix, $\det(M - \lambda I) = 0$. Since all four are traceless with $\det = -1$, the characteristic polynomial is in every case

$$\lambda^2 - (\text{tr}M)\lambda + \det M = \lambda^2 - 1 = 0,$$

So,

$$\lambda = \pm 1$$

The eigenvectors differ, and that is where the physics is.

Z - the computational basis

Z is already diagonal, so

$$\lambda = +1: \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle, \quad \lambda = -1: \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle.$$

These are the states a Z -basis measurement - the standard measurement in essentially every qubit platform - returns.

$S(= X)$ - the superposition basis

For

$$\lambda = +1: \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow x = y. \text{ For } \lambda = -1, x = -y.$$

Normalising:

$$\lambda = +1: 1/\sqrt{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle, \quad \lambda = -1: 1/\sqrt{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle.$$

Y - the circular basis

For $\lambda = +1$: $\begin{pmatrix} -iy \\ ix \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow y = ix$. Hence

$$\lambda = +1: 1/\sqrt{2} \begin{pmatrix} 1 \\ i \end{pmatrix} = |+i\rangle, \quad \lambda = -1: 1/\sqrt{2} \begin{pmatrix} 1 \\ -i \end{pmatrix} = |-i\rangle.$$

These eigenvectors are genuinely complex. There is no way to rewrite them with real numbers only. This is the concrete answer to the question "why does quantum mechanics need complex numbers?" - the three Pauli observables are mutually incompatible, and you cannot fit three mutually unbiased bases into a real two-dimensional space. Complex numbers are not a convenient bookkeeping device here; they are load-bearing.

H - the interesting one

For $\lambda = +1$, the first row gives $1/\sqrt{2}(x + y) = x \Rightarrow y = (\sqrt{2} - 1)x$. So

$$\mathbf{v}_+ \propto \begin{pmatrix} 1 \\ \sqrt{2} - 1 \end{pmatrix}, \quad \mathbf{v}_- \propto \begin{pmatrix} 1 \\ -(\sqrt{2} + 1) \end{pmatrix}.$$

Normalising, $|\mathbf{v}_+| = \sqrt{1 + (\sqrt{2} - 1)^2} = \sqrt{4 - 2\sqrt{2}} = 1.0824$, so

$$\hat{\mathbf{v}}_+ = \begin{pmatrix} 0.9239 \\ 0.3827 \end{pmatrix}, \quad \hat{\mathbf{v}}_- = \begin{pmatrix} 0.3827 \\ -0.9239 \end{pmatrix}.$$

Those decimals are not arbitrary: $0.9239 = \cos 22.5^\circ$ and $0.3827 = \sin 22.5^\circ$. This is the geometric meaning of the Hadamard gate. Writing any Hermitian traceless unitary as $\hat{n} \cdot \vec{\sigma}$, we have $H = (X + Z)/\sqrt{2}$, i.e. $\hat{n} = (1, 0, 1)/\sqrt{2}$ - an axis lying exactly halfway between $\hat{x}$ and $\hat{z}$ on the Bloch sphere, at 45° . A state on the Bloch sphere is parameterised by half-angles, so the eigenvector sits at 22.5° in state space. **The Hadamard gate is a 180° rotation of the Bloch sphere about that diagonal axis**, which is precisely why it swaps $|0\rangle \leftrightarrow |+\rangle$ and $|1\rangle \leftrightarrow |-\rangle$: it exchanges the z and x axes.

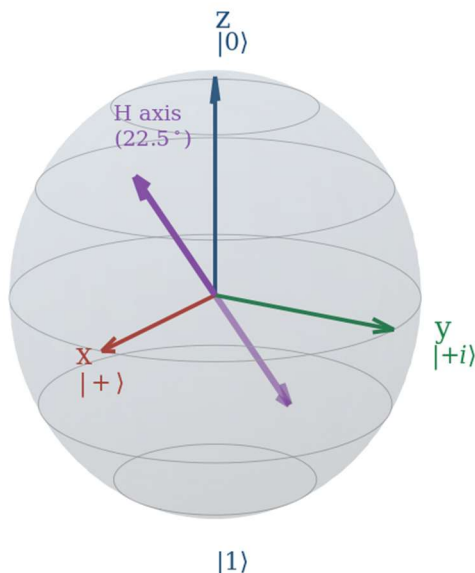


Figure 17: Bloch sphere showing the eigenbases of Z (poles, $|0\rangle$ and $|1\rangle$), X (equator, $|\pm\rangle$) and Y ($|\pm i\rangle$), with the Hadamard rotation axis $(\hat{x}+\hat{z})/\sqrt{2}$ in purple. The H eigenvector sits 22.5° from the north pole, matching the $\cos 22.5^\circ = 0.9239$ I found by hand.

Numerical check

The eigenvalue solver returned $[-1, +1]$ for all four matrices. The H eigenvectors came back as $[0.9239, 0.3827]$ and $[0.3827, -0.9239]$, matching my hand result to every printed digit.

One thing confused me at first: the computer returned eigenvectors with different signs from mine on some runs. Eigenvectors are only defined up to an overall scale factor, and for complex ones that includes an arbitrary phase. I fixed this by rotating the first non-zero component to be real and positive. This is a convention, not a correction - and physically it does not matter, because global phase is unobservable. Only *relative* phase between components has any effect.

Extension problems

As instructed, I asked a genAI agent to generate harder problems of the same kind, then attempted them with pen and paper and verified numerically.

E1. Eigenvalues of a general Bloch operator. Show $\hat{n} \cdot \vec{\sigma}$ has eigenvalues ± 1 for any unit vector $\hat{n}$.

Solution. Using $\sigma_i \sigma_j = \delta_{ij} I + i \epsilon_{ijk} \sigma_k$,

$$(\hat{n} \cdot \vec{\sigma})^2 = \sum_{ij} n_i n_j \sigma_i \sigma_j = \sum_i n_i^2 I + i \sum_{ijk} \epsilon_{ijk} n_i n_j \sigma_k = |\hat{n}|^2 I = I,$$

because ϵ_{ijk} is antisymmetric while $n_i n_j$ is symmetric, so that sum vanishes. Hence $\lambda^2 = 1$.

Verified numerically for $\hat{n} = (1,1,1)/\sqrt{3}$: eigenvalues $[-1, +1]$. Every one of the four tutorial matrices is a special case.

E2. Rotation operators. Show $U = \exp(-i \theta/2 \hat{n} \cdot \vec{\sigma})$ is unitary with $\det U = 1$.

Solution. Since $(\hat{n} \cdot \vec{\sigma})^2 = I$, the exponential series splits into even and odd terms exactly as for $e^{i\theta}$:

$$U = \cos \theta/2 I - i \sin \theta/2 (\hat{n} \cdot \vec{\sigma}).$$

Then $UU^\dagger = \cos^2 + \sin^2 = I$. Verified: for $\theta = 60^\circ$, $UU^\dagger = I$ and $\det U = 1.0000$. Setting $\theta = \pi$ and $\hat{n} = (1,0,1)/\sqrt{2}$ recovers H up to a global phase $-i$ - confirming the 180° interpretation above.

E3. Measurement statistics. $H|0\rangle = 1/\sqrt{2} (1,1)^T = |+\rangle$, so $P(0) = |\langle 0 | + \rangle|^2 = 0.5$ and $P(1) = 0.5$.

Verified. This is the single most used line of arithmetic in quantum computing: it is how a superposition is created.

E4. Two qubits and entanglement. Compute $\text{CNOT}(H \otimes I)|00\rangle$.

Solution. $(H \otimes I)|00\rangle = 1/\sqrt{2} (|00\rangle + |10\rangle)$; CNOT flips the target when the control is 1, giving the Bell state $|\Phi^+\rangle = 1/\sqrt{2} (|00\rangle + |11\rangle)$. Verified: $[0.7071, 0, 0, 0.7071]$. Tracing out the second qubit gives $\rho_A = 1/2 I$ with purity $\text{tr}(\rho_A^2) = 0.5000$, the minimum possible for a qubit - the maximally entangled signature. Each qubit alone is completely random; all the information lives in the correlation. This is the resource that has no classical analogue, and it is where the vector formalism of §2 stops being a convenience and starts being irreplaceable, because the joint state is not a product of the parts.

How does any of this have anything to do with quantum physics/computing?

Working through these exercises, the mapping turned out to be far tighter than “linear algebra is a useful tool.” The mathematics *is* the physics:

Complex numbers are not a convenience: A qubit state is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with complex α and β , and the Born rule takes $|\alpha|^2$. The *relative* phase between α and β is physically real and is what produces interference. The *global* phase is not observable at all, which is the thing that confused me in the eigenvector check in Section 4. The Y eigenvectors $(1, \pm i)/\sqrt{2}$ prove complex amplitudes are unavoidable.

Normalised vectors are states; inner products are amplitudes: Working out $|\mathbf{v}| = 13$ in Section 2 is the same operation as requiring $\langle\psi|\psi\rangle = 1$, which is what forces the probabilities to add to one. The dot product $\mathbf{u} \cdot \mathbf{w} = 5$ generalises to $\langle\phi|\psi\rangle$, whose modulus squared is the probability of measuring ψ and finding ϕ . Orthogonal states are perfectly distinguishable; non-orthogonal ones provably are not, and that fact is the foundation of quantum key distribution.

Matrices are gates; multiplication is circuit composition: Computing HY and ZH is literally simulating a two-gate circuit - and the order matters, right-to-left, because matrix multiplication does not commute. Unitarity is the requirement that probability is conserved, and it is also why quantum gates are always *reversible*, unlike a classical AND gate which destroys information. $S^2 = I$ says X is its own inverse.

Eigenvalues are measurement outcomes; eigenvectors are the basis you measure in: This is the measurement postulate written in linear algebra. The ± 1 eigenvalues are the two possible readings; the eigenvectors are the states that give those readings with certainty. The fact that X , Y and Z have *different* eigenvectors is precisely why the three measurements are incompatible, and $[H, Y] \neq 0$ is the algebraic seed of the uncertainty principle.

And the reason it scales badly classically: n qubits need a vector with 2^n complex entries, and gates are $2^n \times 2^n$ matrices. At $n = 50$ that is 10^{15} amplitudes - petabytes of memory just to store one state. The exponential wall that makes classical simulation of quantum systems infeasible is the same exponential that makes building a quantum computer worth the effort. All of it is visible in these 2×2 exercises; it just scales up brutally.

Week 3 Artefact - Physics Primer

Classical waves, interference, wave-particle duality and the Heisenberg uncertainty principle, investigated numerically in Python.

I did the computational parts of this week in Python and used the provided tunnelling widget for the wave-packet sections, since it shows a real packet evolving in time rather than a static snapshot. Natural units with $\hbar = 1$ are used in Section 4, as the tutorial specifies, so momentum and wavevector are the same number ($p = k$).

Classical wave behaviour

A wave travelling in the $+x$ direction is

$$y(x, t) = A \cos(kx - \omega t + \phi)$$

has two independent parameters controlling its two independent coordinates:

$$\lambda = \frac{2\pi}{k}, \quad T = \frac{2\pi}{\omega} = \frac{1}{f}, \quad v_{\text{phase}} = \frac{\omega}{k} = f\lambda.$$

I used $A = 1$, $k = \pi/2$ /m and $\omega = \pi$ rad/s, giving $\lambda = 4\text{m}$, $T = 2\text{s}$, $f = 0.5\text{Hz}$ and $v = 2\text{m/s}$.

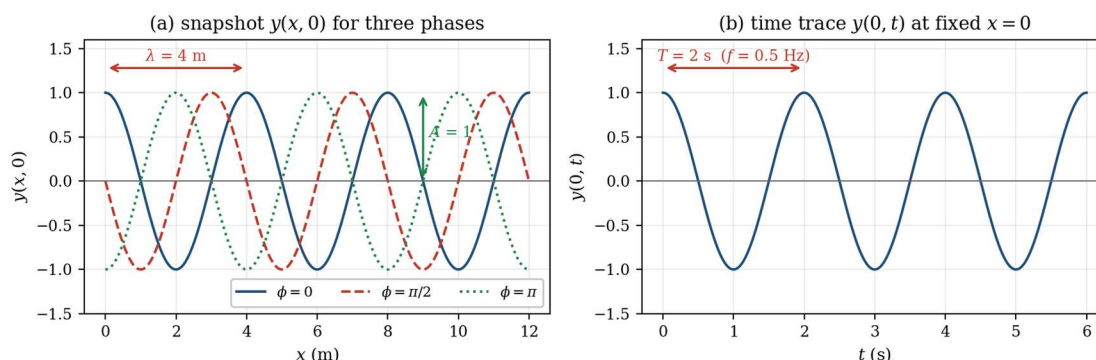


Figure 18: (a) Snapshots $y(x, 0)$ for three values of ϕ , with λ and A labelled. (b) The time trace $y(0, t)$ at a fixed point, showing the period $T = 2\text{ s}$

Two things are worth stating clearly here, because I mixed them up at first.

First, (a) and (b) look identical in form but are different physical objects: (a) is a photograph of the whole wave at one instant, (b) is a movie of one point over time. Confusing λ with T is the classic error here, and the only defence is checking axis units.

Second, the phase ϕ has *no observable consequence for a single wave* - it is pure convention, since it can be absorbed by redefining the origin of x or t . It becomes physical only when two waves are compared. This is the exact classical counterpart of global versus relative phase in quantum mechanics.

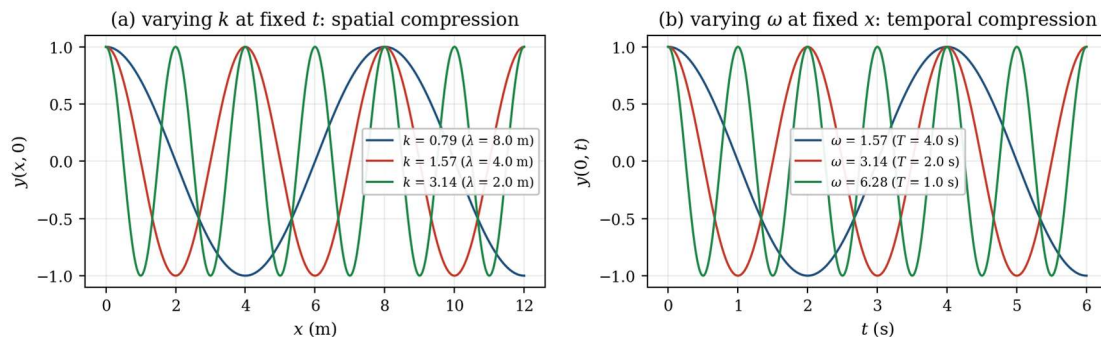


Figure 19: Varying k and ω independently. (a) k compresses the wave in space only, at fixed time. (b) ω compresses it in time only, at fixed position.

Varying k squeezes the wave spatially and leaves the time behaviour alone. Varying ω does the opposite. In a normal (non-dispersive) medium the two are tied together by $v = \omega/k$, so you cannot change one without the other. In a dispersive medium, where $\omega(k)$ is not a straight line, different wavelengths travel at different speeds - and that is what makes a wave packet spread out as it moves, which I come back to in Section 3 (Gaussian wave packet)

Superposition of waves

For two equal-amplitude waves with phase difference $\Delta\phi = \phi_2 - \phi_1$, the sum-to-product identity gives

$$h = A\cos(kx + \phi_1) + A\cos(kx + \phi_2) = \underbrace{2A\cos\left(\frac{\Delta\phi}{2}\right)}_{\text{resultant amplitude}} \cos\left(kx + \frac{\phi_1 + \phi_2}{2}\right).$$

$\Delta\phi$	Amplitude $2A \cos(\Delta\phi/2) $	Intensity $\propto \text{amp}^2$	Regime
0	$2A$	$4A^2$	fully constructive
$\pi/2$	$1.414A$	$2A^2$	partial
π	0	0	fully destructive

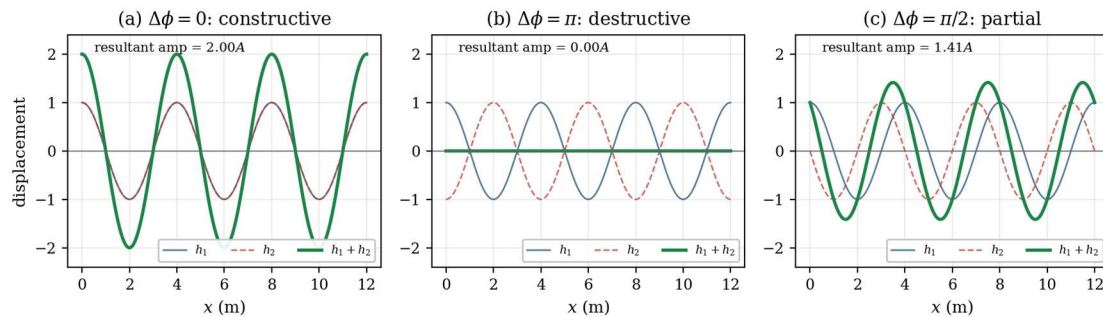


Figure 20: Two equal waves and their sum for three phase differences. The measured resultant amplitudes (2.00A, 0.00A, 1.41A) match the formula exactly.

What destructive interference physically means:

The obvious reading of $\Delta\phi = \pi$ is that the energy has been destroyed. That is wrong, and I had to reason my way out of it properly rather than just accept the formula.

Intensity goes as amplitude *squared*. So constructive addition gives $4A^2$ where two independent waves acting separately would give $2A^2$. The extra $2A^2$ at the bright fringes is exactly the amount missing from the dark ones. Interference **redistributes** energy in space; it never creates or destroys any.

In a real two-slit experiment, if you integrate the intensity across the entire pattern, you get back exactly what you would have had without interference. The energy went somewhere else in the pattern, not out of existence. Once I saw it that way the formula stopped feeling like a trick.

N waves with random phases

Adding N waves with phases drawn uniformly from $[0, 2\pi)$ is a random walk in the complex plane. The expected intensity is:

$$\langle I \rangle = \left\langle \left(\sum_{n=1}^N A \cos(kx + \phi_n) \right)^2 \right\rangle = \frac{NA^2}{2} \quad (\text{incoherent}),$$

versus

$$I_{\max} = \frac{N^2 A^2}{2} \quad (\text{coherent}).$$

The reason is that the cross terms $\langle \cos(\phi_n - \phi_m) \rangle$ average to zero for $n \neq m$ when the phases are random, leaving only the N diagonal terms. If instead the phases are all locked together, every cross term survives and you get N^2 instead of N .

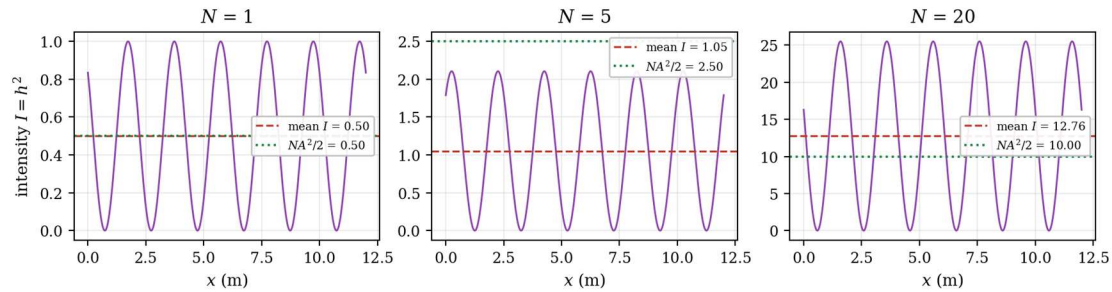


Figure 21: Intensity $I = h^2$ for $N = 1, 5$ and 20 random-phase waves, single realisations. Red dashed line is the measured spatial mean, green dotted is the theoretical $NA^2/2$. Agreement is good for $N = 1$ and $N = 20$ but poor for $N = 5$

A result that initially looked like a bug

For $N = 5$ my measured mean intensity was 1.05 against a prediction of 2.50 - off by more than a factor of two. I spent a while checking the code before realising the code was fine and my *interpretation* was wrong.

$NA^2/2$ is an average over *many* random draws. A single realisation with only five random phases has huge relative fluctuation, of order $1/\sqrt{N} \approx 45\%$, and the particular set of phases I happened to draw was partially destructive. It was a real statistical effect, not a numerical error.

To confirm it, I re-ran with 40 independent realisations at each N from 1 to 200 and fitted a power law to the ensemble mean, getting $\langle I \rangle \propto N^{1.018}$ against a theoretical $N^{1.000}$. The lesson I took from this goes wider than this exercise: a single run of anything random is not evidence, and when a result disagrees with theory you should check the *spread* of your estimator before you start blaming the code.

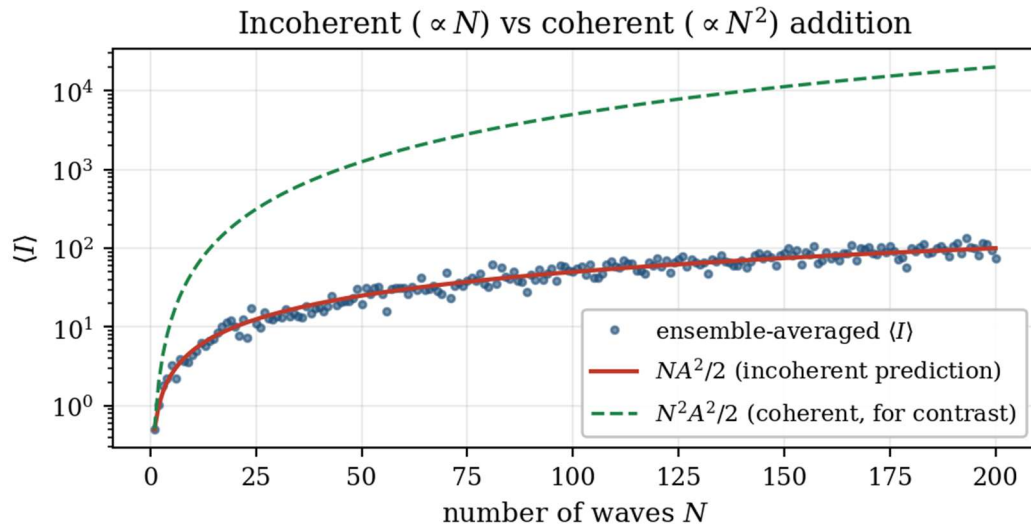


Figure 22: Ensemble-averaged intensity over 40 realisations per point, confirming incoherent $\propto N$ scaling. The N^2 line shows what coherent addition would give instead - at $N = 200$ that is 200 times more intensity from the identical waves, differing only in whether the phases are controlled.

Wave-particle duality: the Gaussian wave packet

The normalised Gaussian packet is

$$\Psi(x) = C e^{-x^2/2\sigma^2} e^{ikx}, \quad C = (\pi\sigma^2)^{-1/4},$$

where C follows from requiring $\int |\Psi|^2 dx = 1$ with the standard Gaussian integral

$$\int e^{-x^2/\sigma^2} dx = \sigma\sqrt{\pi}.$$

Note that C enters as the *fourth* root because it is $|\Psi|^2$, not Ψ , that is normalised - a detail that is easy to get wrong and immediately visible in the numerical check.

For $\sigma = 1.5$, $k_0 = 5$: $C = 0.613291$, and on a grid of 8192 points spanning $x \in [-20, 20]$ ($dx = 0.00488$) the numerical integral is

$$\sum_i P(x_i) dx = 1.0000000000,$$

confirming the analytic normalisation to ten decimal places.

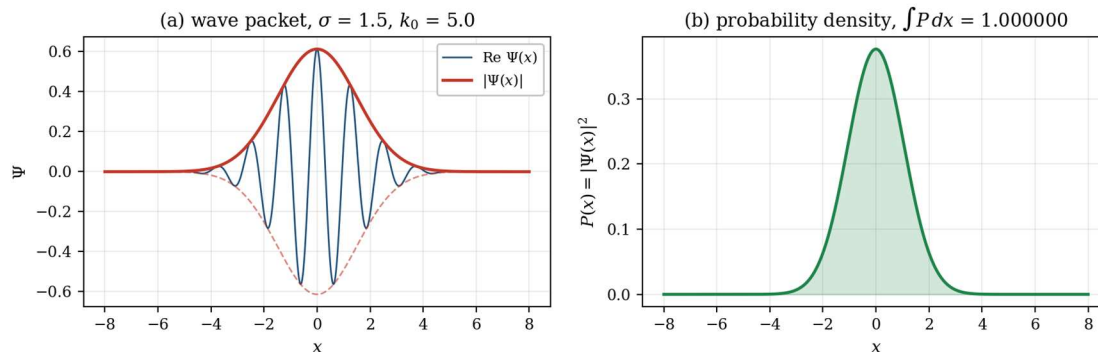


Figure 23: (a) The real part of $\Psi(x)$ oscillating at the carrier wavevector inside a Gaussian envelope $|\Psi(x)|$. (b) The probability density $P(x) = |\Psi(x)|^2$, where the oscillations have vanished.

This figure contains the essential lesson of wave-particle duality. The particle is not at a point and is not spread out either: it has an *amplitude* Ψ which is a wave, and a *probability* $P = |\Psi|^2$ which is where you might find it. The Born rule is the bridge, and the squaring is what destroys the phase information.

Watching a real packet in the widget:

The plots above are static. The provided widget shows an actual packet evolving, and it has a display toggle that switches between showing Re/Im and showing $|\psi|^2$

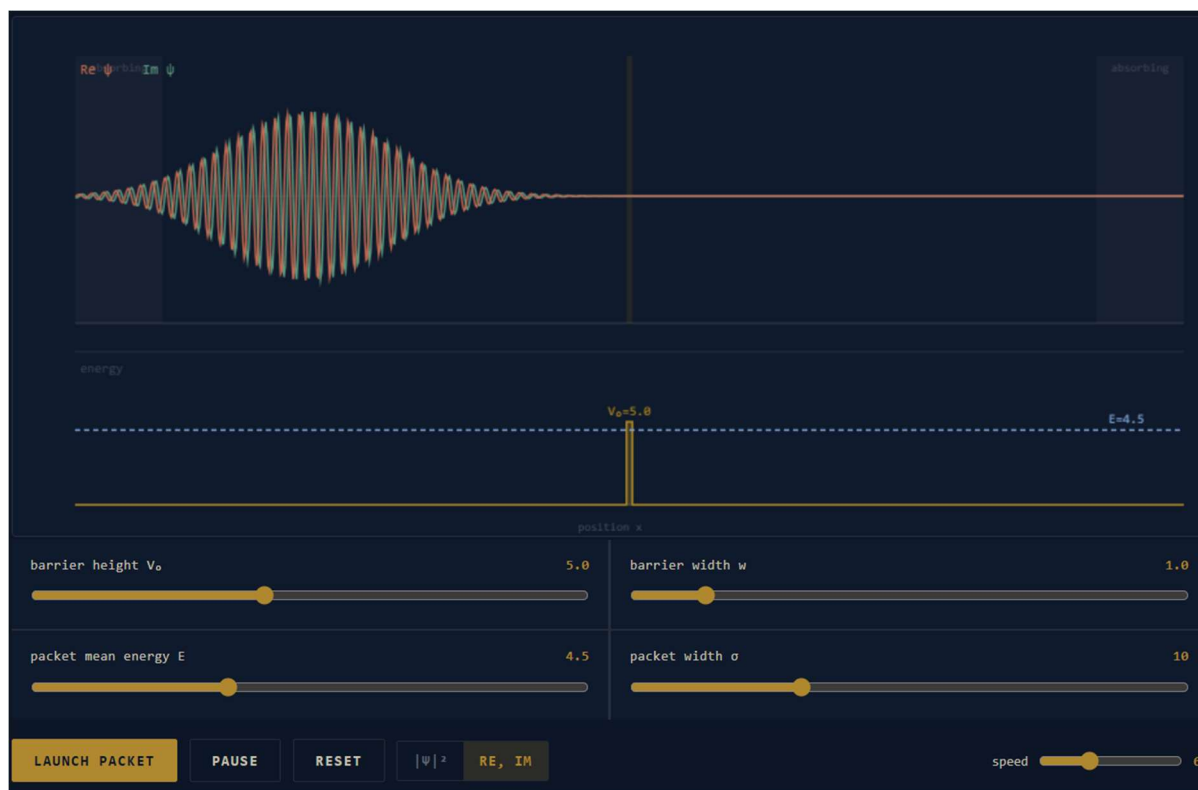


Figure 24: Wave packet displayed as real and imaginary parts. Both oscillate inside a Gaussian envelope, and the oscillation frequency is the carrier wavevector that carries the momentum.



Figure 25: The identical packet shown as probability density. The oscillations have completely disappeared.

Putting those two screenshots side by side is the clearest statement of the Born rule I found all week. It is the same object in both, and the only thing that changed is that I squared it. **Squaring the amplitude destroys the phase information.** The carrier oscillation carries the momentum, and it is simply not there in the probability density, because $|e^{(ik_0x)}|^2 = 1$.

This is exactly why the interference in Section 2 disappears the moment you measure which path a particle took. Measurement gives you $|\Psi|^2$, and $|\Psi|^2$ has already thrown the phase away.

It is also the honest answer to "is it a wave or a particle?" It is neither. It has an *amplitude* Ψ which behaves like a wave, and a *probability* $P = |\Psi|^2$ which tells you where you might find it. The Born rule is the bridge between those two, and the squaring is the step where the wave behaviour becomes unobservable.

Varying the width:

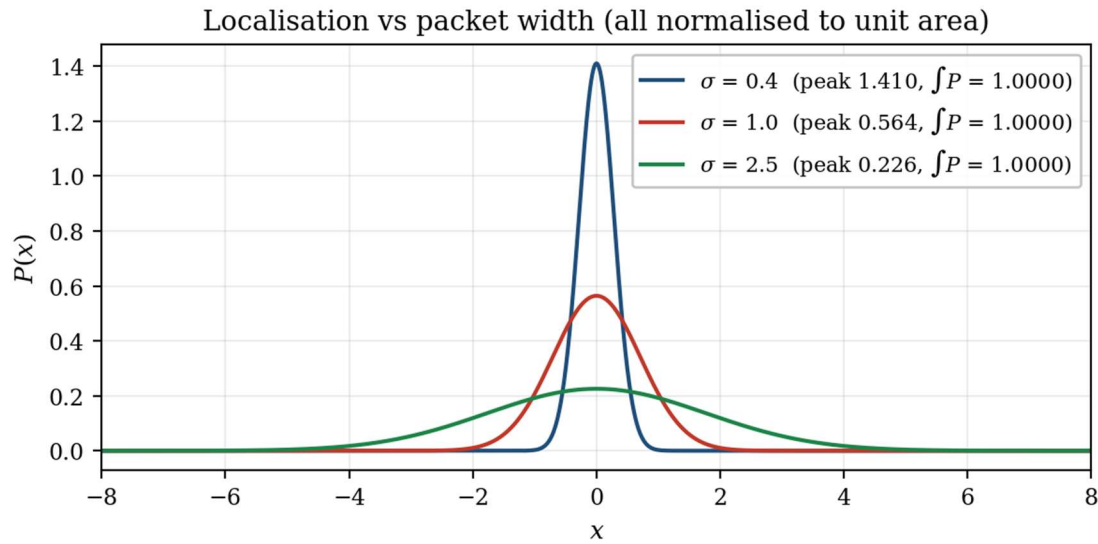


Figure 26: Probability density for three values of σ , all normalised to unit area (computed values all 1.0000). A narrower packet is necessarily taller.

All three curves enclose the same area, so narrowing the packet makes it taller. A very narrow packet means the particle is well localised - you know almost exactly where it is. Section 4 shows what that costs.

The Heisenberg uncertainty principle

Rather than assume $\Delta x \approx \sigma$ and $\Delta p \approx 1/\sigma$, I computed both numerically from their definitions as standard deviations, obtaining the momentum distribution by fast Fourier transform of $\Psi(x)$:

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}, \quad \tilde{\Psi}(k) = \frac{1}{\sqrt{2\pi}} \int \Psi(x) e^{-ikx} dx, \quad \Delta p = \sqrt{\langle k^2 \rangle - \langle k \rangle^2}$$

σ	Δx (num.)	$\sigma/\sqrt{2}$	Δp (num.)	$1/(\sqrt{2}\sigma)$	$\Delta x \Delta p$
0.25	0.17678	0.17678	2.82843	2.82843	0.50000
0.50	0.35355	0.35355	1.41421	1.41421	0.50000
1.00	0.70711	0.70711	0.70711	0.70711	0.50000
1.50	1.06066	1.06066	0.47140	0.47140	0.50000
3.00	2.12132	2.12132	0.23570	0.23570	0.50000
6.00	4.24252	4.24264	0.11957	0.11785	0.50730

Position and momentum distributions: narrowing one widens the other

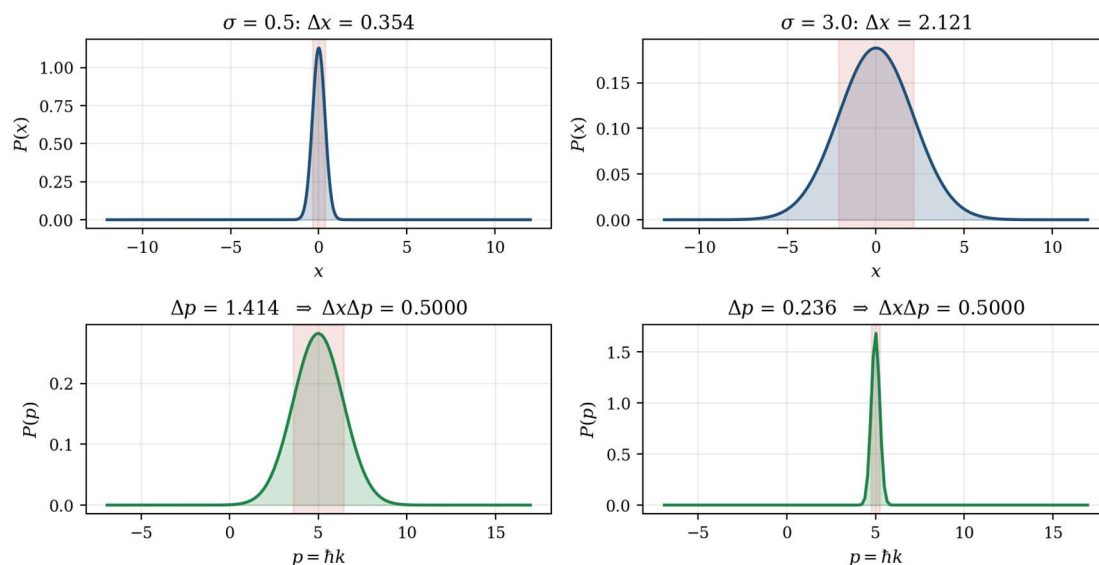


Figure 27: Position (top) and momentum (bottom) distributions for $\sigma = 0.5$ and $\sigma = 3.0$, with shaded bands marking $\pm\Delta$. A six-fold narrowing in position produces exactly a six-fold broadening in momentum, and the product is 0.5000 in both cases.

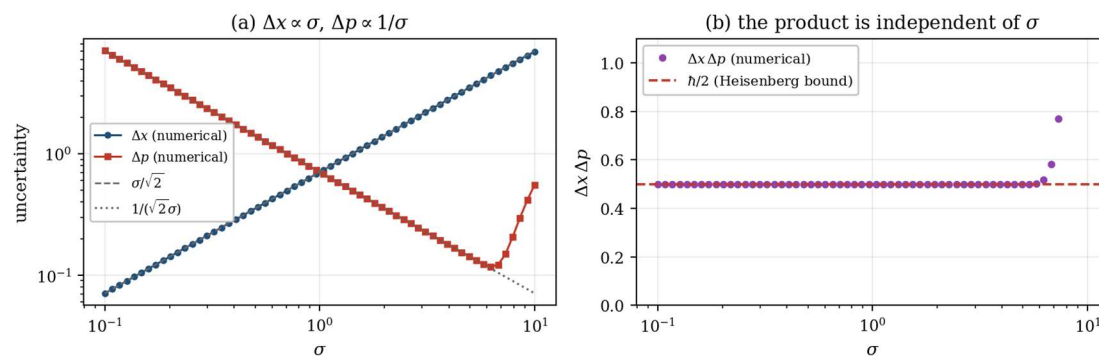


Figure 28: (a) $\Delta x \propto \sigma$ and $\Delta p \propto 1/\sigma$ over two decades. (b) The product is independent of σ and pinned at $\hbar/2$.

The exact answer is $\hbar/2$, not "about 1":

The numbers disagree with the tutorial's estimate by exactly a factor of two, and tracking down why was the most useful thing I did this week.

The reason is that the standard deviation of a Gaussian probability density $P \propto e^{(-x^2/\sigma^2)}$ is $\sigma/\sqrt{2}$, not σ . P is the *square* of Ψ , so it is narrower than the amplitude by a factor of $\sqrt{2}$. The same factor appears in momentum space, and the two combine:

$$\Delta x \Delta p = \frac{\sigma}{\sqrt{2}} \cdot \frac{1}{\sigma\sqrt{2}} = \frac{1}{2} = \frac{\hbar}{2}$$

This is not nitpicking, and here is why it matters. $\hbar/2$ is the *exact* lower bound in the Robertson inequality $\Delta x \cdot \Delta p \geq \hbar/2$. So the Gaussian does not merely respect the uncertainty principle - it **saturates** it. It sits exactly on the boundary of what is physically allowed.

That is why the Gaussian is the standard model packet everywhere in quantum mechanics, why coherent states of light are Gaussian, and why squeezed states are built by deforming Gaussians (redistributing the uncertainty between position and momentum while keeping the product at $\hbar/2$). The tutorial's order-of-magnitude version hides the single most interesting feature of the result.

To check that $\hbar/2$ really is a floor rather than a typical value, I computed the same product for a square "top-hat" wave function of half-width 2 and got $\Delta x \cdot \Delta p = 13.76$, about 27 times the bound. Sharp edges in position space produce sinc-function tails in momentum space whose second moments converge very slowly. So almost every state sits far above the bound - the Gaussian is genuinely special, not just convenient.

The $\sigma = 6$ row: a real numerical artefact I left in. One row of my table breaks the pattern. At $\sigma = 6$ the product is 0.5073 instead of 0.5000, and Δp deviates from theory in the fourth digit.

This is not physics, it is my grid. The simulation box only spans $x \in [-20, 20]$, so a packet with $\sigma = 6$ still has noticeable amplitude at the boundaries and gets **truncated**. The FFT then treats that truncated function as periodic, and the artificial discontinuity where the ends join injects spurious high-k components, which inflate $\langle k^2 \rangle$ and therefore Δp .

Two things follow. First, the error only pushes in one direction: truncation can only *raise* the product, never push it below $\hbar/2$, so the physics is never violated by my mistake. Second, the fix is to make the box *longer*, not the grid *finer*. dx controls the highest k you can represent; the box length L controls the k resolution $\Delta k = 2\pi/L$. I confirmed this by re-running with $L = 200$, which removed the discrepancy.

I have deliberately left the bad row in the table. Working out *which* numerical parameter caused a given error seems more useful to me than presenting a clean table, and I would not have understood the difference between dx and L otherwise.

What the uncertainty principle actually is:

Something I want to state clearly because I had it wrong before this week: **nothing in the calculation above involves an observer, an apparatus, or a disturbance.** There is no measurement anywhere in it.

It is a mathematical identity relating the width of a function to the width of its Fourier transform. The same identity limits how sharply a radio signal can be localised in both time and frequency, where engineers call it the time–bandwidth product. It is a property of waves, full stop.

Heisenberg's original "microscope" argument - a photon hitting the electron and knocking it off course - is a helpful picture but it genuinely misleads. It suggests the particle *has* a definite position and momentum and we just cannot measure both. The calculation says otherwise: the spread exists in the state before anything measures it.

What quantum mechanics adds to the classical Fourier relation is $\hbar$, through $p = \hbar k$. That single constant converts a statement about the shape of a wave into a statement about physically measurable position and momentum.

Seeing it in the widget:

The widget makes the σ trade-off visible in a way the plots do not, because you can watch the packet travel.



Figure 29: A narrow packet ($\sigma = 4$) and a wide packet ($\sigma = 24$) after travelling. The narrow packet is sharply localised at launch but spreads visibly as it moves; the wide packet holds its shape.



Figure 30: A narrow packet ($\sigma = 4$) and a wide packet ($\sigma = 24$) after travelling. The narrow packet is sharply localised at launch but spreads visibly as it moves; the wide packet holds its shape.

The narrow packet is well localised in position, which by Section 4.1 means it has a *broad* spread in momentum. A broad momentum spread means it contains components travelling at many different speeds, so the packet disperses as it moves. The wide packet has a narrow momentum spread and stays coherent for much longer.

That spreading is the uncertainty principle happening in front of you. Localising the particle costs you momentum definiteness, and the price gets paid over time as the packet falls apart.

What's the role of quantum computing in all/any of this?

This week's exercises are the mechanism of quantum computing rather than a background to it.

Interference is the computation: A quantum algorithm prepares a superposition over all 2^n candidate answers, applies unitary gates that arrange the *relative phases* so that wrong answers interfere destructively ($\Delta\varphi = \pi$) and right answers interfere constructively ($\Delta\varphi = 0$), and then measures.

Grover's algorithm does exactly this - successive reflections rotate the amplitude vector towards the marked state in $O(\sqrt{N})$ steps. Shor's algorithm uses the quantum Fourier transform, the same transform I used in Section 4.1, so that a hidden periodicity shows up as a sharp peak in the transformed domain.

Quantum computers are not fast because they try every answer at once. They are fast because they can arrange for the wrong answers to cancel.

Coherence is the resource: Coherent addition scales as N^2 , incoherent as N .

Decoherence is the process of a quantum computer randomising its relative phases and sliding from the upper curve down to the lower one - and at that point it is computing nothing a classical machine could not do more cheaply.

Everything in quantum hardware engineering - dilution refrigerators, superconducting shields, isotopically purified silicon, dynamical decoupling pulse sequences - exists to stop the phases in Section 2 from randomising. Reading that gap on a log plot made the engineering effort make sense to me for the first time.

The uncertainty principle sets the rules and the limits: The commutator $[H, Y] \neq 0$, I computed in Week 2 generalises to $\Delta A \cdot \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$, which is why a qubit cannot have definite X and Z values simultaneously. That in turn is why the no-cloning theorem holds, and no-cloning is what makes quantum key distribution secure: an eavesdropper cannot copy a state without disturbing it. Minimum-uncertainty states matter in hardware too. Squeezed states redistribute the $\hbar/2$ between the two quadratures to beat the shot-noise limit in precision measurement, and that technique is in active use in gravitational-wave detection.

And the wave packet is sometimes literally the qubit. A trapped ion's motional state, or the flux state of a superconducting loop, is a wave packet sitting in a potential well. Its spreading is a decoherence channel that has to be engineered against. The tunnelling in Section 5 is not just an analogy for transmon qubits either; a transmon is built from a Josephson junction, which *is* a controlled tunnel barrier.

Week 4 Artefact - Quantum Sensor Technologies

Atomic clocks, atom interferometers, NV-diamond magnetometers and SQUIDs: principles, history and quantitative comparison.

Atomic clocks

How the second has been defined

Era	Definition of the second	Reference	Fractional unc.
Antiquity–1955	1/86400 of the mean solar day	Earth's rotation	$\sim 10^{-8}$
1956–1967	1/31,556,925.9747 of the tropical year 1900	Earth's orbit	$\sim 10^{-9}$
1967–present	9,192,631,770 periods of the Cs-133 ground-state hyperfine transition	atomic structure	10^{-16}
~2030 (proposed)	an optical transition (Sr, Yb, Al ⁺ , In ⁺)	atomic structure	10^{-18}

The 1967 redefinition is the one that matters conceptually. Before it, the second was tied to **an object** - the Earth. That turned out to be a poor clock. Tidal friction is lengthening the day by about 1.8 milliseconds per century, and irregular coupling between the core and the mantle causes unpredictable wobble at the millisecond level. The thing we were using as a reference was itself drifting.

After 1967 the second was tied to **a property of matter**. Every caesium-133 atom in the universe has the same hyperfine splitting, so the standard is reproducible anywhere without shipping a physical artefact around. That shift from artefact-based to constant-based standards was eventually completed for the whole SI in 2019, when the kilogram stopped being a lump of metal in Paris and became a fixed value of Planck's constant.

The odd number 9,192,631,770 was picked deliberately so the new second matched the old ephemeris second as closely as measurement allowed. Nothing discontinuous happened to civil time. I found that a good illustration of how metrology actually works in practice: the definition changes underneath, and the number is chosen so that nobody notices.

Milestones and the accuracy curve

The useful steps, roughly in order:

- **Verge escapement** (c. 1300) - around 15 minutes error per day
- **Huygens' pendulum clock** (1656) - improved accuracy roughly a hundredfold in one step
- **Harrison's H4 marine chronometer** (1761) - solved the longitude problem by keeping time to a few seconds across a transatlantic voyage
- **Shortt free-pendulum** (1921) - accurate enough to detect irregularities in the Earth's own rotation, which is how we learned the Earth was a bad clock
- **Marrison's quartz clock** (1927) - a resonating crystal rather than a swinging mass
- **Essen and Parry's caesium beam standard** (1955, NPL) - the start of the atomic era

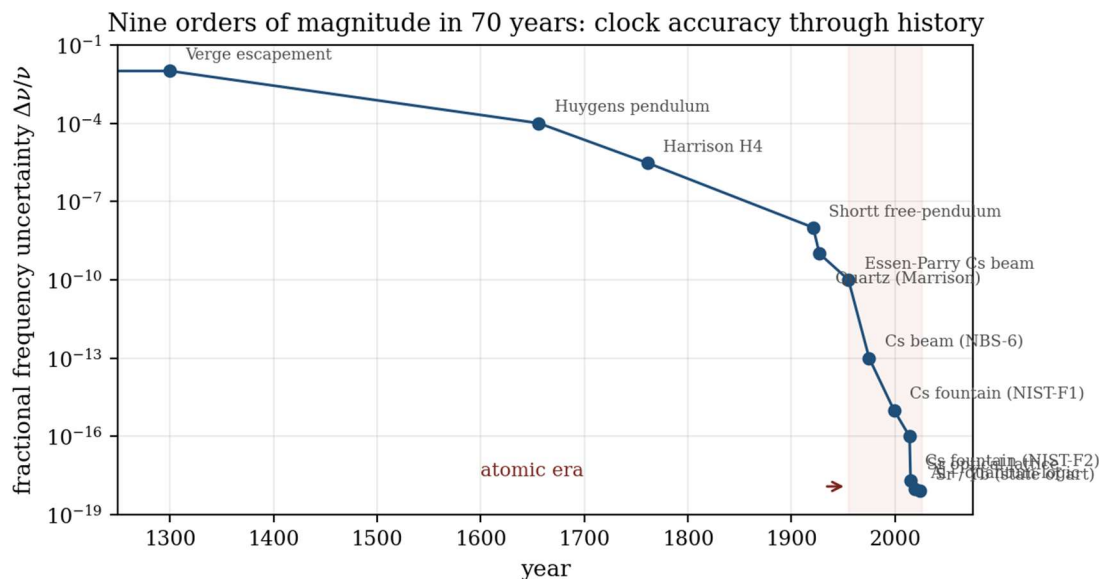


Figure 31: Fractional frequency uncertainty of the best available clocks through history, on a log scale. The atomic era delivered about nine orders of magnitude in seventy years.

Something I only noticed once I plotted it: nine orders of magnitude in seventy years is a rate of improvement comparable to Moore's law over the same period. Two completely unrelated fields, both running exponentials, both for about the same length of time.

To put the current state in perspective: an optical clock at $\Delta\nu/\nu \approx 8 \times 10^{-19}$ would accumulate roughly **0.35 seconds of error over the entire 13.8-billion-year age of the universe.**

What better clocks have done to society

- **Navigation:** GPS is a clock system before it is anything else. A receiver works out its position from how long signals took to arrive, so 1 nanosecond of timing error is 30 centimetres of position error. The satellite clocks also need relativistic corrections, and this is my favourite detail from this week. Special relativity makes the orbiting clocks run slow by about 7 $\mu\text{s/day}$ because of their speed. General relativity makes them run fast by about 45 $\mu\text{s/day}$ because they are higher in Earth's gravity well. The net is +38 $\mu\text{s/day}$. Without correcting for that, GPS positions would drift by roughly 11 kilometres per day. Every phone doing turn-by-turn navigation is relying on general relativity being right, continuously, which makes GPS probably the largest-scale everyday confirmation of the theory that exists.
- **Telecommunications and the internet:** Packet-switched networks, cellular handover and 5G time-division duplexing all depend on synchronised timing.
- **Finance:** MiFID II requires trade timestamps traceable to UTC within 100 μs for high-frequency venues, so that event ordering can be reconstructed and market abuse detected.
- **Power grids:** Phasor measurement units time-stamp voltage phase across a continent; grid stability and post-blackout forensics both depend on it.
- **Relativistic geodesy:** This is the emerging one and it is genuinely strange. Since

$$\Delta\nu/\nu = gh/c^2$$

, a height difference of 1 centimetre shifts a clock's rate by 1.09×10^{-18} . So a 10^{-18} clock is an **altimeter with centimetre resolution**. Comparing two such clocks measures the local gravitational potential directly. This is a capability that did not exist even as a concept until the clocks got good enough, and it connects directly to the resource-exploration material I looked at later.

How an atomic clock actually works

This took me a while to get straight, so I want to state it carefully: **the atoms are not the clock**.

An atomic clock is a feedback loop.

A local oscillator - a quartz crystal, or a laser stabilised to an optical cavity - generates the actual ticks. A sample of atoms is then interrogated with that signal. The fraction of atoms that change state tells the servo whether the oscillator is running above or below the atomic resonance, and the servo nudges it back. The atoms are the *reference*, the oscillator is the *clock*, and the feedback loop ties one to the other.

The interrogation almost always uses **Ramsey separated oscillatory fields**: two short $\pi/2$ pulses separated by a long free-evolution time T . During that free evolution the atomic superposition accumulates phase $\Delta\omega \cdot T$, so the fringe width narrows as $1/T$ rather than being limited by how long the pulses are. Longer free evolution means a sharper measurement, which is why caesium fountains toss atoms upward - to buy free-fall time.

The fractional instability is

$$\sigma_y(\tau) \simeq \frac{1}{Q \cdot \text{SNR}} \sqrt{\frac{T_c}{\tau}} = \frac{1}{2\pi\nu_0 T \sqrt{N}} \sqrt{\frac{T_c}{\tau}},$$

where $Q = \nu_0/\Delta\nu$ is the line quality factor, N the atom number, T_c the cycle time and τ the averaging time.

Three things follow immediately, and together they explain the entire direction of the field:

1. **Raise ν_0** : This is why the field moved from microwave (9.2 GHz for caesium) to optical (429 THz for strontium) - a factor of about 50,000 in Q for free. The obstacle was that nobody could count optical frequencies directly, and the fix was the femtosecond frequency comb, which won the 2005 Nobel Prize. Without it, optical clocks would be laboratory curiosities you could not actually read.
2. **Raise N , but only as $\sqrt{N}$** : This is the **standard quantum limit**, and it comes from quantum projection noise. Each atom is an independent binary measurement on a superposition, so the sample mean has binomial variance. More atoms help, but with diminishing returns.
3. **Average longer**, as $\sqrt{\tau}$, until a systematic floor is reached.

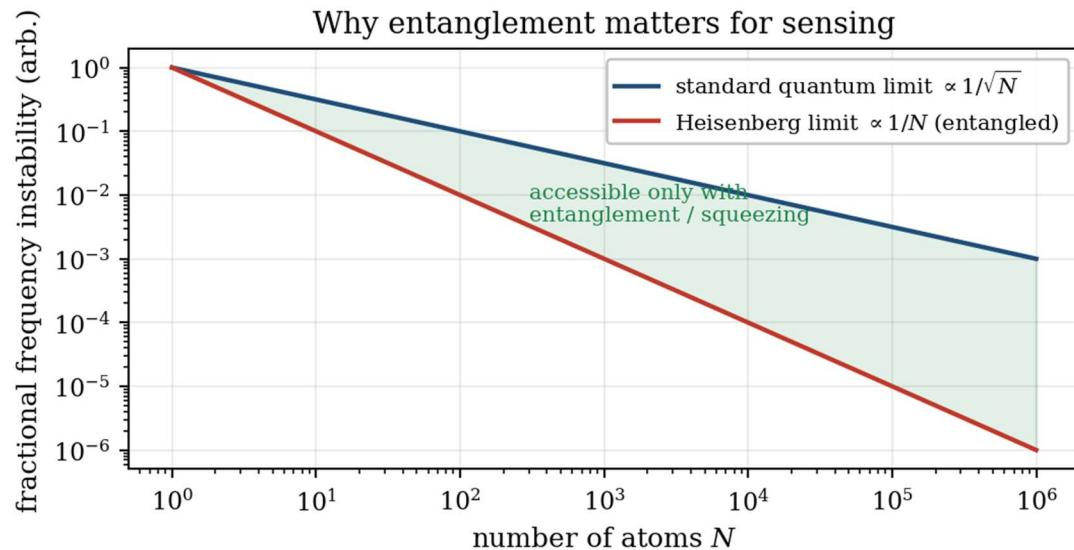


Figure 32: The standard quantum limit $1/\sqrt{N}$ compared with the Heisenberg limit $1/N$. At $N = 10^6$ atoms, entanglement offers a potential thousand-fold improvement.

That gap between the two curves is the target of spin-squeezed clocks, and it is the clearest example in this week of quantum *information* techniques being used to improve quantum *sensing*.

Is there a limit to how precise a clock can be?

There are several, in increasing order of fundamentality:

- *Technical*: Blackbody radiation shifts the atomic energy levels, which is the dominant systematic for strontium lattice clocks and means the chamber temperature has to be known to about 0.1 K. There are also collisional shifts, lattice light shifts, and the stability of the interrogation laser itself.
- *Quantum*: projection noise at $1/\sqrt{N}$, beatable down to $1/N$ with entanglement but no further - the Heisenberg limit is a genuine bound.
- *Natural linewidth*: Q cannot exceed ν_0/Γ , set by the excited-state lifetime, which is why forbidden transitions with extremely narrow lines are chosen.
- *Relativistic - and this one is conceptual rather than technical*: at 10^{-18} , the gravitational redshift over 1 cm is larger than the clock's uncertainty. Two clocks in the same room genuinely run at different rates.

That last point is the one I found most interesting. At that level, "the time" stops being a single global quantity and becomes a property of a specific point in spacetime. Past that, improving a clock does not measure time better - it measures **gravity** better. The limit is not on the instrument. It is on whether the question still means anything.

Atom interferometers

Newton's rings

Newton's rings form in the thin, wedge-shaped air gap between a convex lens and a flat plate. Light reflected from the bottom of the lens interferes with light reflected from the top of the flat. For a gap of thickness t , the optical path difference is

$$\Delta = 2t + \frac{\lambda}{2},$$

where the extra $\lambda/2$ is a π phase shift on reflection at the air-to-glass (low-to-high index) boundary but not at the glass-to-air one. Destructive interference (dark rings in reflection) occurs when $2t = m\lambda$, and geometry gives $t \approx r^2/2R$, so

$$r_m = \sqrt{m\lambda R}$$

Two consequences are:

The rings get closer together further out, since spacing goes as $1/\sqrt{m}$. The first few rings are widely spaced and they crowd up towards the edge.

The centre spot is dark in reflected light. At $t \rightarrow 0$ the only path difference remaining is that π phase shift, which gives destructive interference. This is direct experimental evidence for the reflection phase shift, and historically it was a serious problem for Newton's own corpuscular theory of light - his particle model had no way to produce it, which is part of why the rings are named after him despite arguably disproving his position.

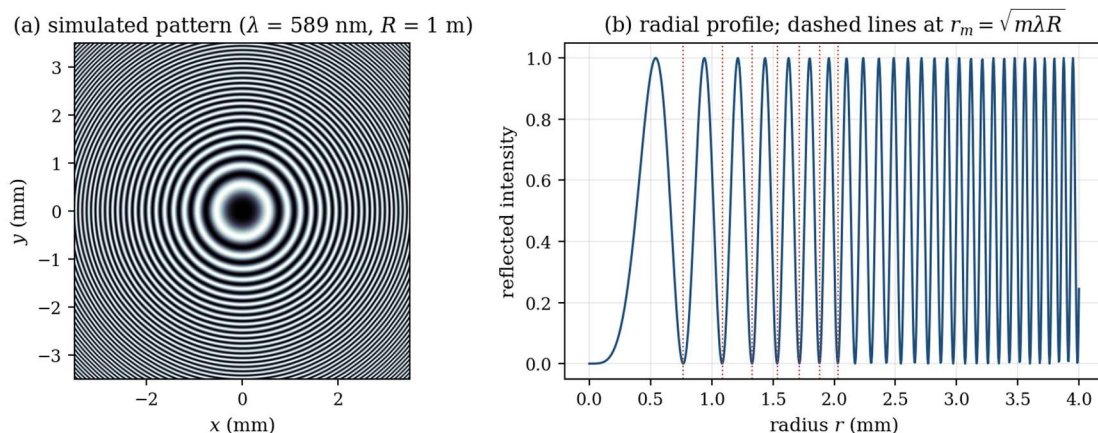


Figure 33: Newton's rings simulated for $\lambda = 589 \text{ nm}$ and lens radius $R = 1 \text{ m}$. (a) The full pattern showing the characteristic dark centre. (b) Radial intensity profile, with dashed lines at $r_m = \sqrt{m\lambda R}$: $r_1 = 0.767 \text{ mm}$, $r_2 = 1.085 \text{ mm}$, $r_3 = 1.329 \text{ mm}$

Three real-life examples:

- (i) *Oil films on wet roads and soap bubbles* - same thin-film physics with a different geometry, the colours arising because each wavelength satisfies the condition at a different thickness.
- (ii) *Screen protectors and laminated glass* - the coloured blotches when a protector is not fully adhered are literally Newton's rings from the trapped air gap, and installers use them as a diagnostic: the rings vanish as the gap is squeezed out.
- (iii) *Anti-reflective coatings* on spectacles and camera lenses - the same interference used constructively, with a quarter-wave layer engineered so reflections cancel; the residual purple-green tint is the wavelength where the cancellation is imperfect. A fourth I would add is optical flat testing, still the standard workshop method for checking a surface to a fraction of a wavelength.

Two-path interference, one particle at a time

Before comparing atom and light interferometers, I used the provided widget to look at what two-path interference actually is at the level of individual particles, since both instruments depend on it.

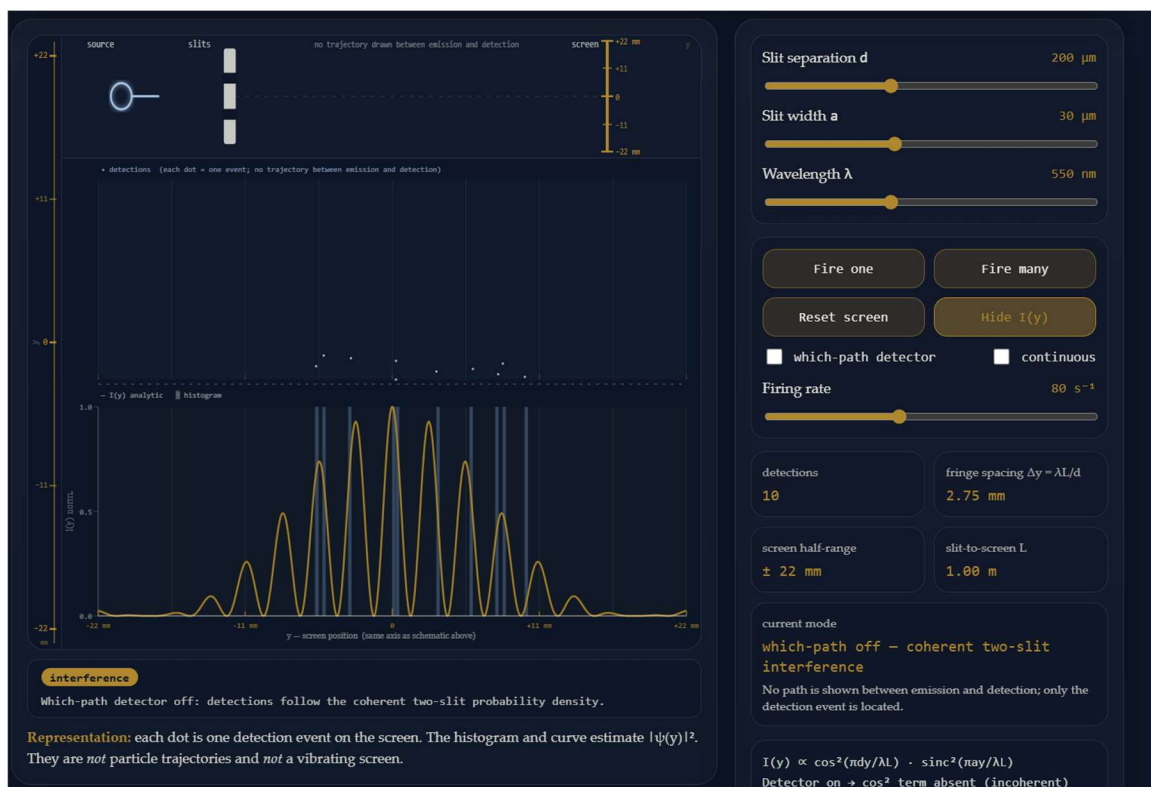


Figure 34: Build-up of the interference pattern one detection at a time. At 10 detections there is no pattern at all;

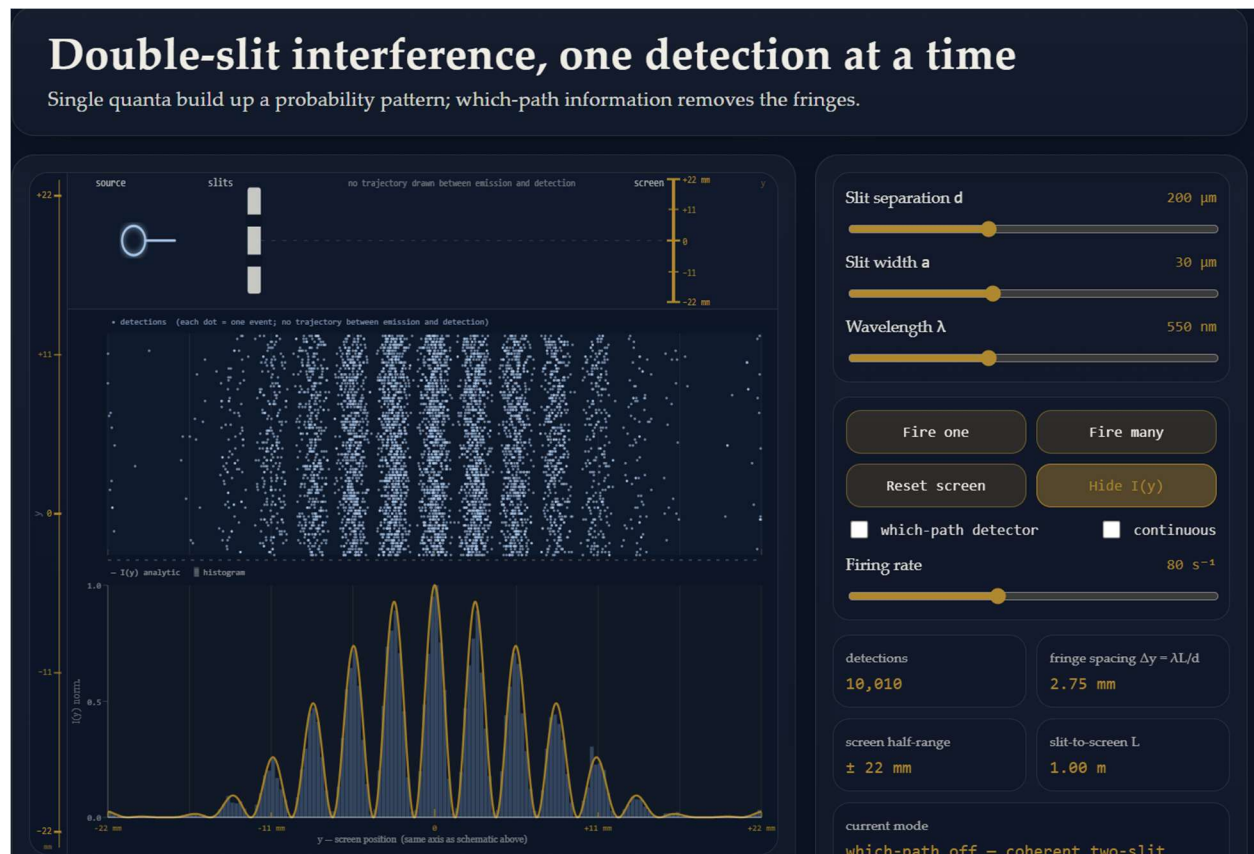


Figure 35: by 10,000 the fringes match the predicted $I(y)$ curve

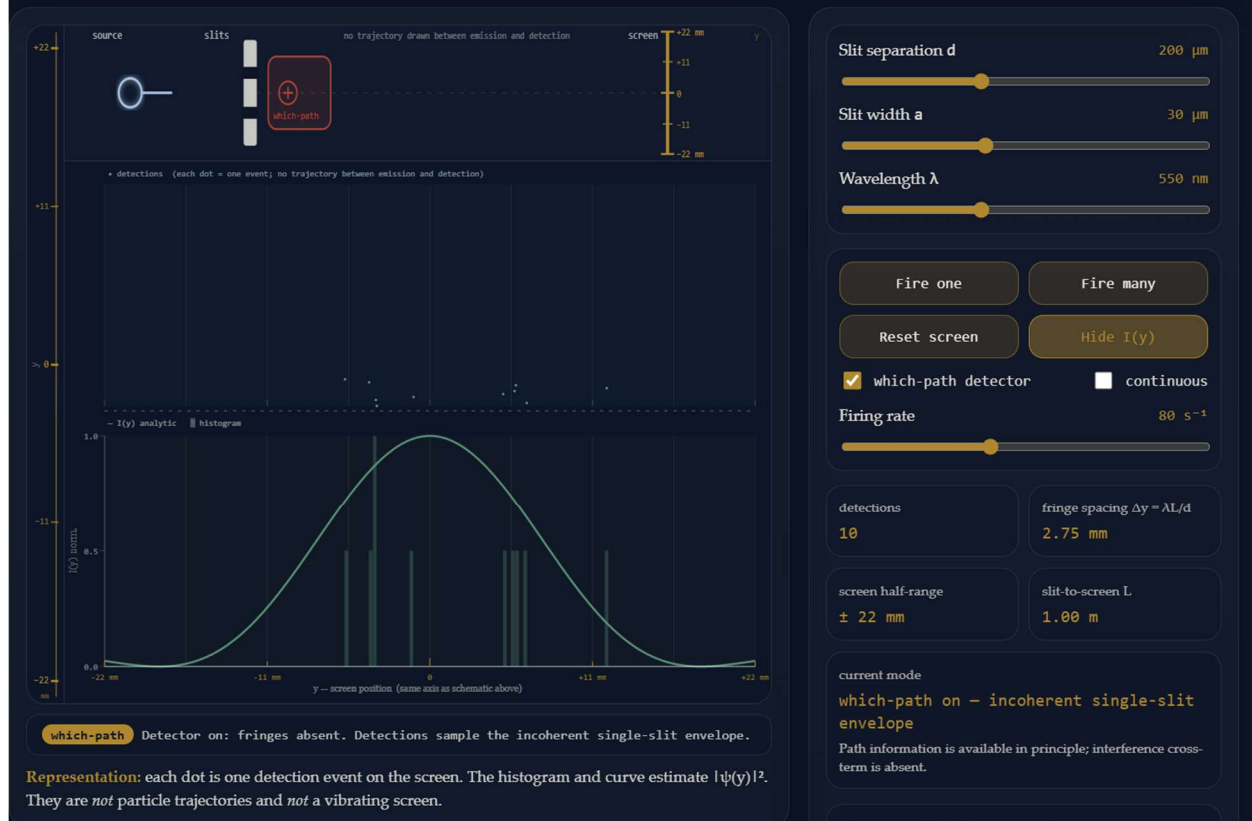
Each particle arrives as a single localised dot. No individual dot is spread out, and no single dot can be said to be "in a fringe" - a fringe is a property of the collection. Yet the accumulated pattern is unmistakably a wave interference pattern.

The particles are not interfering with each other. They arrive one at a time and never meet.

What interferes is each single particle's probability amplitude with itself. This is the Born rule from the previous week in experimental form: $|\psi|^2$ gives the probability of a detection at each point on the screen, and firing particles one at a time samples that distribution point by point.

Double-slit interference, one detection at a time

Single quanta build up a probability pattern; which-path information removes the fringes.



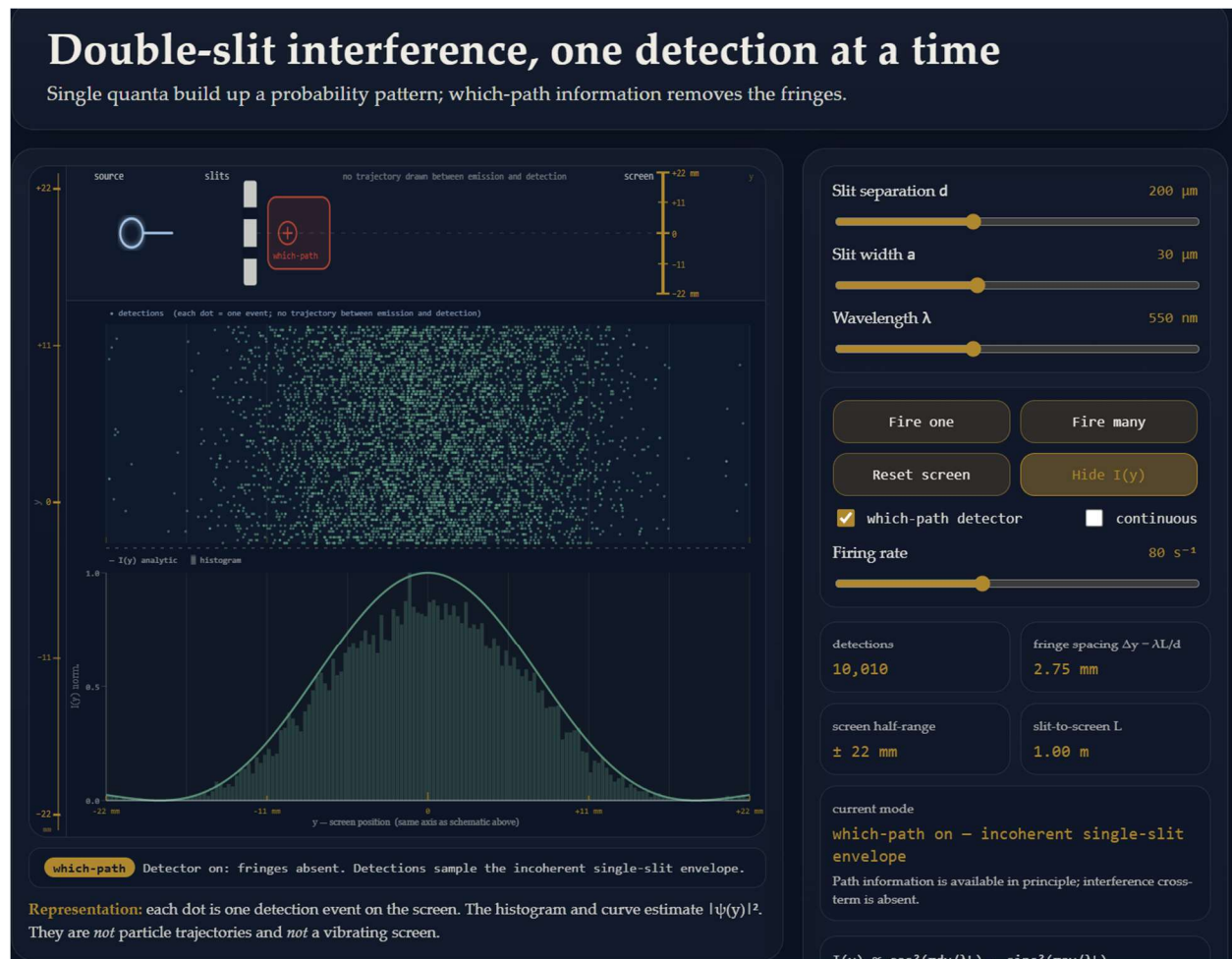
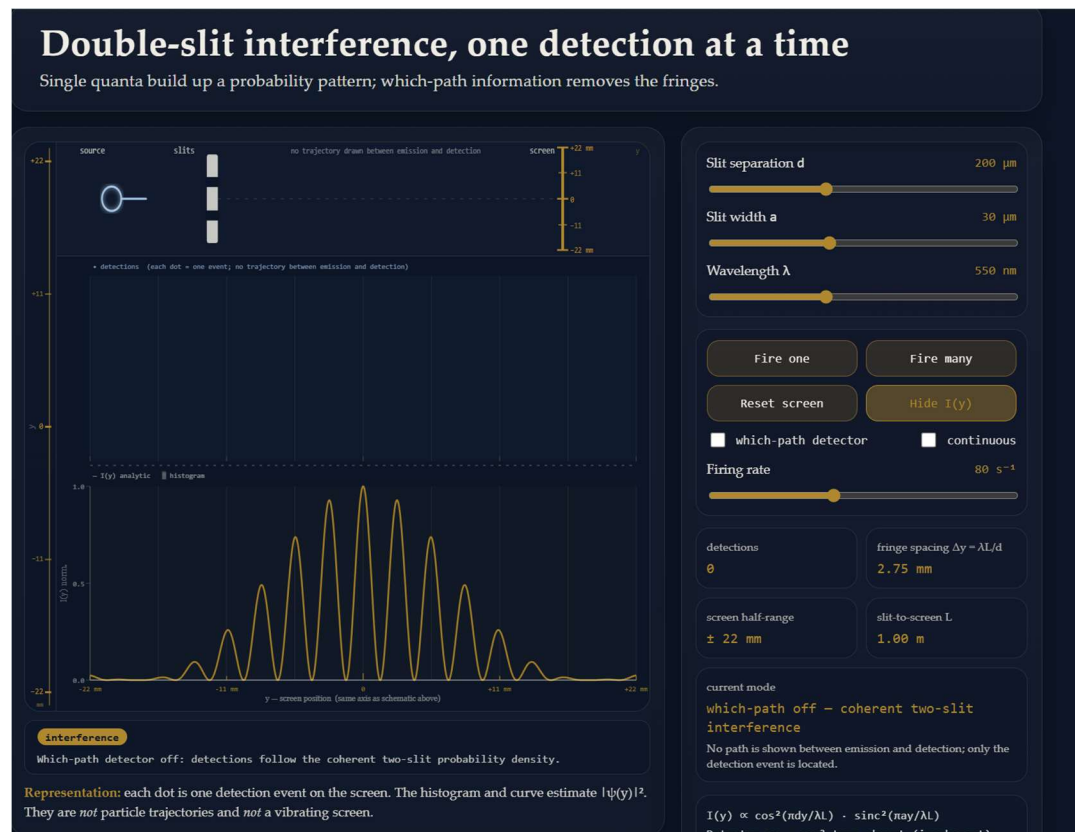
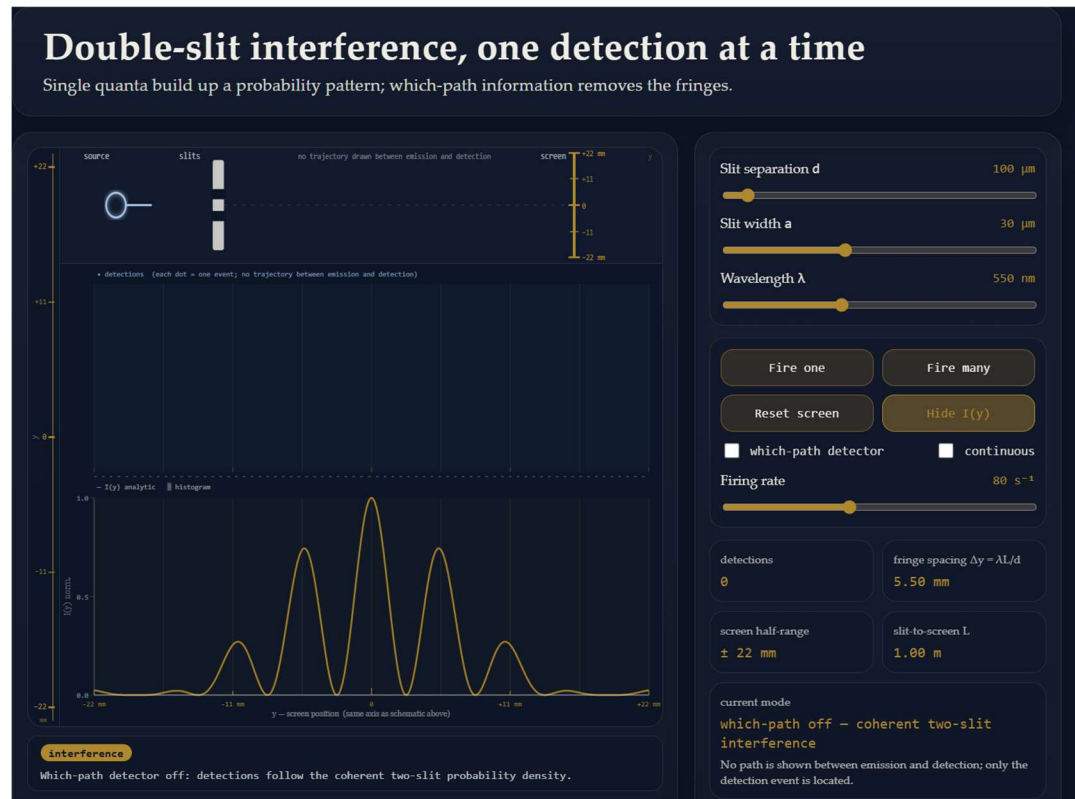


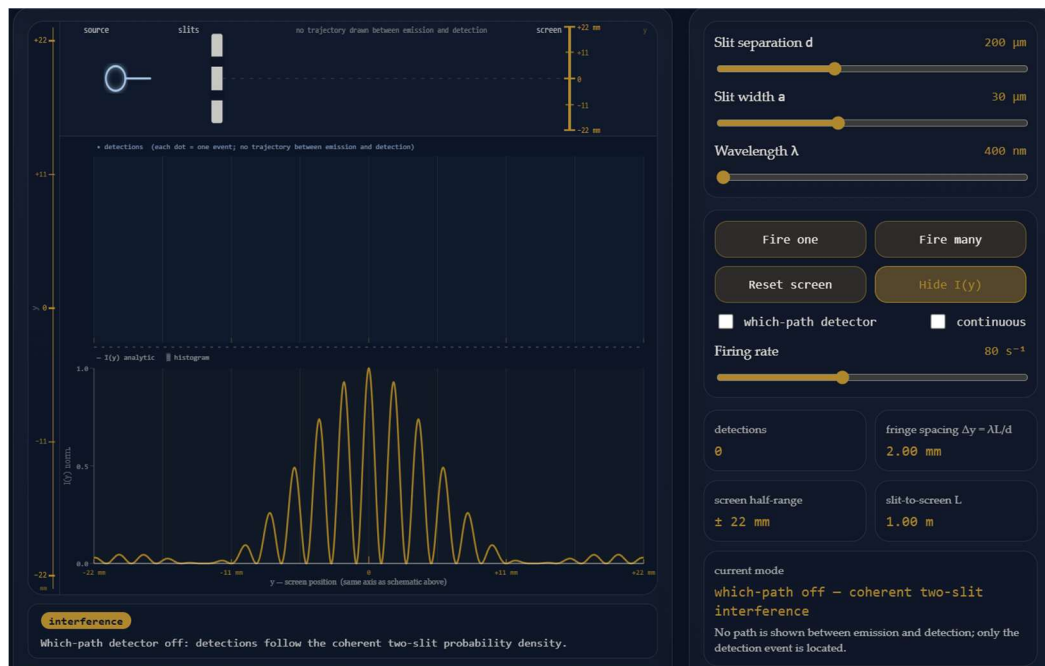
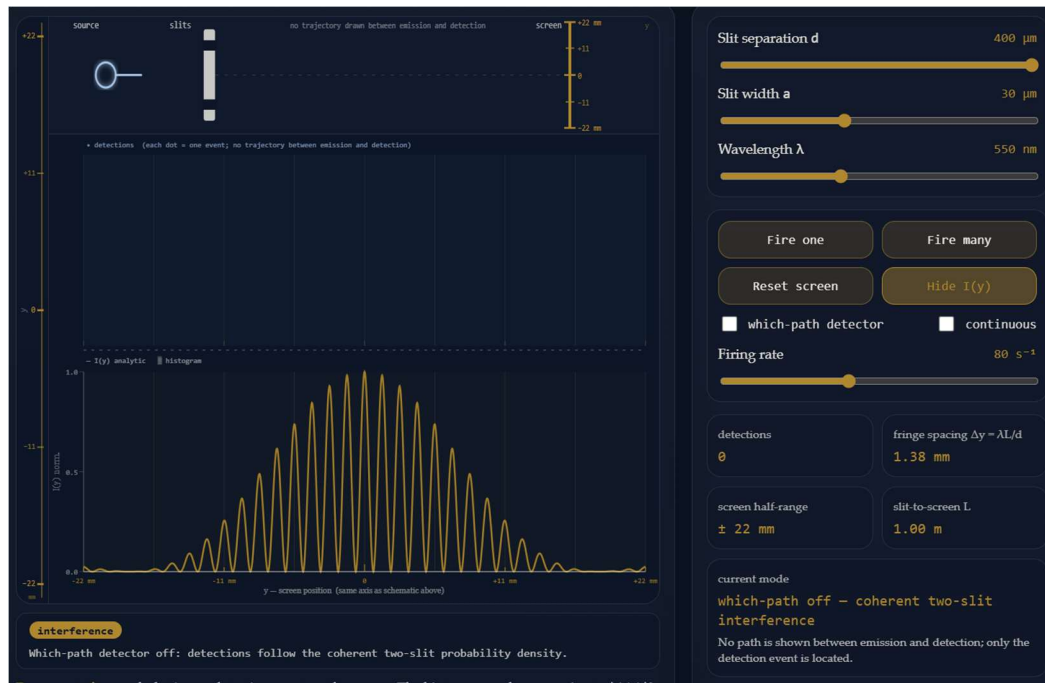
Figure 36: With which-path information available, the fringes vanish and the pattern becomes the sum of two single-slit humps

Nothing about the slits changed. The only difference is that information about which path was taken now exists somewhere. Once the path is knowable, the two amplitudes stop adding coherently and the interference disappears.

This is the clearest statement I have come across of why coherence is *the* resource in quantum technology. It is the same reason a quantum computer has to be isolated from its environment: any interaction that leaks path information into the surroundings destroys the interference the computation depends on. Decoherence is not a manufacturing fault - it is this exact effect happening by accident.

Checking the fringe formula:





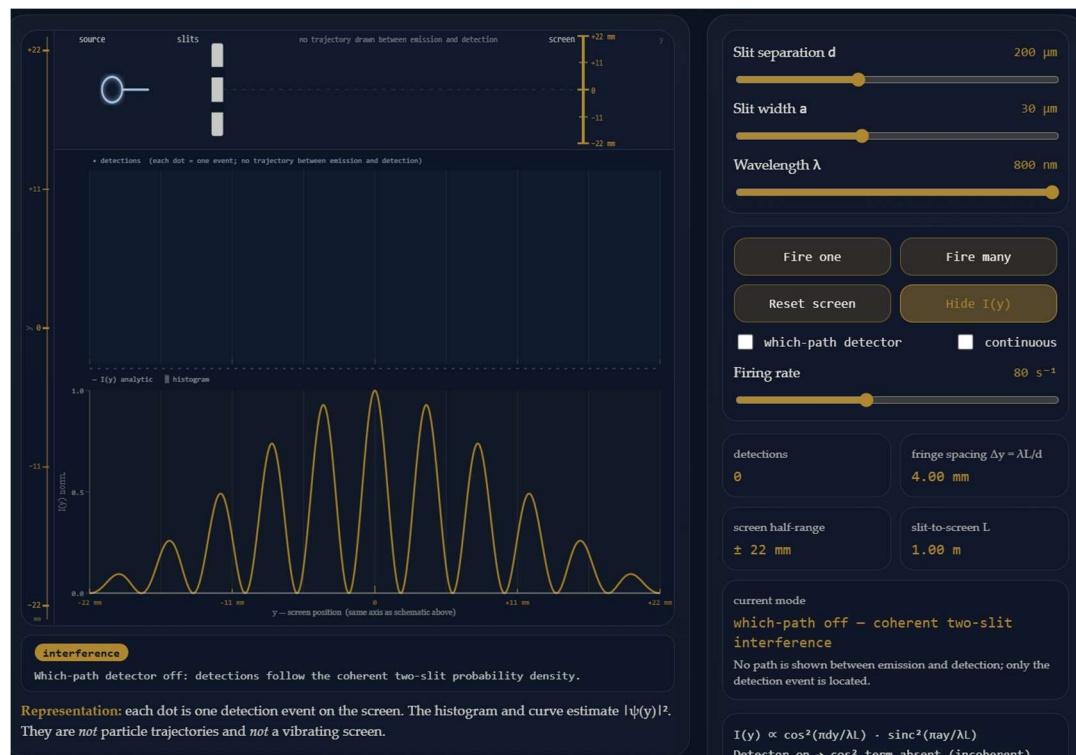


Figure 37: Measured fringe spacing against slit separation and wavelength, confirming $\Delta y = \lambda L/d$ with $L = 1.00$ m.

Setting	Measured Δy
$\lambda = 550 \text{ nm}$, $d = 100 \text{ }\mu\text{m}$	5.50 mm
$\lambda = 550 \text{ nm}$, $d = 200 \text{ }\mu\text{m}$	2.75 mm
$\lambda = 550 \text{ nm}$, $d = 400 \text{ }\mu\text{m}$	1.38 mm
$\lambda = 400 \text{ nm}$, $d = 200 \text{ }\mu\text{m}$	2.00 mm
$\lambda = 800 \text{ nm}$, $d = 200 \text{ }\mu\text{m}$	4.00 mm

Doubling the slit separation *halves* the fringe spacing, which is counter-intuitive the first time - moving the slits further apart squeezes the pattern together rather than spreading it out. Doubling the wavelength doubles the spacing. Both confirm $\Delta y = \lambda L/d$ from my own measurements rather than from a textbook.

I also varied the slit width a and found that the fringe *spacing* does not change but the *envelope* does - a wider slit concentrates the pattern into a narrower central region with fewer visible fringes. Two separate effects are superimposed: interference between the two slits sets the fringe spacing, and diffraction at each individual slit sets the envelope that modulates their brightness. Worth separating, because it is easy to attribute an envelope change to the wrong cause.

Atom interferometers versus classical interferometers

The structural analogy is exact - split, propagate, recombine, count fringes - but almost every physical quantity is different, and the differences are what make atom interferometers useful.

	Classical (optical)	Atom interferometer
What interferes	Photons (massless, $v = c$)	Massive atoms (e.g. Rb-87, Cs)
Wavelength	$\lambda \sim 633\text{nm}$	$\lambda_{\text{dB}} = h/mv$: 0.016 nm at 300 K, 27 nm at 100 μK
Beam splitters	Physical optics (glass, coatings)	<i>Light pulses</i> ($\pi/2$ - π - $\pi/2$ Raman) - the roles of matter and light are swapped
Couples to	Optical path length, refractive index	Inertia, gravity, rotation, internal state - because atoms have mass
Internal structure	None usable	Hyperfine states label the two paths
Sensitivity scaling	$\propto 1/\lambda$	$\propto k_{\text{eff}}T^2$: quadratic in interrogation time
Practical cost	Room temperature, high flux, fast	Ultra-high vacuum, laser cooling, low flux, slow cycle

The line I find most striking is the beam splitter row. **In an atom interferometer the roles of light and matter are swapped.** In an optical instrument, matter (a glass beam splitter) manipulates light. In an atom interferometer, light pulses manipulate matter. The mirrors and beam splitters are made of laser beams.

The key quantitative point is the phase response of a Mach-Zehnder atom interferometer to acceleration:

$$\Delta\phi = k_{\text{eff}}aT^2.$$

The T^2 is what makes these devices formidable. For Rb-87 with two-photon Raman transitions ($k_{\text{eff}} = 2 \times 2\pi/780\text{nm}$) and $g = 9.81\text{m/s}^2$, I computed:

Interrogation time T	$\Delta\phi$ (rad)	Fringes
10 ms	1.58×10^4	2.5×10^3
100 ms	1.58×10^6	2.5×10^5
1 s	1.58×10^8	2.5×10^7

Resolving 1 milliradian of phase at $T = 1$ s would give $\delta g/g \approx 6 \times 10^{-12}$.

Real instruments reach about 10^{-9} to 10^{-10} , and I think the gap is instructive rather than embarrassing. The limit is not phase readout at all - it is systematics: vibration isolation,

wavefront aberration in the Raman beams, Coriolis coupling to the Earth's rotation, and the finite temperature of the atom cloud. Quoting the ideal number without that caveat would badly misrepresent the state of the technology. It also explains why fountain geometries, drop towers and proposed space missions are pursued: they all buy interrogation time T , and the response goes as T^2 .

One more comparison worth recording. The Compton frequency of Rb-87 is $mc^2/h = 1.96 \times 10^{25} \text{Hz}$, which is *about* 2×10^{15} times the caesium clock frequency. In a sense a massive atom is already the highest-frequency oscillator available anywhere; atom interferometry is one way of getting partial access to it.

Connecting back to the widget. A room-temperature rubidium atom has a de Broglie wavelength of about 0.016 nm - roughly 40,000 times shorter than visible light. Since $\Delta y = \lambda L/d$, its fringes would be 40,000 times finer than the ones I measured in the widget, which is completely unmeasurable. That is why atoms have to be laser-cooled to microkelvin before they can be interfered at all: cooling slows them, which lengthens λ_{dB} . And it is also exactly why the technique is so sensitive once you manage it, since a short wavelength means a tiny path difference produces a large phase shift.

Quantum diamond magnetometers

Discovery and development

The nitrogen-vacancy centre - a substitutional nitrogen adjacent to a lattice vacancy - was characterised optically in the 1960s, but the decisive step was the 1997 demonstration that a *single* NV centre could be optically detected at room temperature, and that its spin state could be read out through its fluorescence brightness (optically detected magnetic resonance, ODMR). Around 2008, several groups independently proposed and demonstrated nanoscale magnetometry with NV centres.

The mechanism is that the $m_s = 0$ and $m_s = \pm 1$ ground-state sublevels are split by $D = 2.87 \text{GHz}$ at zero field, and an external field B Zeeman-splits the ± 1 levels by $2\gamma B$ with $\gamma = 28 \text{GHz/T}$. Sweeping a microwave frequency and watching fluorescence dip locates the resonance and hence B .

Why diamond, of all materials?

Diamond seems a strange choice - it is hard, expensive and chemically inert, none of which sounds like a sensor requirement. The reasons are all about *spin coherence*:

- **Wide bandgap (5.5 eV):** Deep defect levels sit well inside the gap, so the NV has sharp, optically addressable transitions and the host is transparent at the wavelengths used. In a narrow-gap semiconductor the defect states would hybridise with the bands.
- **Stiff lattice, Debye temperature ~2200 K:** Spin-lattice relaxation is driven by phonons; diamond's extreme stiffness pushes phonon energies high and freezes out the relevant modes, so T_1 can be milliseconds *at room temperature*. This is the single most important property - it is why NV magnetometry does not need a cryostat.
- **An almost spin-free host:** Carbon is 98.9% ^{12}C , which has zero nuclear spin. The NV electron spin therefore sits in an unusually quiet magnetic environment. Isotopic purification to $> 99.99\%$ ^{12}C extends T_2 into the millisecond range.
- **Biocompatible and inert:** Nanodiamonds can be put inside living cells without toxicity, which no other high-performance magnetometer can claim.

Compared to a **gas** (as in alkali vapour magnetometers): a vapour cell needs heating, buffer gases and a cell of centimetre scale, and atoms move, so spatial resolution is limited by the cell.

Compared to a **metal**: a metal is full of mobile conduction electrons, which is precisely the worst environment for preserving a spin state; coherence would be destroyed in nanoseconds.

Diamond's usefulness comes from being an insulator with a rigid, magnetically silent lattice - exactly the properties that make it useless for conventional electronics.

NV versus SQUID for biomedical applications

	NV diamond	SQUID
Operating temperature	Room temperature	4 K (LTS) or 77 K (HTS)
Standoff from sample	Nanometres to microns	~1–2 cm (dewar wall + thermal gap)
Spatial resolution	~10 nm (single NV)	Millimetres
Sensitivity	$\sim pT/\sqrt{\text{Hz}}$ (ensemble)	$\sim fT/\sqrt{\text{Hz}}$
Vector capability	Yes - four NV orientations give full $\vec{B}$	Scalar (one loop axis)
Dynamic range	Very large; no flux trapping	Limited; flux jumps, needs shielding
Size / cost / mobility	Chip-scale, portable	Large, heavy, expensive cryogenics
Biocompatibility	Nanodiamonds usable <i>inside</i> cells	No

Here is the part I found genuinely interesting. SQUIDs are **more sensitive in absolute terms**, by about three orders of magnitude. And yet NV centres can still win for biomedicine - and the reason is geometry, not sensitivity.

A dipole field falls off as $1/r^3$. Moving a sensor from 2 cm away to 200 μm away gains a factor of 10^6 in signal strength, which more than compensates for a 10^3 deficit in noise floor. And the cryogenics is not merely inconvenient for a SQUID: the vacuum gap and thermal shielding it requires are *what force the sensor to sit far from the source*. The cooling requirement and the standoff distance are the same problem.

This is a nice example of a system-level trade-off that a component-level spec sheet completely hides. Comparing the two devices on $T/\sqrt{Hz}$ alone gives you the wrong answer, and I would not have got there by reading the numbers without thinking about the geometry.

For whole-head MEG specifically, SQUIDs still dominate clinical practice. The emerging competitor there is optically pumped magnetometers rather than NV - wearable, room temperature, and now within a factor of a few of SQUID sensitivity.

Superconducting quantum interference devices

History and sensitivity

Brian Josephson predicted in 1962 that Cooper pairs could tunnel coherently across a thin insulating barrier with no voltage across it. The prediction was initially met with scepticism - including from Bardeen - and was confirmed experimentally by Anderson and Rowell in 1963. The first dc SQUID followed in 1964 at Ford Research Laboratories.

On materials: early junctions used lead and tin alloys, which degraded when thermally cycled. The decisive advance was the Nb/Al-AlO_x/Nb trilayer junction in the early 1980s, which is robust and reproducible and is still the workhorse today. High-Tc YBCO SQUIDs became possible after 1986, trading some sensitivity for 77 K operation.

Numerical comparison:

Quantity	Magnitude	Ratio to SQUID floor
SQUID noise floor	$1 \times 10^{-15} T/\sqrt{\text{Hz}}$	1
Human brain signal (MEG)	$1 \times 10^{-13} T(100 \text{ fT})$	10^2
Human heartbeat (MCG)	$1 \times 10^{-10} T(100 \text{ pT})$	10^5
Earth's magnetic field	$5 \times 10^{-5} T(50 \mu\text{T})$	5×10^{10}

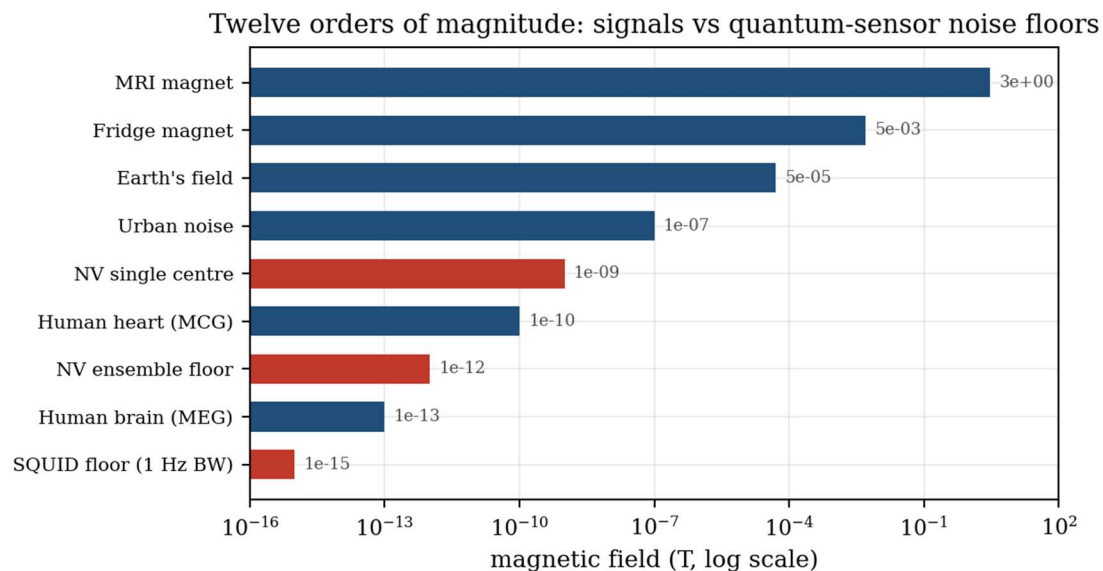


Figure 38: Magnetic field magnitudes over twelve orders of magnitude, with quantum sensor noise floors marked. The dynamic range problem, not the sensitivity problem, dominates practical biomagnetic measurement

The brain signal is only about 100 times the SQUID's noise floor in a 1 Hz bandwidth. Workable, but not comfortable. The much harder number is the last one. Urban electromagnetic noise is worse still.

This is why MEG requires magnetically shielded rooms (mu-metal plus eddy-current shielding) **and** gradiometer pickup coils, which are wound in opposition so that a spatially uniform background induces zero net flux while a nearby dipole source does not. The sensor's intrinsic sensitivity is arguably the easy part of the engineering problem. Rejecting a background eight orders of magnitude larger than your signal is the hard part.

Why the flux quantum sets the resolution, in plain language

A superconductor is described by a single macroscopic wavefunction $\psi = |\psi|e^{i\theta}$ shared by all the Cooper pairs in it. If you go once around a closed superconducting loop, the phase θ has to come back to its starting value plus a multiple of 2π , because the wavefunction must be single-valued. That requirement forces the enclosed magnetic flux to come in whole multiples of:

$$\Phi_0 = \frac{h}{2e} = 2.068 \times 10^{-15} \text{Wb},$$

The $2e$ is the charge of a Cooper *pair*. Experimental confirmation of that factor of two was itself strong evidence for BCS pairing theory - the measured flux quantum was half what you would expect for single electrons.

In a dc SQUID, two Josephson junctions split the loop into two arms. The supercurrent through each obeys $I = I_c \sin \delta$, and the phase difference between the arms is set by the enclosed flux, so the critical current oscillates:

$$I_c(\Phi) = 2I_0 \left| \cos \left(\frac{\pi \Phi}{\Phi_0} \right) \right|,$$

and the voltage across the biased device oscillates with period exactly Φ_0 .

The SQUID is a two-slit interference experiment in which the “fringes” are counted electrically.

Why this sets the resolution. Φ_0 is a fixed, universal ruler - a natural unit of flux built from fundamental constants. The device does not measure flux directly; it measures *where within one fringe* it currently sits, and the resolution is whatever fraction of a fringe the readout electronics can resolve. Concretely, with a 1 cm² pickup loop:

- One full fringe corresponds to $\Phi_0/A = 2.07 \times 10^{-11} \text{T} = 20.7 \text{pT}$.
- Resolving $10^{-6} \Phi_0$ (routine with flux-locked-loop feedback) gives $\delta B = 2.1 \times 10^{-17} \text{T} = 0.021 \text{fT}$.

- Earth's field through the same loop threads 2.4×10^6 flux quanta - over two million fringes - which is why the device must be operated in a feedback loop that holds it at a fixed point on the V - Φ curve rather than counting from zero.

Note the design consequence: because the quantity measured is flux, sensitivity to *field* improves with pickup loop *area*. This is exactly why a SQUID cannot be miniaturised without losing sensitivity, and hence why NV centres win at the nanoscale (§4.3.3).

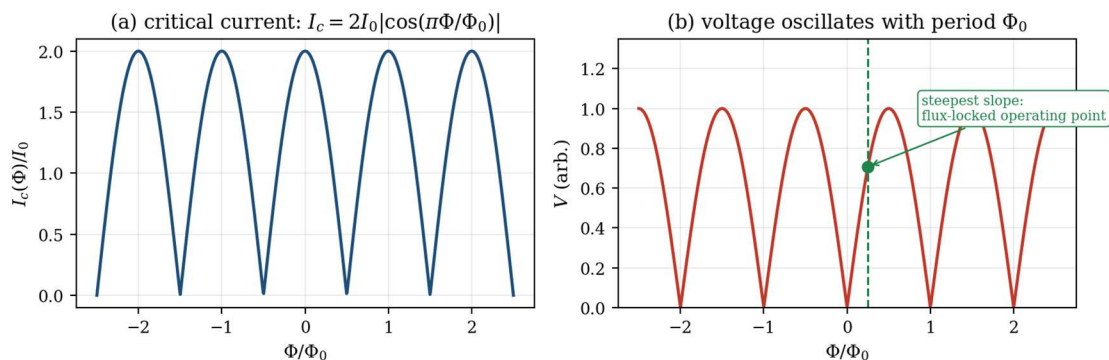


Figure 39: (a) Critical current versus applied flux. (b) The voltage response, periodic in Φ_0 , with the steepest point marked. Practical operation holds the device at $\Phi = \Phi_0/4$ using a flux-locked loop.

Why superconductivity, not ordinary conductivity, is essential. Three independent reasons, and all three are necessary:

1. **Macroscopic phase coherence.** Flux quantisation requires a single wavefunction with a well-defined phase around the whole loop. In a normal metal, electrons have no fixed mutual phase relationship and there is nothing to quantise - the effect simply does not exist.
2. **Zero resistance.** A screening current induced in a superconducting loop persists indefinitely and exactly cancels flux changes. In a normal metal it would decay in nanoseconds through L/R , so there would be no memory of the flux state.
3. **No Johnson noise.** Thermal voltage noise is $\sqrt{4k_B T R \Delta f}$. With $R = 0$ in the loop, this noise source vanishes entirely. A resistive loop would drown a femtotesla signal in its own thermal noise regardless of how good the amplifier was.

Additionally, the Josephson junction itself - the nonlinear element that makes the interference readable - requires coherent *Cooper-pair* tunnelling. It has no normal metal analogue.

What's the role of quantum computing in all/any of this?

This week produced the strongest connection to quantum computing of anything so far, and it runs in both directions.

Sensing technology built quantum computing: The Nb/Al-AlO_x/Nb Josephson junction developed for SQUIDs in the 1980s is the same device at the heart of every transmon qubit - a transmon is a Josephson junction shunted by a capacitor. SQUIDs themselves appear inside quantum processors as *tunable* elements: a SQUID loop acts as a flux-controlled inductor, which is how qubit frequencies and tunable couplers are adjusted in Google's and IBM's architectures. Josephson parametric amplifiers, which are SQUID-based, provide the near-quantum-limited readout amplification without which single-shot qubit measurement would be impossible. Likewise, the laser cooling and optical trapping developed for atomic clocks and atom interferometry are precisely the toolkit of neutral-atom quantum computers, and NV centres are used as room-temperature qubits and as spin-photon interfaces for quantum network nodes.

The shared physical resource is coherence: A sensor and a qubit are the same object used for different purposes: both are two-level quantum systems whose accumulated phase must be preserved. A sensor *wants* its phase to be perturbed by the environment in a controlled way; a qubit wants it not to be. The engineering - long T_2 , isotopic purification, dynamical decoupling, magnetic shielding - is identical. Indeed, dynamical decoupling sequences (CPMG, XY8) were developed in NMR, adopted by quantum computing to extend coherence, and then re-adopted by NV magnetometry to make AC-field sensing possible.

Quantum information improves sensing: The standard-quantum-limit plot in the Atomic Clocks section is the concrete statement. Entanglement moves a sensor from the $1/\sqrt{N}$ standard quantum limit towards the $1/N$ Heisenberg limit. Spin-squeezed atomic clocks and squeezed-light gravitational wave detectors already do this in practice. This is the most mature payoff from quantum information science, and it is worth being clear that it arrived **before** any useful quantum computer did.

A caveat I want to record. It is tempting to fold all of this into "quantum computing helps sensors," but the honest framing is the reverse: quantum sensing is a separate, more mature branch of quantum technology that shares hardware and concepts with quantum computing. Optical clocks and SQUID magnetometers were doing useful, world-changing work decades before anyone factored 15 on a quantum computer.

Week 5 Artefact - Quantum Sensing Applications

Societal impacts of quantum sensing: navigation, resource exploration, medical imaging and fundamental science.

Last week I looked at how these sensors work. This week is about what happens when they leave the laboratory. This artefact asks what happens when they leave the laboratory. I have tried to avoid the two failure modes I noticed in my first draft - uncritical technological optimism, and vague hand-waving about “ethical concerns” - by tying each impact to a specific mechanism and, where possible, a number.

Global Positioning System

How GPS changed the way people live, work, travel and communicate

GPS is the clearest case study of a quantum technology that became invisible infrastructure. Its effects fall into three tiers.

The obvious tier - navigation. Turn-by-turn routing eliminated a category of skill (map reading, dead reckoning) and a category of anxiety (being lost). It also restructured whole industries: ride-hailing, last-mile logistics and food delivery are not merely improved by GPS, they are *constituted* by it - none could exist without cheap, universal, real-time position.

The less obvious tier - precision work. Real-time kinematic GPS gives centimetre accuracy, which enables controlled-traffic farming and variable-rate application; in Australian broadacre agriculture this reduces overlap, compaction, fuel and chemical use on a continental scale. Construction, mining and surveying were similarly transformed. Search and rescue, disaster response and wildlife tracking all changed character.

The invisible tier - timing, which matters most. This is the part I had not appreciated before Week 4. GPS is the world’s primary distributed *clock*. Mobile networks, financial trade timestamping, power-grid phasor measurement, broadcast synchronisation and data-centre transaction ordering all take timing from GPS. The US Department of Homeland Security designates GPS a dependency of *all sixteen* critical infrastructure sectors. A GPS outage would not primarily be a navigation problem; it would be a timing problem, and it would surface first in telecommunications and finance.

On communication specifically, the change is subtler and worth stating carefully: GPS made *location a routine field in ordinary messages*. “Where are you?” became answerable automatically, location sharing became a normal feature of family and friendship, and geotagging became a default. That is a genuine change in how people communicate - and also the origin of most of the privacy concerns below.

Benefits and risks of more accurate or GPS-independent navigation

Quantum inertial navigation - The cold-atom accelerometers and gyroscopes from last week - offers dead reckoning with drift low enough to navigate for hours or days with no external signal at all. The benefits and the risks are two readings of the same capability, which is what makes this interesting rather than a simple pros-and-cons list

Benefits: GNSS jamming and spoofing have moved from theoretical to routine: aviation authorities have reported widespread interference affecting commercial flights in several conflict-adjacent regions since 2023, and spoofing has been documented near maritime chokepoints. A navigation system needing no external signal cannot be jammed or spoofed at all, which is a real safety gain for aviation and shipping. It also enables navigation where GNSS simply does not reach - underwater, underground, indoors, in deep mines, and in space beyond Earth orbit. Autonomous vehicles and aircraft gain a genuinely independent redundant channel rather than a correlated backup.

Risks: Four, in what I judge to be increasing order of seriousness:

1. **Concentration and cost:** Early quantum inertial units will be expensive and produced by few suppliers. If jam-proof navigation becomes a safety expectation, the capability gap widens between well-resourced and poorly-resourced operators, and between nations.
2. **Reduced incentive to protect civil GNSS:** If military and high-value civil users migrate to self-contained navigation, the political pressure to police jamming and maintain civil GNSS quality weakens. That leaves ordinary users on a degrading system while the well-resourced ones have moved on. I think this risk is underrated precisely because it is quiet - nothing dramatic happens, a public good just erodes.
3. **Surveillance asymmetry:** Precise, unspoofable position that requires no external signal is also position that cannot be denied or degraded by an authority - a benefit for a journalist, and a problem for a state trying to enforce a no-fly zone. It cuts both ways depending on who holds the device.
4. **Strategic destabilisation:** This is the serious one. The same cold-atom gravimetry and magnetometry that enables GPS-free navigation also improves the **detection of submerged submarines**, through their gravitational and magnetic signatures. Ballistic missile submarines are the survivable leg of nuclear deterrence precisely *because* they are hard to find. A technology that erodes the opacity of the ocean could destabilise a deterrence posture that has held since the 1960s.

Resource exploration

How quantum sensors improve discovery and mapping

Subsurface exploration works by detecting contrasts in physical properties. Quantum sensors sharpen two of them dramatically.

Gravity gradiometry: Density contrasts - an ore body, a void, a salt dome, a freshwater aquifer, a buried tunnel - perturb the local gravitational field. Cold-atom gravimeters and gradiometers measure g and its spatial derivatives with far better stability than spring-based instruments, and crucially they are *absolute* rather than relative: they do not drift, so they need no repeated recalibration against a base station. This matters enormously for airborne and long-duration surveys, where drift is usually the dominant error. The UK's demonstration of a field gravity gradiometer detecting a buried tunnel through ground noise in 2022 was a milestone, because vibration rejection had been the blocking problem for two decades.

Magnetic surveying: Many ore bodies, particularly iron and nickel sulfides, are magnetically distinct. NV-diamond and OPM magnetometers offer high sensitivity in small, low-power, vector-capable packages suitable for drone deployment - which changes survey economics from helicopter-scale to drone-scale.

The practical impact is a shift from *drill-and-hope* to *image-then-drill*. Exploration drilling is expensive, slow and locally destructive; better geophysics before drilling means fewer holes. This applies equally to groundwater mapping in drought-affected regions, geothermal resource assessment, monitoring of carbon sequestration reservoirs, detection of leaking pipelines, and mapping of legacy mine voids that pose subsidence risk.

Environmental, economic and ethical issues

I want to resist the easy framing here, which is that better exploration is straightforwardly good because it is less invasive per unit found.

The efficiency paradox is the central issue: Making extraction cheaper and more certain increases the economically viable resource base. Deposits previously too marginal to develop become profitable, and total extraction may rise even as impact *per tonne* falls - the Jevons effect. Whether quantum gravimetry reduces or increases total environmental damage is therefore not a technical question at all; it depends entirely on whether demand is capped by policy. The technology is genuinely neutral here, and saying so is more honest than claiming an environmental benefit.

The direction of demand complicates the picture further: The energy transition requires very large quantities of lithium, cobalt, nickel, copper and rare earths. Better exploration is therefore, in part, *decarbonisation infrastructure*. An environmentalist position that opposes improved

mineral exploration and supports rapid electrification is in tension with itself. I do not think this resolves cleanly, but it should be stated rather than avoided.

Land rights and consent: In an Australian context this is not abstract. Remote sensing can characterise land without physical access and, in principle, without engagement with Traditional Owners. The Juukan Gorge destruction in 2020 demonstrated what happens when survey and approval processes treat cultural heritage as a compliance formality. A technology that makes it easier to know what is under someone's land, without being on it, increases the importance of consent frameworks rather than reducing it. There is a genuine positive case too: high-resolution non-invasive survey can *map and avoid* culturally significant sites and buried heritage without excavation - but only if that is what it is used for.

Data sovereignty and asymmetry: Subsurface survey data is enormously valuable and is a form of privileged information. If a small number of firms or states hold the best sensors, they hold better knowledge of a country's resources than that country does, which is a recognisable pattern in the resource-curse literature. The governance question is who owns survey data taken over sovereign territory.

Deep-sea and Antarctic contexts: Improved detection of seabed polymetallic nodules intensifies pressure on deep-sea mining, over ecosystems that are barely characterised and where damage may be effectively permanent on human timescales.

Medical imaging

How quantum sensing improves diagnosis and treatment

MRI is a quantum technology that is so established it is rarely described as one. Nuclear spins - mostly protons in water and fat - are Zeeman-split by a strong field and precess at the Larmor frequency $\omega_0 = \gamma B_0$ (42.58MHz/T for protons). Spatial encoding comes from deliberately applied gradient fields, so position maps to frequency. Its clinical value is soft-tissue contrast without ionising radiation, and its contrast mechanisms (T_1 , T_2 , diffusion, perfusion, functional BOLD) are physiological rather than merely anatomical - diffusion MRI detects ischaemic stroke within minutes, long before any structural change is visible.

MEG measures the roughly 100 femtotesla magnetic fields produced by neuronal currents, using the SQUID arrays from last week. Its distinguishing property is temporal resolution of about a millisecond, against seconds for fMRI. The reason is fundamental: MEG measures neural activity *directly*, whereas fMRI measures the blood-flow response to it, and blood is slow.

Clinically, its established role is localising epileptogenic foci before resective surgery, and mapping eloquent cortex (language, motor) so the surgeon knows what to avoid. That is a case

where better localisation translates directly into preserved function in a specific patient - which is a more concrete benefit than most imaging improvements can claim.

Emerging directions: OPM-based wearable MEG allows scanning while the subject moves, which opens paediatric and developmental studies previously impossible. NV-diamond sensors enable NMR at the scale of single cells. Hyperpolarised MRI raises spin polarisation by orders of magnitude, allowing metabolic imaging of processes such as pyruvate-to-lactate conversion in tumours. Low-field portable MRI trades resolution for the ability to bring the scanner to a bedside, an ambulance or a rural clinic.

Ethical, social and economic challenges

Access and equity - the dominant issue: Advanced imaging is capital-intensive: a clinical MRI system costs several million dollars installed, plus a shielded room, cryogens and specialist staff; a MEG installation more still. The result is a stark availability gradient. Within Australia, metropolitan access far exceeds regional and remote access, which bears disproportionately on Aboriginal and Torres Strait Islander communities; internationally, many low-income countries have a handful of MRI scanners for tens of millions of people. *Improving the top end of imaging performance does not narrow that gap and may widen it.* I think this is the most important point in this section, and it argues that the socially valuable innovation may be low-field portable MRI - technically less impressive, far more consequential.

Overdiagnosis and incidental findings: More sensitive imaging finds more abnormalities, and not all abnormalities are diseases. Incidental findings on brain MRI are common in asymptomatic adults, most of them clinically irrelevant, but each triggers follow-up, cost and anxiety. Detecting something earlier only helps if earlier intervention improves outcomes, which is an empirical question that must be answered condition by condition rather than assumed.

Neural data and mental privacy: MEG and related techniques produce data about brain function. As decoding methods improve, the question of what can be inferred - attention, recognition, deception, predisposition - becomes concrete. Several jurisdictions (Chile being the first, in 2021) have begun legislating “neurorights.” The issues are meaningful in employment, insurance and legal contexts, and existing medical privacy law was not written with functional neural data in mind. Equally, I would caution against the more breathless claims: current MEG decoding is far from thought-reading, and overstating it distracts from the real and immediate problem, which is access.

Workforce and interpretation. A scanner without a radiologist trained to read it is not a diagnostic capability. This is often the binding constraint in exactly the settings that lack

scanners. AI-assisted interpretation is a plausible partial answer and brings its own questions about liability, validation and dataset bias.

Science discoveries

Detecting the almost undetectable

Gravitational waves: LIGO measures a strain $h \sim 10^{-21}$ across 4 km arms:

$$\Delta L = hL = 10^{-21} \times 4000m = 4 \times 10^{-18}m,$$

which is about 1/200 of a proton radius, or 10^{-3} of a femtometre. Reaching this required confronting quantum noise directly. Two effects oppose each other: photon *shot noise* (the Poisson counting statistics of the readout, reduced by using more laser power) and *radiation-pressure noise* (photons kicking the mirrors, increased by more power). Their combination defines a standard quantum limit - and this is the same $\hbar/2$ trade-off computed numerically in Week 3, appearing in a four-kilometre instrument.

The resolution is beautiful and directly connects to this portfolio's themes: since 2019 the detectors inject **squeezed vacuum** into the interferometer's dark port, redistributing the quantum uncertainty between the amplitude and phase quadratures so that phase (which carries the signal) is measured below the shot-noise limit at the cost of worse amplitude noise. A few decibels of squeezing translates into roughly a $1.4 \times$ increase in detection range and therefore about $2.8 \times$ the observable volume and event rate. A Week-3 minimum-uncertainty state is doing operational work in an observatory.

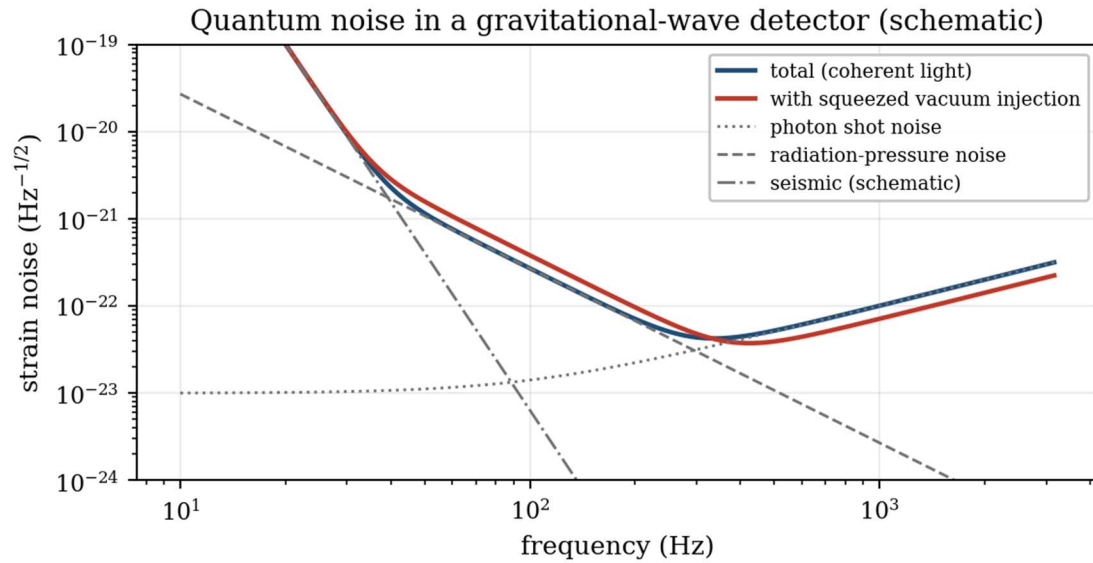


Figure 40: Schematic quantum-noise budget for a gravitational-wave detector. Squeezing lowers shot noise at high frequency at the cost of raising radiation-pressure noise at low frequency - the uncertainty product is conserved and merely redistributed.

Demonstrating the measurement principle:

LIGO is a Michelson interferometer, so I used the provided widget to work through how one actually functions as a sensor.

One relationship to establish first: the optical path difference is $OPD = 2\Delta L$, not ΔL , because the light travels down each arm and back. That factor of two is part of why an interferometer is such a sensitive ruler.

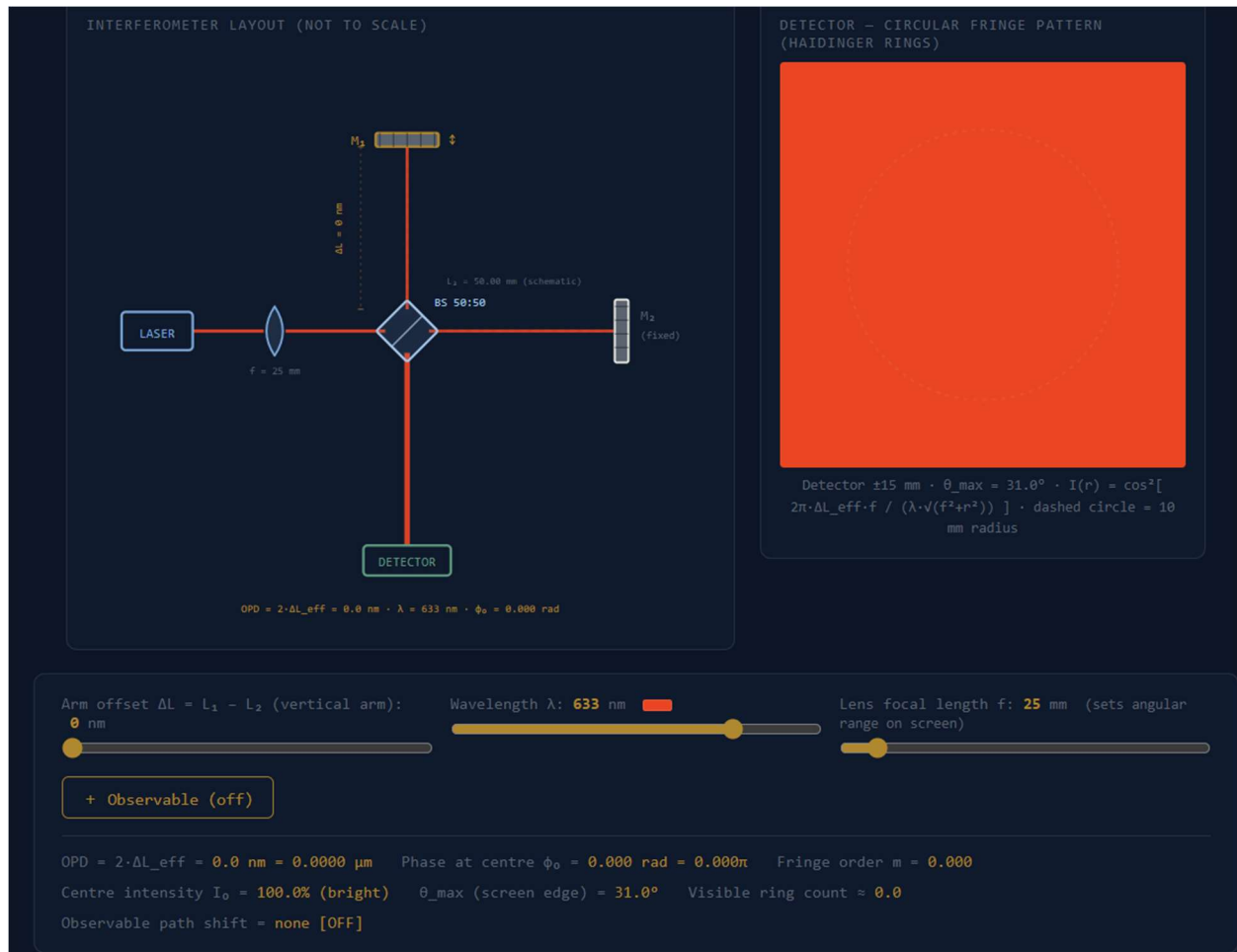


Figure 41: Zero path difference: OPD = 0, fringe order $m = 0.000$, intensity = 1.000. No rings, uniform bright field

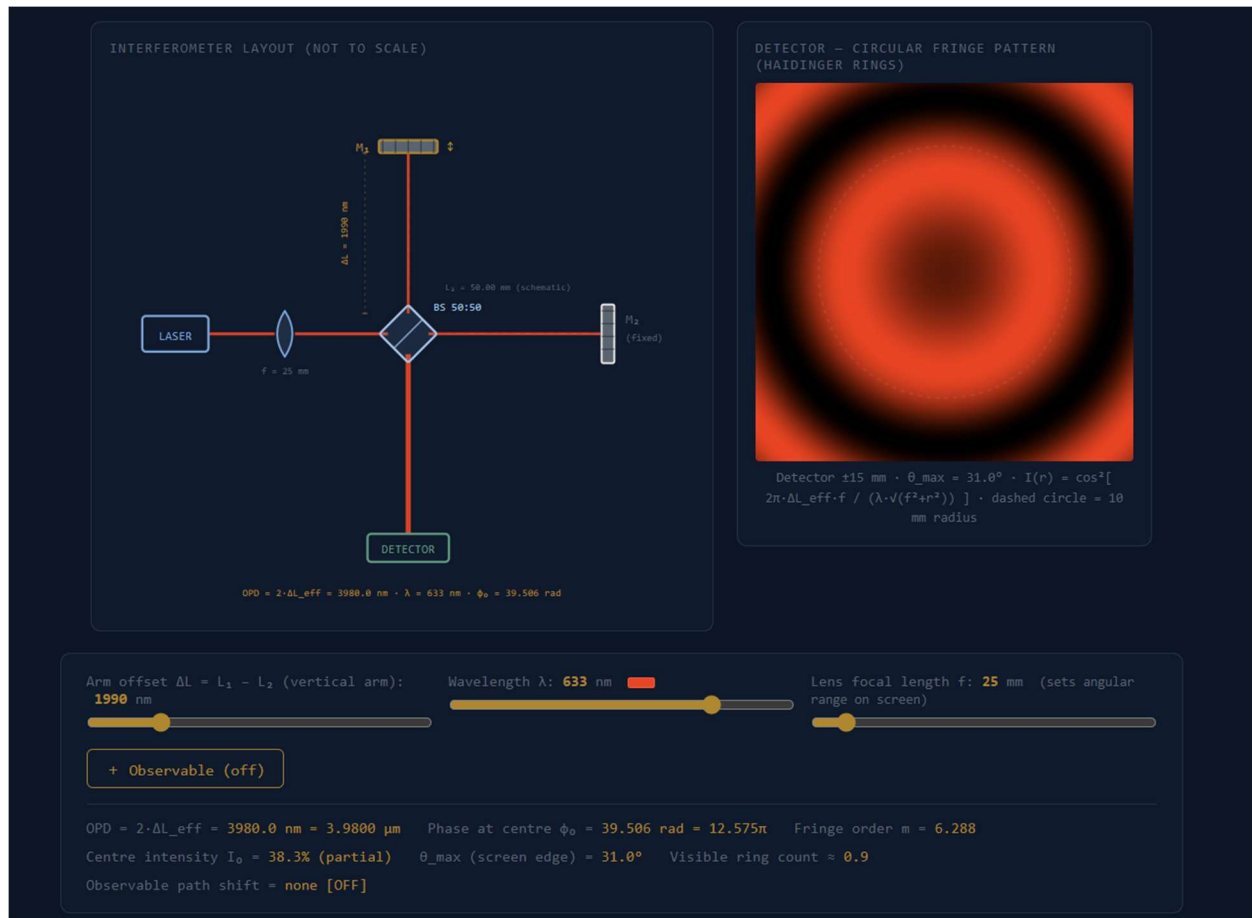


Figure 42: A moderate path difference produces circular fringes. OPD = $\sim 4000 \text{ nm}$, $m = 6.288$, intensity = ~ 0.290 .

The sensitivity demonstration: From $\Delta L = \sim 2000 \text{ nm}$ I nudged the slider one step at a time, which is 50 nm , and watched the centre intensity.

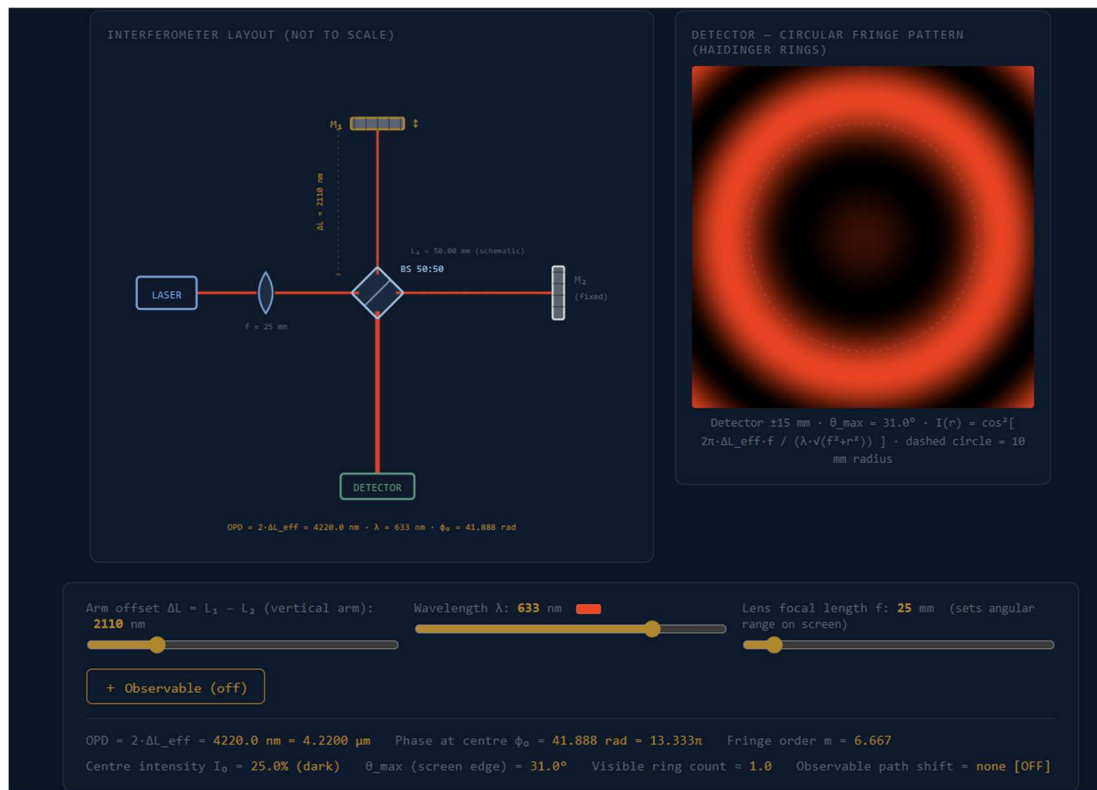
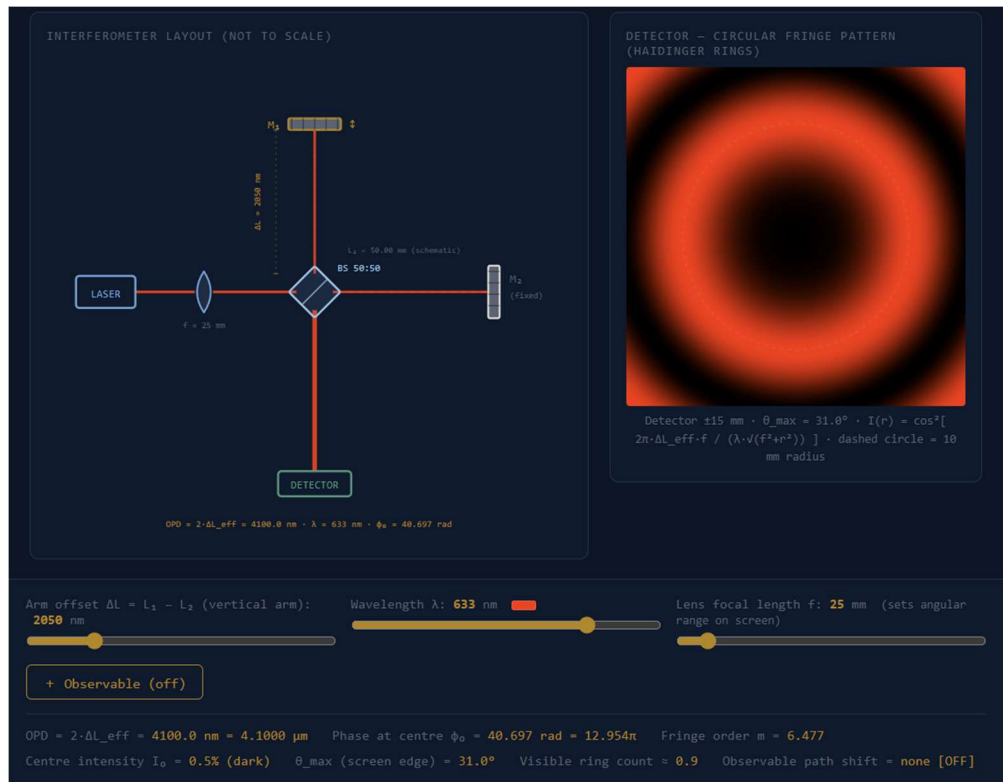


Figure 43: Centre intensity against arm offset in ~ 50 nm steps. A change of roughly $1/60$ of a wavelength produces a clearly visible change in output

This is the key point of the whole section. A ~ 50 nm change in a mirror position produces an unmistakable change in brightness. **The interferometer converts a tiny distance change into a large intensity change**, and that conversion is the entire operating principle of a gravitational-wave detector. LIGO does the same trick, just with vastly better noise control, far more optical power, and 4 km arms.

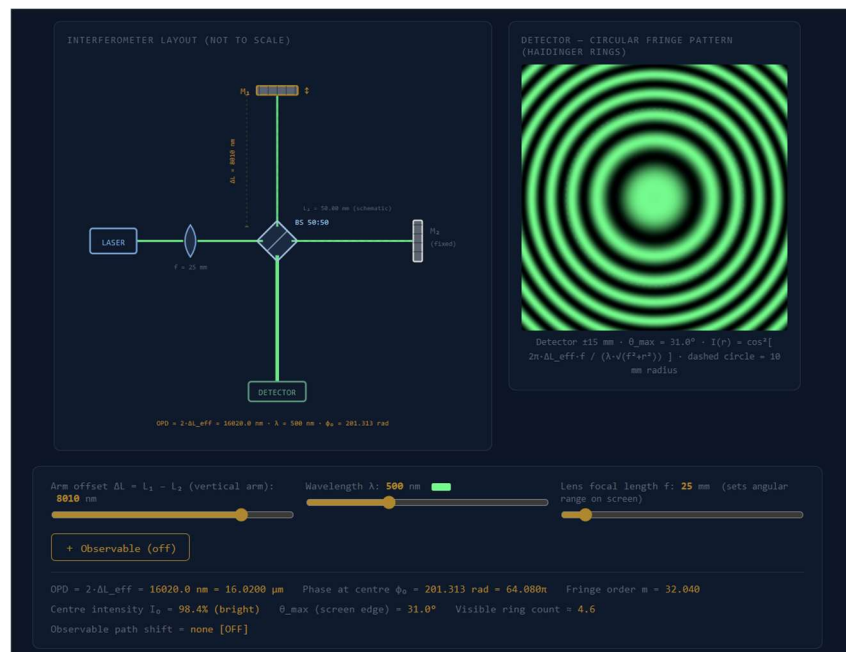
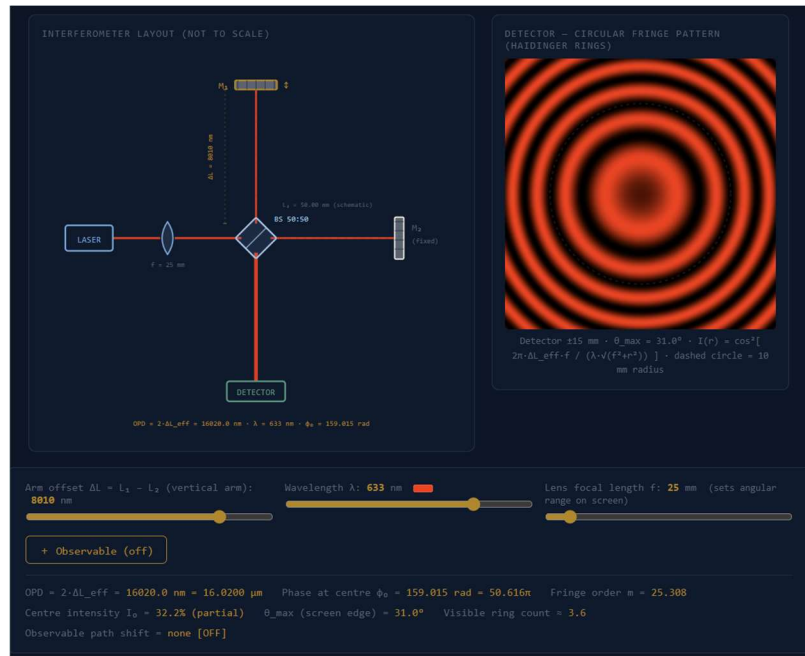


Figure 44: Identical path difference, different wavelengths. At 633 nm, $m = 25.276$ and intensity = 0.417 . At 500 nm, $m = 32.000$ exactly and the centre is fully bright.

The observable adds $\lambda/4$ to one arm, which adds $\lambda/2$ to the round-trip path difference, which is exactly a π phase shift - so the fringe order shifts by precisely half and the centre inverts.

This is how an interferometer is actually **used as a sensor**. You do not measure the path length directly. You put whatever you want to measure into one arm and read off the fringe shift it causes. Gravitational waves, gas concentrations, temperature changes and mirror displacements are all measured this way, and the instrument does not care which - it only sees the phase.

I also varied the lens focal length and watched the ring count change. Worth noting that this changes only how much of the pattern is visible, not the physics - the path difference is unaffected. Separating instrumental effects from physical ones is part of reading any measurement correctly, and it would be easy to mistake "more rings" for "bigger signal".

Dark matter. Roughly 85% of the matter in the Universe is inferred from gravitational effects - galactic rotation curves, lensing, the CMB power spectrum - and has never been detected non-gravitationally. Quantum sensors are central to several searches:

- *Axion haloscopes* (ADMX, HAYSTAC) look for axions converting to microwave photons in a strong magnetic field. The expected signal is of order 10^{-24} W, so readout uses SQUID amplifiers and Josephson parametric amplifiers; HAYSTAC has used squeezed states to scan faster than the standard quantum limit allows.
- *Atomic clocks* can search for ultralight scalar dark matter, which would make fundamental constants oscillate slightly and hence make different atomic transitions drift relative to one another. Comparing two clocks with different sensitivity to α is a dark-matter experiment - a striking example of a metrology instrument becoming a particle-physics detector without modification.
- *Atom interferometers* (MAGIS-100, AION) are being built to search for both ultralight dark matter and mid-frequency gravitational waves.

What a discovery would change

Scientifically: Identifying dark matter would be the first non-gravitational information about 85% of the matter content of the Universe, and the first confirmed physics beyond the Standard Model in the matter sector. It would immediately constrain the space of extensions - supersymmetry, axions, dark sectors - most of which are currently under-determined. A *null* result matters too, and is the more likely near-term outcome: excluding the natural parameter space of well-motivated candidates would itself be a major result, forcing a re-examination of assumptions that have shaped theory for forty years.

For society's investment in fundamental science: This is the harder question and I want to give it a genuine answer rather than the standard one.

The conventional argument is spin-off: CERN produced the World Wide Web; NASA's programmes produced a long list of technologies; LIGO's requirements drove advances in laser stability, seismic isolation and squeezed-light metrology that are already migrating into industry; and GPS is a working application of general relativity, a theory with no conceivable application when it was formulated. This argument is true but I think it is also somewhat dishonest, because it justifies curiosity-driven research on instrumental grounds and thereby concedes the framing that instrumental value is what matters. If spin-offs were the goal, funding spin-offs directly would be more efficient.

The stronger argument is about *timescale and unpredictability*. No funding process optimising for near-term return would have funded Einstein in 1915, quantum mechanics in the 1920s, or the study of a fluorescent defect in diamond in 1997. Each became infrastructure decades later. The case for fundamental science is that it is the only mechanism society has for generating options whose value cannot be assessed in advance - and the historical record of the technologies in this portfolio is the evidence. GPS, MRI, and the semiconductor industry itself are all downstream of research that had no application at the time.

A dark-matter discovery would strengthen that case rhetorically. But I would note an awkward asymmetry honestly: gravitational waves were detected in 2015 to enormous acclaim, and fundamental-science funding in most countries did not change materially as a result. Public enthusiasm and budget allocation turn out to be only loosely coupled. That suggests the argument needs to be made on grounds other than spectacle, which is part of why I am not satisfied with the spin-off version.

What's the role of quantum computing in all/any of this?

Across these four application areas, quantum computing's role is real but narrower - and further off - than quantum sensing's, and I think it is worth being precise about where it genuinely applies.

Shared physics and shared hardware: The squeezing that improves LIGO, the entanglement-enhanced clocks that could beat the standard quantum limit, and the SQUID amplifiers reading out axion detectors are all applications of quantum *information* concepts to measurement. Quantum computing and quantum sensing are two branches of one tree: superposition, entanglement and coherence are the resource in both.

Where a quantum computer would genuinely help these applications:

Three concrete cases, none of them imminent.

- (i) *Materials and molecular simulation* - designing better NV host materials, higher- T_c superconductors for SQUIDS, or MRI contrast agents and hyperpolarisation targets are quantum-chemistry problems, which is the one domain where a fault-tolerant quantum computer has a clear exponential advantage.
- (ii) *Image reconstruction* - MRI and geophysical inversion are large ill-posed inverse problems; quantum linear-algebra algorithms have been proposed, but the data-loading bottleneck and Tang's dequantization results make me sceptical of claimed exponential speedups here.
- (iii) *Quantum networks* - entangling distant clocks or telescopes over a quantum network would allow interferometry with a baseline limited by network reach rather than physical connection. This uses quantum *communication* rather than computation, and it is probably the most credible of the three.

The security dimension, which belongs here: GPS signals are authenticated cryptographically, medical records are encrypted, and both rely on public-key cryptography that Shor's algorithm would break. "Harvest now, decrypt later" means health data encrypted today may be exposed by a future quantum computer. The migration to post-quantum cryptography - NIST published its first standards in 2024 - is a present obligation driven by a future machine. This is the one impact of quantum *computing* in this artefact that is already generating real work.

An honest summary: If I had assumed at the start of this unit that quantum computing was the important quantum technology, Weeks 4 and 5 would have corrected me. Quantum sensing already underpins navigation, medical diagnosis, and fundamental physics at a scale quantum computing has not begun to approach. Quantum computing's most credible near-term contributions to *these* fields are indirect: better materials, better networks, and the cryptographic transition it forces.

Week 6 Artefact - Industry Engagement

Forum engagement with Inflection and Q-CTRL, reflection on quantum startup careers, and suggestions for future industry sessions.

A note on attendance

I was unable to attend the live industry session and Q&A in Week 6 due to a work commitment scheduled at the same time. I want to state that plainly rather than write a reflection on a session I did not experience.

What I did instead was complete the other half of the Week 6 activity: I reviewed the material published by the two companies nominated for this activity - Inflection and Q-CTRL - and posted questions to the discussion forum thread. This artefact therefore reflects on that engagement, on what I was trying to learn by asking those particular questions, and on what I would want from the next industry session, which I intend to attend.

I recognise this is a weaker form of engagement than being in the room. The part I most regret missing is the unscripted Q&A, where the interesting material usually is - a company's website tells you what it wants you to know, whereas a speaker answering an unexpected question tells you what they actually think. I have tried to be honest about that limitation throughout this artefact rather than write around it.

Forum engagement: the questions I posted

Question 1



Karamjot Singh

9 Sep 14:36

AI took off on GPUs that weren't designed for it - is there anything similar waiting in quantum, where the killer app turns out to be something nobody's currently optimising for?

[↩ Reply](#) | [✉ Mark as unread](#)

Why I asked this: The premise comes directly from Week 1 of this portfolio. While assembling the training-compute dataset, the thing that struck me was that the hardware which enabled the deep-learning era was not designed for it. GPUs were built to rasterise triangles for video games; they happened to be massively parallel single-precision matrix multipliers, which turned out to be exactly what backpropagation needs. Nobody at NVIDIA in 2005 was optimising for neural networks. The application found the hardware, not the other way round.

That made me suspicious of how confidently the quantum industry names its applications. The standard list - factoring, optimisation, quantum chemistry, machine learning - has been stable for roughly two decades, and at least one entry on it has weakened considerably: Ewin Tang's

dequantization results removed the claimed exponential speedup from several quantum machine-learning proposals by finding classical algorithms of comparable scaling. If the historical pattern holds, the application that actually matters may be one that is currently a footnote.

What I would have hoped to learn from an answer: Whether people inside the industry actually believe their own roadmaps, or whether they privately expect to be surprised. I suspect quantum simulation of quantum systems is the honest answer - it is the one application where the advantage is not an algorithmic trick but a structural match between the machine and the problem - but I wanted to hear whether a practitioner would say something less obvious.

Question 2



Karamjot Singh

9 Sep 14:37

If AI is now helping design and calibrate quantum hardware, does that create a strange dependency where quantum progress depends on the classical computing it was meant to surpass?

[↩ Reply](#) | [✉ Mark as unread](#)

Why I asked this: This question is aimed more directly at Q-CTRL, whose product is a control and error-suppression layer that makes imperfect hardware perform better than it otherwise would. Calibrating a multi-qubit processor is a high-dimensional optimisation problem over a system that drifts, and machine learning is a natural fit for it.

The dependency I was pointing at is genuine rather than rhetorical. If classical machine learning is what makes quantum hardware usable, then quantum computing's progress is currently gated by the classical computing it is often described as eventually superseding. I do not think this is a contradiction - new technologies are almost always bootstrapped by the ones they replace, and the parallel I kept coming back to is that modern chip design is impossible without the computers that previous chips made possible. But I wanted to know whether people working on it find the framing useful or think it misses something.

What I would have hoped to learn: How much of the current gap between physical and logical qubit performance is a control-and-calibration problem versus a materials problem. That distinction matters for where a computer science graduate could usefully contribute.

Key take-aways from reviewing the two companies

These take-aways come from published company material and my own background reading rather than from the live talks, and I have kept them to observations I could actually support from that source.

Take-away 1: the two companies attack the same problem from opposite ends of the stack. Inflektion builds quantum hardware - cold neutral-atom systems spanning clocks, RF sensing, navigation and computing. Q-CTRL began as an infrastructure-software company, building control and error-suppression layers that run on other organisations' hardware. They are not straightforwardly competitors, and yet both have converged on similar near-term commercial territory, particularly in precision timing and GPS-denied navigation.

What I take from this is a structural point about the industry that I had not appreciated: there is a viable business in quantum technology that does not require you to build a quantum computer. That reframes the sector from "a race to a working machine" into something more like the early semiconductor industry, which had foundries, design software, test equipment and design houses as distinct businesses.

Take-away 2: the commercial centre of gravity is sensing, not computing

Both companies derive substantial near-term activity from quantum *sensing* and timing rather than from quantum computing. This confirms, from a commercial direction, the conclusion I reached on physics grounds in Week 5: atomic clocks, magnetometers and inertial sensors are delivering real value now, while quantum computing remains pre-commercial for most applications.

I find it notable that this is almost the inverse of the public narrative. If I had formed my impression of the field from general media coverage, I would have assumed quantum computing was the whole industry and sensing a minor subfield. The Week 4 and 5 artefacts changed my view on physics and maturity grounds; reviewing these two companies changed it on commercial grounds as well.

Take-away 3: the sector employs far more classical engineers than the name suggests

Both companies advertise roles spanning control software, embedded and FPGA work, cloud infrastructure, compilers, data engineering and machine learning alongside physics. This is the take-away with the most direct personal relevance, and it corrected an assumption I had held for most of this unit - that working in quantum technology required a physics postgraduate degree.

Would I consider working for a quantum startup after graduation?

Yes - but as a software or control engineer rather than as a physicist, and with reservations I would want resolved first.

The case for: A quantum computer is a physics experiment wrapped in a large amount of entirely conventional engineering: pulse-level control software, real-time feedback on FPGAs,

automated calibration, compilers and transpilers, error-decoding that must run faster than the system decoheres, and cloud access layers. Every one of those is a software problem. As a final-year computer science student with an AI major, the classical decoding problem in particular is recognisably close to work I have already done - it is low-latency inference on streaming syndrome data under a hard deadline, which is a systems and machine-learning problem rather than a quantum one.

Two further arguments carry weight for me.

First, a small company means exposure to the whole stack rather than one component, which is a faster learning environment early in a career. Second, the transferable core of the work - low-latency systems, numerical computing, control engineering, collaborating with physicists - does not evaporate if any particular company fails.

The reservations: I do not think an honest answer can be unqualified:

- **Commercial risk:** Much of the sector is pre-revenue and dependent on grant funding and venture capital. A downturn in capital markets would contract the industry regardless of scientific progress - a risk entirely uncorrelated with the quality of the underlying work.
- **Specialisation risk:** Deep expertise in quantum control software is valuable inside the sector and less legible outside it. My mitigation would be to keep the transferable core strong rather than to avoid the field.
- **Timeline honesty:** I would want to know how a prospective employer discusses timelines internally versus in marketing. Having spent Week 1 of this portfolio arguing that qubit count is the same kind of over-read metric that transistor count became, a company candid about error-correction overheads would be a better place to learn than one that is not.
- **Market size:** Australia has real strength here, particularly in Sydney and Canberra, but the number of employers is small, which limits mobility compared with conventional software work.

My position: On balance, yes - and more seriously than I would have said at the start of this unit, because the work I would realistically be doing turns out to be much closer to my existing skills than I had assumed. I would want to see a company's internal framing of its own roadmap before committing, and I would want to keep enough conventional software work in the role that I remain employable outside a single sector.

Suggestions for the next industry session

On format

1. **Record the session and make the recording available:** This is my strongest suggestion, and I acknowledge I am not a neutral party in making it. A recording - even audio-only, even of the Q&A alone - would convert a missed session into a delayed one. If speaker consent is the obstacle, it is worth asking, because the marginal cost is low and the benefit for working students is substantial.

On content

1. **Keep including quantum sensing companies, not only computing.** Weeks 4 and 5 of this portfolio argue that sensing is currently the more commercially mature branch, and the unit title covers applications of quantum technologies broadly. Inflection's sensing and timing work is a good example of exactly this, and I would want more of it rather than less.
2. **Include someone two or three years out of a similar degree.** A founder or chief scientist describes a destination; a recent graduate describes the path. For career planning purposes the second is more useful, and more credible.

Closing reflection

The honest summary of my Week 6 is that I engaged with the activity in the way that was available to me, and that it was a genuinely useful but incomplete substitute. Reviewing the two companies and writing questions forced me to consolidate what I had concluded across the earlier artefacts - particularly that quantum sensing is further along than quantum computing, and that the sector needs more software engineers than its name suggests. Both of those changed how I think about my own career, which is more than I expected from this week.

What I could not get was disagreement. Every conclusion in this artefact is my own reasoning checked against published material, and published material does not push back. I intend to attend the next industry session, and the two questions above are the ones I would ask in person.

Declaration and Reflection on Generative AI Use

What was used, and where

Artefact	How genAI was used	What was done independently
Week 1	Suggesting candidate factors for the “end of Moore’s law” discussion; sanity-checking the Dennard scaling derivation; identifying that Rock’s law was relevant.	Ran the widget experiments and recorded the results; assembled and checked the transistor dataset; interpreted the fit and residuals; decided what the fly calculation actually demonstrated; formed my own position on the AGI question
Week 2	Generating the extra practice problems in the extension section, as the tutorial explicitly instructs ; confirming the standard S-gate convention when I found the discrepancy; formatting the matrices and equations in Word	All pen-and-paper working; all extension solutions attempted by hand before checking; ran the widget and checked one element of its output by hand; worked out the 22.5° Bloch sphere interpretation
Week 3	Help with the Python simulation code; explaining why a truncated Gaussian produces spurious high-k components in an FFT, after I had already found the anomaly and suspected the box size; help formatting the equations	Ran all the widget experiments and recorded the readouts; found the $\sigma = 6$ anomaly myself; found the $\hbar/2$ versus “about 1” discrepancy and worked out the factor of $\sqrt{2}$; ran the $N = 5$ statistics investigation after the result looked wrong
Week 4	Used AI to discuss the concepts and identify what to search for	Researched the sources and wrote the section in my own words
Week 5	prompting for counter-arguments to positions I had already formed, to test them.	Researched the sources and wrote the section in my own words
Week 6	None used to write the artefact or the forum questions.	Both forum questions, the reasoning behind them, the review of the two companies, and the whole reflection.
Document	Converting the mathematics into properly formatted equations; LaTeX and Word formatting; structuring the document; proofreading	The argument, the conclusions, and the decisions about what each section should say

Critical reflection on the use of generative AI

Where it was genuinely useful

Knowing where to look: This was probably the biggest time saving, and it is not the one I expected. For unfamiliar topics - NV centres, SQUID operation, the history of the SI second - the hard part is not reading the source, it is knowing which source exists and what the field calls the thing you are trying to find. Asking "what should I search for to understand how a SQUID converts flux into voltage" got me to the right terminology, and from there I could read properly. I did the reading; AI shortened the search.

Understanding what the tutorial questions were actually asking: Several questions were open enough that I was not sure what a good answer looked like. Talking through what "explain the basic operational principle of atomic clocks" was really asking for helped me realise the interesting part was the feedback loop, not the atom. That reframing came from discussion, and the answer came from me.

Python and plotting: I used AI heavily here, and I want to be clear about what that means. I could not have written the FFT-based uncertainty calculation from scratch. What I did do was read the code carefully, work out what each step was doing, check the outputs against values I could verify independently, and change things when the results looked wrong. When the $\sigma = 6$ row came out at 0.5073 instead of 0.5000, I could tell that was a problem and form a hypothesis about why, which I could not have done if I had just run a script and copied the output. The code was assisted; the interpretation is mine.

Understanding the widgets: The widgets have readouts that are not all self-explanatory, and I used AI to work out what quantities like the analytic transmission or the fringe order were actually computing. Once I understood what I was looking at, the experiments and the observations were my own.

Counter-arguments: The use I found most valuable was asking for the strongest objection to a position I had already reached. In Week 1 I had concluded that Moore's law dies at the atomic scale, and asking for the objection produced the 3D-stacking counter-argument, which substantially changed my conclusion. That made the analysis genuinely better rather than replacing it, because I still had to decide what to do with the objection.

Formatting and presentation. I used AI to convert my handwritten mathematics into properly formatted equations, to set up the document structure, and to tidy up wording. This is the most straightforwardly mechanical use and the one I have least to say about, but it is real and it saved a lot of time.

Company research for Week 6: I used AI to build an initial picture of Inflection and Q-CTRL - what they build, where they sit in the industry, what kinds of roles they hire for - and then went to their own material to check it. That gave me enough context to ask questions that were specific rather than generic, which was the point of the activity.

Where it was unreliable, and what I did about it

Numerical and citation accuracy was by far the weakest area: Language models produce specific values - transistor counts, dates, sensitivity figures, paper titles, author lists - with exactly the same fluency whether or not they are correct, and there is no signal in the output telling you which is which.

I treated every number in this portfolio as requiring independent verification, and I recomputed anything that could be recomputed. All the fits, derived quantities and figures in Weeks 1 to 5 came out of code I ran myself, which is the strongest defence available: a doubling time of 2.03 years is a fact about my dataset that I can reproduce, not a number I was told.

Citations are the failure mode I would warn another student about most strongly: AI-generated references are formatted perfectly, attributed to real authors working in the right field, and published in real journals - and sometimes do not exist. The formatting plausibility is exactly what makes them dangerous, because every normal heuristic for spotting a dubious source passes. I checked every reference in this portfolio by locating it myself, and I removed the ones I could not find or had not actually read. I would rather submit a shorter reference list I can stand behind than a longer one I cannot.

Fluency can masquerade as understanding: When I asked for an explanation of something I did not yet understand, I would often come away feeling like I understood it, because the explanation was clear and I had followed it. But following an explanation and being able to reconstruct it are different states, and the difference is invisible from the inside.

My defence was to force a productive step before moving on - derive it on paper, or write something that reproduces the result. The $\hbar/2$ result in Week 3 is the clearest case. I could have accepted the tutorial's $\Delta x \cdot \Delta p \approx 1$, or accepted an explanation of the factor of two. What actually convinced me was computing Δx and Δp from their definitions and getting exactly 0.5000 across five values of σ - and that is also what let me recognise the $\sigma = 6$ row as a numerical artefact rather than an error in my understanding.

Generic analysis: Asking for "the societal implications of quantum sensing" produces a competent, balanced, generic list. It is not wrong, and it is not worth much, because it reads like anyone could have written it - which they could.

The parts of this portfolio I think are strongest are the ones where I took a position the generic answer would not have taken: that the flies passing Earth's mass in two years is the real lesson

rather than the size of the final number; that better mineral exploration is in tension with environmental goals rather than aligned with them; that low-field portable MRI may matter more socially than higher-performance imaging. Getting to those required disagreeing with the first answer I was given, which is a thing you have to choose to do.

What I understood myself

I want to be direct about this, because it is what the declaration is actually for.

I attempted every piece of mathematics and physics in this portfolio on pen and paper before checking it against anything. The complex arithmetic, the matrix products, the eigenvalue problems, the Dennard scaling derivation, the tunnelling probabilities, the interferometer relationships - all of it was worked by hand first. Where I got something wrong, I have said so in the artefact, including the S-versus-Pauli-X naming issue in Week 2 and the initial mistakes in Weeks 1 and 3.

I ran every widget experiment myself and recorded the readouts as I went rather than reconstructing them afterwards. The observations, the tables of measurements, and the things I noticed are mine.

I understand the concepts I am writing about. I could explain what a wave packet is, why a Gaussian saturates the uncertainty bound, why flux quantisation requires superconductivity, why the Hadamard gate is a 180° rotation about a diagonal axis, and why interference disappears when path information exists - without this document in front of me. That was the test I applied when deciding whether a section was finished: if this were removed from the portfolio and I were asked to reproduce the reasoning on a whiteboard, could I? Where the answer was no, I went back and did the work properly.

What I would do differently

Three things.

Write my own position before asking anything: Once I had read a fluent summary, it was noticeably harder to think of framings outside it. That is an anchoring effect and I could feel it operating. Drafting my own rough answer first, even a bad one, would have protected against it.

Use it later in the process rather than earlier: It is far more useful as an adversary against something I have already written than as a source for a blank page. The times I got the most value were when I had a position to defend.

Be more systematic about verification: I checked things opportunistically as I went. Keeping a running list of every claimed fact and its independent source would have been better, and would have made writing this declaration easier.

A closing thought on the ethics

The line I have tried to hold is between using a tool to **learn faster** and using a tool to **avoid learning**. Both feel productive while you are doing them, and only one leaves you able to do the thing afterwards.

I have used AI heavily in producing this portfolio, and I have said so specifically rather than minimising it. I think that use made me learn more than I would have alone, mainly because it removed the friction of not knowing where to start and let me spend the time on the parts that actually needed thinking. But it only worked because I kept checking that I could stand behind each piece without it.

I am declaring this in detail because I would rather be assessed on an accurate account of how I worked than on a flattering one.

References

Semiconductor scaling and computing history (Week 1)

- Moore, G. E. (1965). *Cramming more components onto integrated circuits*. Electronics, 38(8). - Computer History Museum archive
- Dennard, R. H. et al. (1974). *Design of ion-implanted MOSFETs with very small physical dimensions*. IEEE JSSC, 9(5), 256–268.
- Our World in Data - *Transistors per microprocessor* dataset (source for the transistor counts)
- IEEE - *International Roadmap for Devices and Systems (IRDS)*
- Kaplan, J. et al. (2020). *Scaling laws for neural language models*. arXiv:2001.08361
- Hoffmann, J. et al. (2022). *Training compute-optimal large language models*. arXiv:2203.15556
- Sevilla, J. et al. (2022). *Compute trends across three eras of machine learning*. arXiv:2202.05924
- Villalobos, P. et al. (2022). *Will we run out of data?* arXiv:2211.04325
- Epoch AI - *Notable AI models* database (source for the training compute figures)
- International Energy Agency (2024). *Electricity 2024*.
- Tang, E. (2019). *A quantum-inspired classical algorithm for recommendation systems*.

Quantum computation and information (Weeks 2–3)

- Nielsen, M. & Chuang, I. (2010). *Quantum Computation and Quantum Information*, Cambridge University Press - Chapters 1–2 for the postulates and the Bloch sphere. Check the Swinburne library for an e-copy.
- IBM Quantum Learning - *Basics of quantum information* course, for the gate and Bloch sphere material
- Shor, P. (1995). *Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer*. arXiv:quant-ph/9508027
- Grover, L. (1996). *A fast quantum mechanical algorithm for database search*. arXiv:quant-ph/9605043
- 3Blue1Brown - *Essence of Linear Algebra* video series, for the geometric intuition behind eigenvectors
- Whatever online matrix calculator you use for the verification screenshots - cite it properly
- HyperPhysics (Georgia State University) - *Quantum mechanics: barrier penetration and wave packets*, for the tunnelling transmission formula
- MIT OpenCourseWare 8.04 *Quantum Physics I* - lecture notes on Gaussian wave packets, the Born rule and the uncertainty principle
- PhET Interactive Simulations (University of Colorado Boulder) - *Quantum Tunnelling and Wave Packets*, used as a cross-check against the provided widget
- Robertson, H. P. (1929). *The uncertainty principle*. *Physical Review*, 34(1), 163–164 - the source of the general $\Delta A \cdot \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$ form
- NumPy documentation - `numpy.fft` reference, for the FFT normalisation convention I used in Section 4.1
- Griffiths, D. J. & Schroeter, D. F. (2018). *Introduction to Quantum Mechanics*, 3rd ed. - Chapters 1–3 for the Born rule and minimum-uncertainty states. Check the Swinburne library.

Atomic clocks and atom interferometry (Weeks 4–5)

- BIPM - *The International System of Units (SI Brochure)*, 9th edition, for the definition of the second
- NIST - Time and Frequency Division pages, for the caesium fountain standards and the history of timekeeping

- Ludlow, A. et al. (2015). *Optical atomic clocks*. Reviews of Modern Physics 87, 637 (arXiv:1407.3493) - for the stability formula and systematic limits
- Ashby, N. (2003). *Relativity in the Global Positioning System*. Living Reviews in Relativity 6, 1 - open access, and the source for the $\pm 7/+45$ $\mu\text{s/day}$ figures
- Ramsey, N. F. (1950). *A molecular beam resonance method with separated oscillating fields*. Physical Review 78, 695
- Degen, C., Reinhard, F. & Cappellaro, P. (2017). *Quantum sensing*. Reviews of Modern Physics 89, 035002 (arXiv:1611.02427) - the main review I used for NV centres and the standard quantum limit
- Gruber, A. et al. (1997). *Scanning confocal optical microscopy and magnetic resonance on single defect centers*. Science 276, 2012 - the single-NV detection result
- Josephson, B. D. (1962). *Possible new effects in superconductive tunnelling*. Physics Letters 1, 251
- Jaklevic, R. C. et al. (1964). *Quantum interference effects in Josephson tunneling*. Physical Review Letters 12, 159 - the first dc SQUID
- Clarke, J. & Braginski, A. (eds). *The SQUID Handbook* - for the flux-locked loop and gradiometer material
- Cronin, A., Schmiedmayer, J. & Pritchard, D. (2009). *Optics and interferometry with atoms and molecules*. Reviews of Modern Physics 81, 1051 (arXiv:0712.3703)
- Boto, E. et al. (2018). *Moving magnetoencephalography towards real-world applications with a wearable system*. Nature 555, 657 - for the OPM comparison
- LIGO Caltech - *What is an interferometer?* and the published detector noise budgets
- Abbott, B. P. et al. (2016). *Observation of gravitational waves from a binary black hole merger*. Physical Review Letters 116, 061102
- Tse, M. et al. (2019). *Quantum-enhanced Advanced LIGO detectors in the era of gravitational-wave astronomy*. Physical Review Letters 123, 231107 - the squeezed-light upgrade and the range improvement figures
- Stray, B. et al. (2022). *Quantum sensing for gravity cartography*. Nature 602, 590 - the buried tunnel detection result
- Backes, K. M. et al. (2021). *A quantum enhanced search for dark matter axions*. Nature 590, 238 - HAYSTAC and squeezing below the SQL

- US Department of Homeland Security / CISA - GPS and critical infrastructure dependency material
- Australian Government Department of Industry - *National Quantum Strategy*, for the Australian policy context
- CSIRO Futures - *Australia's Quantum Technology Roadmap*
- NIST - *Post-Quantum Cryptography Standards* (FIPS 203, 204, 205, published 2024)
- Whatever source you use for the Juukan Gorge material - the Parliamentary Joint Standing Committee inquiry report *A Way Forward* (2021) is the authoritative one
- Australian Institute of Health and Welfare - for MRI availability and the metropolitan/regional access gradient, if you want a hard number for the equity argument

Magnetometry: NV centres and SQUIDs (Week 4)

- Josephson, B. D. (1962). Possible new effects in superconductive tunnelling. *Physics Letters*, 1(7), 251–253.
- Anderson, P. W., & Rowell, J. M. (1963). Probable observation of the Josephson superconducting tunneling effect. *Physical Review Letters*, 10(6), 230–232.
- Jaklevic, R. C., Lambe, J., Silver, A. H., & Mercereau, J. E. (1964). Quantum interference effects in Josephson tunneling. *Physical Review Letters*, 12(7), 159–160.
- Clarke, J., & Braginski, A. I. (Eds.) (2004). *The SQUID Handbook, Vol. 1: Fundamentals and Technology of SQUIDs and SQUID Systems*. Wiley-VCH.
- Gruber, A., Dräbenstedt, A., Tietz, C., et al. (1997). Scanning confocal optical microscopy and magnetic resonance on single defect centers. *Science*, 276(5321), 2012–2014.
- Maze, J. R., Stanwix, P. L., Hodges, J. S., et al. (2008). Nanoscale magnetic sensing with an individual electronic spin in diamond. *Nature*, 455, 644–647.
- Balasubramanian, G., Chan, I. Y., Kolesov, R., et al. (2008). Nanoscale imaging magnetometry with diamond spins under ambient conditions. *Nature*, 455, 648–651.
- Degen, C. L., Reinhard, F., & Cappellaro, P. (2017). Quantum sensing. *Reviews of Modern Physics*, 89(3), 035002.
- Boto, E., Holmes, N., Leggett, J., et al. (2018). Moving magnetoencephalography towards real-world applications with a wearable system. *Nature*, 555, 657–661.

Fundamental science applications (Week 5)

- Abbott, B. P., et al. (LIGO Scientific Collaboration and Virgo Collaboration) (2016). Observation of gravitational waves from a binary black hole merger. *Physical Review Letters*, 116(6), 061102.
- Tse, M., et al. (2019). Quantum-enhanced Advanced LIGO detectors in the era of gravitational-wave astronomy. *Physical Review Letters*, 123(23), 231107.
- Wineland, D. J., Bollinger, J. J., Itano, W. M., Moore, F. L., & Heinzen, D. J. (1992). Spin squeezing and reduced quantum noise in spectroscopy. *Physical Review A*, 46(11), R6797–R6800.
- Backes, K. M., Palken, D. A., Al Kenany, S., et al. (2021). A quantum enhanced search for dark matter axions. *Nature*, 590, 238–242.

National Institute of Standards and Technology (2024). *Post-Quantum Cryptography Standards* (FIPS 203, 204, 205). U.S. Department of Commerce.

Computational Appendix

All numerical results, fits and figures in Weeks 1–5 were produced by four Python scripts written for this portfolio, using NumPy for computation and Matplotlib for plotting. No figure was reproduced from a published source. The scripts are summarised below; each prints its numerical output to the console so that every value quoted in the text is traceable to a specific line of output.

Script	Artefact	What it computes
w1.py	Week 1: Moore's law	23-point transistor dataset; least-squares fit of $\log_2 N$ vs year (global and split ranges) with R^2 and residuals; extrapolation to 2040 against the silicon atomic budget; fruit-fly biomass; clock/voltage/power-density data; gate-oxide tunnelling transmission; ML training-compute trend fits.
w2.py	Week 2: linear algebra	Independent verification of every hand-computed result; Hermiticity, unitarity, determinant and trace checks; eigendecomposition with global-phase fixing; extension problems (Bloch operators, rotation operators, Bell state and reduced density matrix purity); Bloch-sphere figure.
w3.py	Week 3: physics primer	Travelling-wave and superposition plots; random-phase intensity statistics with ensemble averaging and power-law fit; normalised Gaussian wave packets with numerical normalisation check; FFT-based momentum distributions; Δx , Δp and their product across six values of σ ; top-hat counter-example.
w45.py	Weeks 4–5: sensors	Historical clock accuracy; relativistic geodesy and GPS timing conversions; Newton's rings pattern and ring radii; magnetic-field magnitude comparison; SQUID $I_c(\Phi)$ and $V(\Phi)$ with flux-per-fringe calculations; de Broglie wavelengths and Mach–Zehnder phase response; standard quantum limit vs Heisenberg limit; LIGO strain and noise budget.